

where  $n$  must be divisible by 4. These matrices give rise to a class of symmetric block designs called Hadamard configurations.

**HADAMARD PRODUCT:** The matrix of the entry by entry products of two given matrices of the same dimensions; that is, where  $C$  is the Hadamard product of  $A$  and  $B$ ,  $c_{ij} = a_{ij} b_{ij}$ .

**HADAMARD, JACQUES SALOMON (1865-1963):** French mathematician who proved the prime number theorem and made major contributions to complex function theory, differential geometry and partial differential equations.

**HADEAN (HADEAN EON):** The geological time period during which Earth formed, from the time of the accretion of planetary material, around 4600 million years ago, to the date of the oldest known rocks, about 3800 million years ago.

**HADLEY CELL:** A vertical circulation of air in the atmosphere, caused by convection. It often occurs in the tropics. It is a direct, thermally driven and zonally symmetric circulation. It consists of air rising up into the atmosphere at or near the equator, flowing toward the poles above the surface of the Earth, returning to the Earth's surface in the subtropics and flowing back towards the equator. This flow of air occurs because the Sun heats air at the Earth's surface near the equator. The warm air rises, creating a band of low pressure at the equator. Once the rising air reaches the top of the troposphere at approximately 10-15 kilometres above the Earth's surface, the air flows toward the north and south poles. The Hadley cell eventually returns air to the surface of the Earth in the subtropics, near 30 degrees north or south latitude.

**HADRON:** A subatomic particle consisting of mesons and baryons that is built from quarks and expresses a strong nuclear force. All hadrons have short lives and are produced in high-energy collision of subatomic particles and are subject to gravitation. Charged hadrons are subject to electromagnetic forces. Some hadrons break up by way of the weak force, others decay via the strong and electromagnetic forces.

**HADRON-ELECTRON RING ACCELERATOR (HERA):** A particle accelerator built in a tunnel in Hamburg, Germany, which is 6.3 km in length and is the only particle accelerator that is capable of bringing about collisions between beams of electrons or positrons and beams of protons. It began operations in 1992 and is capable of accelerating protons to energies of 820 GeV (billion electron volts) and electrons to 30 GeV.

**HAEM:** The iron-containing group of haeme proteins such as haemoglobin and the cytochromes.

**HAEMATITE:** A dense darkened mineral containing iron(III) oxide ( $\text{Fe}_2\text{O}_3$ ), with Mohs hardness 5.5 to 6.5 and specific gravity 4.2-5.25. It is the most abundant ore of iron. It is widely distributed all over the world, occurring in rocks of all ages. It also occurs in nature in red to reddish-brown earthy masses and in steel-grey to black crystalline forms. Haematite that has a metallic lustre is called **specular haematite** or **specular iron**. The red powdered haematite is used as a pigment or ochre and as rouge in polishing.

**HAEMATOCRIT:** The relative volume of blood occupied by erythrocytes (red blood cells). The hematocrit is abbreviated as Hct and expressed in % (volume).

**HAEMATOLOGY:** A branch of medicine concerned with blood diseases, such as anaemia, leukaemia and blood clotting disorders.

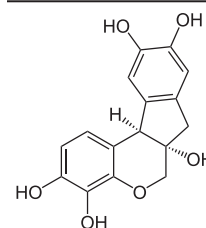
**HAEMATOMA:** A pool of clotted or partially clotted blood in an organ, tissue or body space, usually caused by a broken blood vessel. It may occur almost anywhere in the body and, depending on the site and amount of accumulated blood, may vary in severity from a minor to a potentially fatal disorder.

**HAEMATOPHAGOUS:** Referring to organisms which feed on blood. Examples include certain leeches, vampire bats and anopheles mosquitoes.

**HAEMATOPOIETIC STEM CELL:** An unspecialised precursor cell that will develop into a mature blood cell.

### HAEMATOXYLIN (NATURAL BLACK 1; 7,11b-DIHYDROINDENO[2,1-C]CHROMENE-3,4,6a,9,10(6H)PENTOL):

A compound with the molecular formula  $\text{C}_{16}\text{H}_{14}\text{O}_6$  extracted from the wood of the logwood tree. When oxidised it forms haematein, a compound that forms strongly coloured complexes with certain metal ions, notably Fe(III) and Al(III) salts. It is used in its oxidised form as a blue dye in optical microscopy, particularly, for staining smears and sections of animal tissue.



**Chemical formula of Haematoxylin**

**HAEME:** A subunit of the haemoglobin molecule, with the molecular formula  $\text{C}_{34}\text{H}_{32}\text{FeN}_4\text{O}_4$ . It consists of an organic part, called protoporphyrin IX and an iron atom. The organic part is made up of four rings that are linked together to form a tetra structure. The iron atom is in the centre of the four rings and binds to oxygen. Structurally it resembles haematin with the iron in the Fe(II) state and with no chlorine. Haeme combines with the protein globin to form the blood pigment haemoglobin. Haematic describes anything resembling, containing or acting on blood.

**HAEMIN TEST (TEICHMANN TEST):** A test for blood hydrochloride of haematin, derived from haemoglobin. The crystals are readily recognisable under the microscope.

**HAEMOCHROMOGEN:** A bright red compound formed from or related to haemoglobin with a nitrogenous base, as a globin or pyridine, typically with two absorption bands in the green part of the visible spectrum. **Haemochromogen test** or **Takayama test** is a microcrystal test which uses the characteristic appearance of haemochromogen crystals observed under a microscope in the identification and confirmation of the blood samples or stains.

**HAEMOCOEL:** The body cavity of arthropods and molluscs, which acts as a hydrostatic skeleton and filled with a combination of blood, lymph and interstitial fluid called **haemolymph**, circulating within it.

**HAEMOCYANIN:** The oxygen-carrying substance like haemoglobin with a bluish pigment found in the blood or haemolymph of some arthropods and molluscs that gives them a bright blue tinge and also responsible for transporting oxygen. Haemocyanins are respiratory proteins in the form of metalloproteins containing two copper atoms that reversibly bind a single oxygen molecule ( $\text{O}_2$ ). Oxygenation causes a colour change between the colourless Cu(I) deoxygenated form and the blue Cu(II) oxygenated form. Unlike the haemoglobin in red blood cells found in vertebrates, haemocyanins are not bound to blood cells but are instead suspended directly in the haemolymph.

**HAEMOCYTOMETER:** A device used to estimate the concentration of cellular components of a known volume of blood viewed with a microscope. It consists of a glass slide with a shallow depression divided into fine-meshed grid.

**HAEMODIALYSIS:** A dialysis process in which undesired metabolites and toxic by-products, such as urea and creatine, are removed from blood.

**HAEMODYNAMICS:** The study of the flow of blood through the circulatory system. It uses and applies the principles of fluid dynamics to the special conditions pertaining to a living circulatory system that contains a fluid such as blood with unusual properties and vessels that can change their shape.

**HAEMOERYTHRIN:** A red iron-containing respiratory pigment responsible for oxygen ( $\text{O}_2$ ) transportation that occurs in the blood of certain worms, especially sipunculids, priapulids, brachiopods and in a single annelid worm, magelona.

**HAEMOGLOBIN:** Groups of iron-containing proteins or red pigment found in red blood corpuscles, used by all vertebrates and some invertebrates for oxygen transport because the two substances combine reversibly.

**HAEMOGLOBIN S, Hb S (SICKLE CELL HAEMOGLOBIN):** An abnormal form of haemoglobin produced in sickle cell disease, in which valine has replaced glutamic acid causing the haemoglobin to become less soluble under decreasing oxygen concentrations and to polymerise into crystals that distort the red blood cells into a sickle shape.

**HAEMOGLOBINIC ACID:** A very weak acid formed inside red blood cells when hydrogen ions combine with haemoglobin. The presence of the hydrogen ions, which are produced by the dissociation of carbonic acid encourages oxyhaemoglobin to dissociate into haemoglobin and oxygen. The oxygen diffuses into the tissue cells and the haemoglobin acts as a buffer for the excess hydrogen ions, which it takes up to form haemoglobinic acid.

**HAEMOGLOBINURIA (HAEMOGLOBINURIA):** The presence of haemoglobin in the urine. Haemoglobin is mainly contained within red blood cells, although a small amount is free in the blood plasma. Excessive haemolysis or breakdown of red bloods may be caused by heavy exercise, cold weather, falciparum malaria (blackwater fever) and haemolytic anaemia, increasing the concentration of free haemoglobin in the plasma. The excess of free haemoglobin in the plasma is excreted in the urine.

**HAEMOLYMPH:** Circulatory fluid containing water, sugars, salts, amino acids and white cells in the circulatory system of those molluscs and arthropods that have an open circulatory system. It also contains haemocyanin, a copper-based protein which turns blue when oxygenated and when not oxygenated, quickly loses its colour and appears grey.

**HAEMOLYSIS:** The breakdown of red blood cells or the loss of haemoglobin from red blood cells. It is a normal occurrence, chiefly in the blood sinuses of the spleen, when the cells are ageing after 3-4 months in circulation, but also as a result of damage to the cell membrane. The damage may be due to the action of disease-causing micro organisms, osmosis, poisons and antibodies in mismatched blood transfusions or certain allergic reactions, resulting in anaemia and jaundice.

**HAEMOLYTIC ANAEMIA:** A rare form of anaemia in which red blood cells (erythrocytes) are destroyed and removed from the bloodstream before their usual lifespan is up. Healthy red blood cells usually live about 120 days in the bloodstream before the body removes them. In haemolytic anaemia, the body breaks down and removes red blood cells faster than it can replace them. The breakdown of red blood cells is called **haemolysis**.

**HAEMOPHILIA:** Any of the several hereditary genetic disorders linked to a recessive gene on the X-chromosome, in which normal blood clotting or coagulation is impaired. It results in prolonged bleeding externally and internally bleeding from even minor injuries. It is a sex-linked character in humans, occurring almost exclusively in men and boys.

**HAEMOPOIETIC TISSUE:** The tissue that gives rise to all the blood cell types in the process of haemopoiesis (hematopoiesis); constituting 1:10.000 of cells in myeloid tissue. The haemopoietic tissue of the embryo and foetal stage of vertebrates is the bone marrow, liver, thymus, spleen, yolk sac and lymph nodes but after birth, haemopoiesis occurs in the red bone marrow.

**HAEMOPOIESIS:** The production of all types of blood cells (white blood cells, red blood cells and platelets) including formation, development and differentiation in the bone marrow. It begins with the most basic blood cell, the stem cell. Prenatally, haemopoiesis occurs in the yolk sack, then in the liver and lastly in the bone marrow. In the normal situation, in adults it occurs in the bone marrow and lymphatic tissues.

**HAEMORRHAGE:** An acute internal or external bleeding from the blood vessels in the body. Trauma or injury to major arteries, veins or the heart may lead to massive haemorrhage.

**HAEMORRHOIDS (PILES):** Dilated or enlarged veins at the junction of the rectum and anus, which may bleed or come down through the anal canal, usually on defecation and which are made worse

by constipation and straining. Internal haemorrhoids are located above the dentate line, are covered by mucosa and do not have sensory innervations. External haemorrhoids are located below the dentate line, are covered by squamous epithelium and have sensory innervation.

**HAEMOSTASIS:** The process of stopping of bleeding or haemorrhage. This can occur naturally or by surgical procedure. In the natural mechanism, it can occur by coagulation; by the vessel walls closing temporarily, called vasoconstriction; or by an abnormal obstruction caused by, for example, a plaque.

**HAEMOTHORAX:** Blood in the pleural cavity, usually due to injury. If the blood is not drained dense fibrous adhesions occur between the pleural surfaces, which can impair the normal movement of the lungs. The blood may also become infected.

**HAFELE-KEATING EXPERIMENT:** A 1971 test of the predictions of time dilation in which four caesium atomic beam clocks were flown in aircrafts, east and west, around the world and then compared to the one that remained fixed in the same location on Earth at the United States Naval Observatory. All the three sets of clocks did not register the same time. The clock flying eastward, being the fastest relative to the surface of the Earth, recorded the smallest elapsed time. The experiment also had to take into account the effects of general relativity since the clocks were at different heights in a gravitational field. Joseph C. Hafele, a physicist, and Richard E. Keating, an astronomer, performed the experiment as a test of the theory of relativity.

**HAFNIUM:** A silvery metallic, tetravalent transition metal with the symbol Hf, atomic number 72 and relative atomic mass 178.49. They absorb neutrons better than most metals and are used in the control rods of nuclear reactors, as well as for light-bulb filaments.

**HAHN DECOMPOSITION:** A partition (P, N) of a sigma-ring, X, equipped with a signed measure,  $\mu$ , such that  $P \perp N = X$ ,  $P \perp N = X$  and, for all A in the sigma-ring on X,  $\mu(A \cap P) \geq 0$  and  $\mu(A \cap N) \leq 0$ .

**HAHN-BANACH THEOREM:** The extension theorem proving that a linear functional defined on a subspace of a vector space and dominated thereon by a sublinear function defined on the entire space has a linear extension still dominated by the sublinear function. The geometric form of the Hahn-banach theorem is known as the separation theorem of Mazur.

**HAHN, OTTO (1879-1968):** German chemist, who studied in London under William Ramsay, and in Canada with Ernest Rutherford before returning to Germany in 1907. In 1917, together with Lise Meitner, he discovered protactinium. In 1938, with Fritz Strassmann (1902-80) they bombarded uranium with slow neutrons and found the first chemical evidence of nuclear-fission products, one of which was barium. Meitner interpreted the process as nuclear fission. In 1944, he received the Nobel Prize for Chemistry for his discovery of nuclear fission.

**HAIL:** A precipitation in the form of pellets of ice common in the mid-latitudes with diameter of between 5 mm and 10 cm. It can cause damages to buildings and crops. Hail forms as a result of the strong updrafts common in severe weather systems. When a strong convective cell forms, warm air rises and cool air sinks. If there is a sufficient amount of supercooled water, accumulation of ice can begin in the clouds. Rising air often reaches a point in the atmosphere that is below freezing and ice forms. The ice is suspended in the air by the strong updrafts and will later fall down.

**HAIR:** 1. A multicellular thread-like structure consisting of keratinised cells growing from follicles found in the dermis of the mammalian skin. In humans, the development of hair begins in the embryo. Except for albinos, hair contains pigment and has neither blood vessels nor nerves, but is nourished by the blood vessels in a papilla at the root of the hair. 2. Any of the various thread-like structures on plants.

**HAIR ANALYSIS:** A diagnostic technique in which a sample of hair is analysed in a procedure called atomic-emission spectroscopy. This is done to determine levels of potentially toxic elements such as molecules of lead compound and also for nutritional purposes

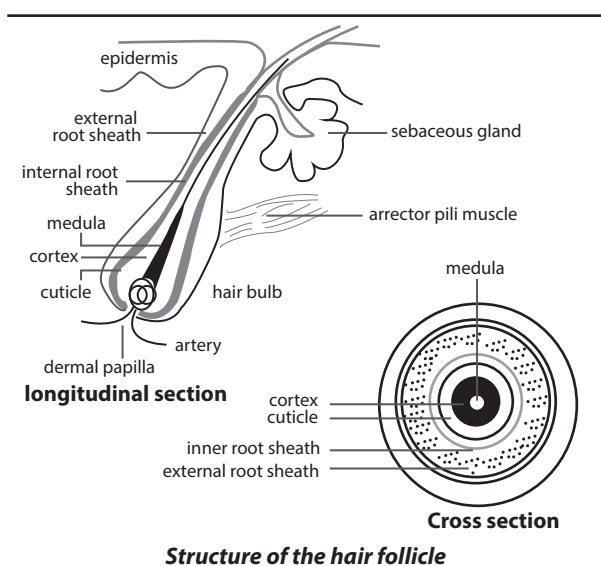
to determine levels of essential elements necessary for healthy body function.

**HAIR BALLS:** A small collection of hair or fur formed in the stomach of animals, making digestion difficult. It is formed in the stomach when pieces of hair and fur are ingested, when animals groom themselves by licking. They can cause fits and convulsions in very young calves and vision may be slightly impaired.

**HAIR BULB:** The base of a hair, which is enlarged and softer in texture than the shaft. It contains cells that divide mitotically to produce columns of hair cells.

**HAIR CELL:** The sensory receptors of both the auditory system and the vestibular system in all vertebrates, equipped with hair-like cilia and adapted to detecting movement of the surrounding medium, whether water or air; hence it transduces mechanical stimuli into electrical stimuli, in the form of nerve impulses in associated sensory nerves. In mammals, the auditory hair cells are located within the organ of Corti on a thin basilar membrane in the cochlea of the inner ear.

**HAIR FOLLICLE:** A narrow tubular depression of the epidermis in mammalian skin containing the root of a hair.



**HAIR ROOT:** The portion of a hair that is enclosed in a hair follicle and extends from the skin's surface to the hair bulb.

**HAIR SHAFT:** The portion of a hair that extends above the skin's surface.

**HAIRWORM:** 1. A very thin worm of the species *Capellaria*, which infests poultry. 2. Any parasitic nematode worm of the genus *Trichostrongylus* that infests the stomach and small intestine of cattle, sheep and related animals. 3. A long slender worm, phylum Nematomorpha, found in water or damp soil whose larvae live as parasites on arthropods.

**HAIRY BALL THEOREM:** In mathematics, the term refers to the theorem that any continuous tangent vector field on the sphere must have a point where the vector is zero. This theorem also means that somewhere on the Earth's surface there has to be a point where the horizontal wind speed is zero, even if it's windy everywhere else.

**HAKE BAR:** An attachment which links a trailed plough to the tractor.

**HALBARD (HASTATE) SHAPED:** Shaped like a battle axe, with the basal lobes turned outward.

**HALE-BOPP:** A large and exceptionally active comet, which in March 1997 made its closest fly-by to Earth since 2000 BC, coming within 190

million km. It has a nucleus of approximately 40 km and an extensive gas coma (when close to the Sun Hale-Bopp released 10 tonnes of gas every second). Unusually, Hale-Bopp has three tails: one consisting of dust particles, one of charged particles and a third of sodium particles. It was named after two US amateur astronomers, Alan Hale and Thomas Bopp, who independently discovered it in July 1995.

**HALF LONG:** A sheep produced by crossing a Cheviot ram with a Blackface ewe.

**HALF-ADDER:** A combinational circuit that performs the addition of two bits. The input variables designate the augend and addend bits. The output variables produce sum (s) and carry (C).

**HALF-ANGLE FORMULA:** A trigonometric formula that expresses function of half-angle in terms of the full angle.

It is given by  $\sin \frac{1}{2}x = \pm \sqrt{\frac{1 - \cos x}{2}}$ ,  $\cos \frac{1}{2}x = \pm \sqrt{\frac{1 + \cos x}{2}}$ , and

$$\tan \frac{1}{2}x = \pm \sqrt{\frac{1 - \cos x}{1 + \cos x}} \text{ or } \tan \frac{1}{2}x = \frac{1 - \cos x}{\sin x} = \frac{\sin x}{1 + \cos x}.$$

**HALF-BREED:** An animal with only one parent of known pedigree or an animal or plant that is a hybrid product of two distinct types. It is also used to describe anyone who is of mixed racial parentage.

**HALF-CELL:** An electrode in contact with a solution of ions forming part of a cell. Chemical reactions generate electric charges between the electrode and the electrolyte, resulting in a potential difference between the electrode and the electrolyte. Different types of half cells exist with the simplest consisting of a metal electrode immersed in a solution of metal ions.

**HALF-CHAIR:** The conformation of a six-membered ring structure in which four contiguous atoms are in a plane and the other two atoms lie on opposite sides of the plane.

**HALF-DUPLEX:** In telecommunications, it is data transmission that can be used in either direction, but not at the same time.

**HALF-EQUATION:** A separate equation showing which substance gains (reduction) or losses (oxidation) an electron at the electrodes of a chemical cell or in electrolysis in a redox reaction. For example, in the equation  $\text{CuSO}_4 + \text{Zn} \rightarrow \text{ZnSO}_4 + \text{Cu}$ , the oxidation half-reaction is given by  $\text{Zn} \rightarrow \text{Zn}^{2+} + 2e^-$ ; the reduction half-equation is given by  $\text{Cu}^{2+} + 2e^- \rightarrow \text{Cu}$ . The standard way of representing them is the oxidised form +  $ne^-$  reduced form. A half-equation is balanced by adding or taking away, a number of electrons equal to the total number of charges on the ions in the equation: the number of atoms of each element on the left hand side of the equation must equal the number of atoms of each element on the right hand side of the equation; and the sum of the charges on the left hand side of the equation must equal the sum of the charges on the right hand side of the equation.

**HALF-HARDY:** Referring to a plant that is able to survive slight frost without damage. Example of half-hardy perennial is chrysanthemums. Half-hardy annuals include cosmos, nemesia, marigolds and tobacco plants, which complete their life-cycle in one season.

**HALF-INFERIOR:** Referring to an ovary which is attached midway at the point of insertion by other floral parts. The ovary is surrounded by the receptacle. It occurs in flowers of the Lythraceae family such as Crape Myrtles. More often, half-inferior ovaries are grouped with either the superior or inferior ovaries.

**HALF-LIFE (HALF LIFE;  $t_{1/2}$ ):** The time required for half of a sample of a radioactive isotope to break down by radioactive decay to form a daughter isotope. Also, for a drug, it is the time it takes for the concentration of the drug in the plasma or the total amount in the body to be reduced by 50%.



**HALF-LINE:** A straight line that has one end point and extends indefinitely in one direction. It is a ray without its endpoint.

**HALF-OPEN (HALF-CLOSED):** An interval that contains one endpoint but not the other. It is of the form  $[a, b)$  or  $(a, b]$ .

**HALF-PEAK POTENTIAL:** In linear-sweep voltammetry, triangular-wave voltammetry, cyclic triangular-wave voltammetry and similar techniques, the potential of the indicator electrode at which the difference between the total current and the residual current is equal to one-half of the peak current. This potential is attained in the interval in which the rate of the charge-transfer process and hence the (absolute value of the) current, increase monotonically with time.

**HALF-PLANE:** A planar region consisting of all points on one side of an infinite straight line and no points on the other side. For example, the line  $y = mx + c$  separates the coordinate plane into two half planes; the one above is the graph of  $y > mx + c$  and the one below is the graph of  $y < mx + c$ . It is the set of points  $(x, y)$  where  $ax + by \geq c$  or  $ax + by \leq c$  for closed half-planes or corresponding open half-planes or a subset when the line is parallel to an axis, where it is defined as  $x \geq a$ ,  $x \leq a$ ,  $y \geq b$  and  $y \leq b$ . A **closed half-plane** is a half-plane that includes the points on the line. If the points on the line are not included, it is called **open half-plane**.

**HALF-REACTION:** A reduction or oxidation reaction redox reaction, which must occur in multiples of at least two simultaneously so that the electron released by one reactant is accepted by another in order to complete the reaction. It shows the total number of electrons lost or gained. For example, in the equation  $\text{CuSO}_4 + \text{Zn} \rightarrow \text{ZnSO}_4 + \text{Cu}$ , the oxidation half-reaction is given by  $\text{Zn} \rightarrow \text{Zn}^{2+} + 2\text{e}^-$ ; the reduction half equation is given by  $\text{Cu}^{2+} + 2\text{e}^- \rightarrow \text{Cu}$ .

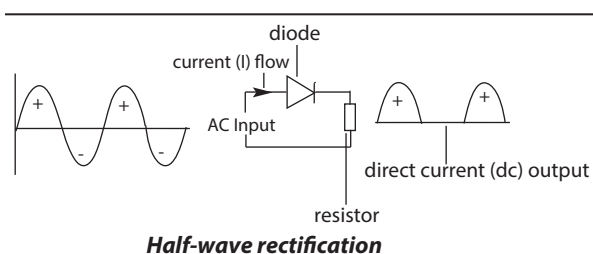
**HALF-SPACE:** All points on one side of a hyperplane in a Euclidean or more general space. For example, half Euclidean space is given by the three-dimensional region satisfying  $x > 0$ ,  $-\infty < y < \infty$ ,  $-\infty < z < \infty$ . If the points on the line are excluded, it is called **open half-plane**; if points on the line are included, it is called **closed half-plane**; although it may be neither.

**HALF-THICKNESS (HALF THICKNESS; HALF-VALUE LAYER; HALF-VALUE THICKNESS):** The thickness of a specified substance which, when introduced into the path of a given beam of radiation, reduces the value of a specified radiation quantity by one half.

**HALF-WAVE PLATE (WAVE PLATE; RETARDER):** An optical device that alters the polarisation state of a light wave travelling through it. A wave plate works by shifting the phase between two perpendicular polarisation components of the light wave. A typical wave plate is simply a birefringent crystal with a carefully chosen orientation and thickness.

**HALF-WAVE POTENTIAL:** The potential of a polarographic or voltammetric indicator electrode at the point, on the rising part of a polarographic or voltammetric wave, where the difference between the total current and the residual current is equal to one-half of the limiting current. The quarter-wave potential, the three-quarter-wave potential, etc., may be similarly defined.

**HALF-WAVE RECTIFICATION:** The type of rectification which makes use of one diode to block the input signal of one half cycle of the alternating current (a.c.) The a.c., thus, is converted to alternating current (d.c.).



**HALF-WIDTH (FULL WIDTH AT HALF MAXIMUM, FWHM):** The full width of a spectral band at a height equal to half of the height at

the band maximum. The dimension of band width should be either inverse length (wavenumbers) or inverse time (frequencies) so that the values give an indication of the energies.

**HALFTONE PROCESS:** The technique used in printing by breaking up an image into a series of dots to reproduce the full range of tones in a photograph or illustration. It is done by placing a glass screen printed with a tight grid of lines over the plate being exposed. The grid breaks up the image into hundreds of tiny dots, each of which is read by the camera as either black or white or, in the case of colour art, as either a single printing colour or white.

**HALIDE:** A binary compound, of which one part is a halogen atom that has become an ion or a compound that contains a halogen or any compound produced by the combination of a halogen, such as chlorine or iodine with a less electronegative element. They may be formed by ionic bonds or by covalent bonds.

**HALITE:** The naturally occurring mineral sodium chloride, NaCl or common salt or rock salt. When pure it is colourless and transparent but it is often pink, red or yellow. It is soft and has a low density.

**HALITOSIS (BAD BREATH):** Foul-smelling or unpleasant odours exhaled in breathing. Halitosis may be caused by poor oral hygiene, a disease of the mouth, throat, nose or lungs or digestive disturbances.

**HALL EFFECT:** The production of a voltage or emf called the **Hall voltage** within or across a conductor or semiconductor carrying an electric current placed at right angle to a surrounding magnetic field or when there is a strong transverse magnetic field. The voltage is developed because the magnetic field distorts the flow of electrons or other charge carriers that constitute the current, pushing the charged particles to one side of the conductor. Discovered in 1879 by Edwin H. Hall (1855 -1938), it can be used to measure certain properties of current carriers as well as to detect the presence of a current on a magnetic field.

**HALL EFFECT THRUSTER (PLASMA THRUSTER):** A small rocket engine that uses a powerful magnetic field to accelerate low density plasma and so produce thrust. It is a form of electrostatic propulsion, which in turn is a form of ion propulsion (a category of electric space propulsion). Like gridded ion engines, such as XIPS, Hall thrusters are classified as electrostatic thrusters. Both utilise an inert gas, commonly xenon, as a propellant.

**HALL PROBE:** An instrument used to measure magnetic field strength. It is a type of proximity sensor that generates voltage, when the sensor is perpendicular to a magnetic field. The stronger the magnetic field, the more voltage it generates.

**HALL SUBGROUP:** A subgroup H of a finite group G that has the property that  $H$  and the number of left cosets of H in G,  $G:H$ , are relatively prime.

**HALL-HEROULT PROCESS:** An industrial process using electrolytic cell to extract aluminium from bauxite. The bauxite is first purified by dissolving it in sodium hydroxide and filtering off insoluble constituents. The oxide (alumina) is dissolved in molten cryolite ( $\text{Na}_3\text{AlF}_6$ ) called a cell and the molten mixture with a quantity of aluminium fluoride,  $\text{AlF}_3$  added, reduces the melting point of the cryolite-alumina mixture, which is electrolysed by passing a direct electric current through it using graphite anodes. The cathode is the lining of the cell, also of graphite. The electrochemical reaction causes liquid aluminium metal to be deposited at the cathode as a precipitate, while the oxygen from the alumina combines with carbon from the anode to produce carbon dioxide,  $\text{CO}_2$ . The electrolyte is kept in a molten state (about  $850^\circ\text{C}$ ,  $1123\text{ K}$ ) by the current. Molten aluminium collects at the bottom of the cell and can be tapped off. Oxygen forms at the anode and gradually oxidises it away. The current through each cell can range from 220 kA to 340 kA. Since direct current is needed for the process, alternating current (AC) supplied to the factory is converted into direct current (DC), which is connected to a number of Hall-Heroult cells connected in series.

**HALL'S MARRIAGE THEOREM (MARRIAGE THEOREM):** The theorem that a perfect matching exists for a bipartite graph of which all the vertices have the same non-zero degree. It gives a necessary and sufficient condition for finding a matching that covers at least one side of the graph. The theorem also deals with a collection of finite sets, giving a necessary and sufficient condition for selecting a distinct element from each set.

**HALLEY, EDMUND (1656-1742):** British astronomer and mathematician, who published a catalogue of southern stars in 1679, made improvements to barometers and investigated the optics of rainbow. In 1705, he accurately calculated the orbit of Halley's Comet and in 1718 discovered the proper motion of the stars. He was made Astronomer Royal in 1721 and began an 18-year study of the complete revolution of the Moon through its ascending and descending nodes.

**HALLEY'S COMET:** A comet with an elliptical orbit around the Sun every 76 years, named after Edmund Halley, who calculated its orbit. It is the brightest and most conspicuous of the periodic comets. It was last observed in 1986.

**HALLUCINATION:** An experience similar to a normal perception but with the difference that sensory stimulus is either absent or too minor to explain the experience satisfactorily. Certain abnormal mental conditions produce hallucinations, as does taking of hallucinogenic drugs. Hallucinations may also result from exhaustion or fever or may be experienced while falling asleep (hypnagogic) or waking (hypnopompic) and also by individuals under hypnosis. A **negative hallucination** is the lack of perception despite adequate stimulus. A **mass hallucination** is hallucination shared by the members of a group.

**HALLUCINOGEN (PSYCHOTOMIMETIC):** Any of a group of substances that cause alteration in perception, mood or thought. Examples are lysergic acid diethylamide (LSD), phencyclidine (PCP) and psilocybin (magic mushrooms).

**HALLUX (BIG TOE; GREAT TOE; THUMB TOE):** The innermost digit on the hind limb of a tetrapod vertebrate. In man, it is the big toe, which in some people is longer than the second or index toe. In most birds it is directed backwards or opposed, which allows for grasping and perching.

**HALO (ATMOSPHERIC HALO):** Any of two large luminous rings that are sometimes observed around the Sun or the Moon. It is caused by diffraction of their light by particles of small, hexagonal ice crystals falling slowly through the Earth's atmosphere.

**HALO BLIGHT:** A bacterial disease caused by *Pseudomonas phaseolicola*, which affects the pods of peas and beans, making them brown and withered.

**HALO NUCLEUS:** A type of atomic nucleus, in which nucleon (neutrons or protons) are not mixed in the same proportion throughout its volume, but has more of one species than the other. Halo nucleus often contains more neutrons than protons in the stable isotopes of an element. Sometimes, a few of the extra neutrons are only weakly bound to the rest of the nucleus and are relatively far from the centre of the nucleus. Halo nuclei are highly unstable; examples include beryllium-11 and carbon-19.

**HALOALKANES (HALOGENOALKANES; ALKYL HALIDES):** Hydrocarbons containing only singly bonded carbon atoms in which one or more hydrogen atoms of an alkane have been replaced by a halogen atom. Haloalkanes can be formed by direct reaction between alkanes and halogens using ultraviolet radiation or by reaction of an alcohol with a halogen carrier. Examples are chloromethane ( $\text{CH}_3\text{Cl}$ ) and dibromoethane ( $\text{CH}_2\text{BrCH}_2\text{Br}$ ). They are used as flame retardants, in fire extinguishers, refrigerants, propellants, solvents and pharmaceuticals.

**HALOCARBONS:** Compounds that contain one or more carbon linked by covalent bonds with one or more halogen atoms such as fluorine, chlorine and hydrogen. The simplest are compounds such as tetrachloromethane ( $\text{CCl}_4$ ) and tetrabromomethane ( $\text{CBr}_4$ ).

**HALOCHROMISM:** The colour change which occurs on addition of acid (or base or a salt) to a solution of a compound. A chemical reaction such as ion formation transforms a colourless compound into a coloured one.

**HALOENZYME:** A complex in its active form consisting of an enzyme molecule and its cofactor, which acts with and is essential to the activity of an enzyme.

**HALOFORM REACTION:** A reaction specific to methyl carbonyl compounds and methyl alcohol compounds which occurs, when treated with a halogen dissolved in sodium hydroxide or potassium hydroxide. The reaction can be used to produce chloroform ( $\text{CHCl}_3$ ), bromoform ( $\text{CHBr}_3$ ) or iodoform ( $\text{CHI}_3$ ). The reaction is used to identify alcohols and carbonyls in which a methyl group ( $\text{CH}_3$ ) is directly attached to the carbon bearing the hydroxyl group or the carbonyl carbon.

**HALOGEN (HALOGEN ELEMENT):** Any of a group of five poisonous, non-metallic elements or a non-metal that is a member of Group 7 in the Periodic Table, comprising fluorine (F), chlorine (Cl), bromine (Br), iodine (I) and astatine (At). All the elements contain all the three familiar states of matter at standard temperature and pressure.

**HALOGENATION:** A chemical reaction that involves a bond between a halogen atom and another atom, for example, the chlorination of benzene to form chlorobenzene. Some of the products include plastic and polymer intermediates, insecticides, fumigants, sterilants, refrigerants, additives for gasoline and materials used in fire extinguishers.

**HALOMORPHIC SOIL:** It refers to a soil that has been strongly affected by either neutral or alkali salts or both.

**HALON:** Organic chemical compound containing one or two carbon atoms, together with bromine and other halogens.

**HALONIUM IONS:** Ions of the form  $\text{R}_2\text{X}^+$ , where X is any halogen such as bromonium ions ( $\text{Br}^+$ ), chloronium ions ( $\text{Cl}^+$ ), fluoronium ions ( $\text{F}^+$ ) and iodonium ions ( $\text{I}^+$ ). They may be open-chain or cyclic.

**HALOPHILES:** Microbes, including bacteria and archaea, which live, grow and multiply in highly saline environments. They are a type of extremophile. Extreme halophiles, all of which are archaea, apart from the green alga *Dunaliella salina*, inhabit water that is up to 10 times more saline than ordinary seawater.

**HALOPHYTE:** A plant that can tolerate a high concentration of salt in soils such as mangrove swamps, marshes and sloughs and seashores. An example is the rice grass or the salt marsh grass, *Spartina alterniflora*.

**HALOSERE:** A pioneer plant community that develops in a low-lying region of land originally beneath the sea and subsequently silted up with mud, silt and shingle. The plants must be adapted to survive inundation by the tide twice a day. A major pioneer plant is glasswort (*Salicornia europaea*), which gradually gives way to salt-marsh grasses (*Puccinellia*) or sea aster (*Aster tripolium*). When the salinity decreases, other species such as thrift (*Armeria maritima*) and seablites (*Suaeda*), grow. The region becomes stabilised to form mudflats and salt marshes.

**HALOTHANE (2-BROMO-2-CHLORO-1, 1, 1-TRIFLUOROETHANE):** A colourless non-flammable oily liquid with the chemical formula  $\text{CHBrClCF}_3$ , which smells like trichloromethane and has a sickly burning taste. It is used widely as general anaesthetic, often administered with oxygen and dinitrogen oxide. It produces a deep level of unconsciousness when inhaled.

**HALOTHANE GENE:** A recessive gene found in some breeds of pigs, which determines the animal's susceptibility to stress and can lead to porcine stress syndrome and also the risk for having pale exudative meat at slaughter.

**HALOTOLERANT:** An organism that can survive in a saline environment, but does not require it for growth. This is not the same

as halophiles which are organisms that require the salinity to survive, while halotolerants do not necessarily require salty environments for growth or for survival.

**HALT:** Of a Turing machine, to reach a designated final state.

**HALTER:** A rope with a noose for holding and leading horses or cattle.

**HALTING PROBLEM:** The proof by Alan Turing in 1936 that it is not possible to decide whether a Turing machine, when presented with a given input string arbitrary program will eventually halt or run forever.

**HALTERES:** Two knob-like structures found in two-winged insects that take the place of functioning hind wings. When in flight the insect vibrates the halteres rapidly so that they act like gyroscopes, helping the insect maintain stability and manoeuvre precisely. Examples occur in houseflies and mosquitoes. Crane flies have particularly visible halteres.

**HAM:** The thigh of the back leg of a pig; or meat from this part of the pig usually cured in brine and dried in smoke.

**HAM RADIO (AMATEUR RADIO):** A popular hobby and service that brings people, electronics and communication together. People use ham radio to talk across town, around the world or even into space, all without the Internet or cell phones.

**HAM SANDWICH THEOREM:** A theorem that given three volumes in Euclidean three-space, there is at least one plane that simultaneously bisects the three volumes. In other words, the volumes of any  $n$ -dimensional solids can always be simultaneously bisected by a  $(n-1)$ -dimensional hyperplane.

**HAMATE:** 1. Hooked-shaped. 2. **Hamate bone** is a bone at the carpal (wrist) which is shaped like a hook.

**HAMEL BASIS:** 1. A basis for a vector space, especially when the basis vectors are orthogonal. 2. In particular, the non-countable basis for the real numbers regarded as a vector space over the rationals, in terms of which every non-zero real can be written uniquely as

$\sum_{i=1}^n x_i b_i$ , where the  $x_i$  are non-zero rationals and the  $b_i$  are elements of the basis.

**HAMEL DIMENSION:** The minimal number of mutually independent vectors that generate the given space, that is, in terms of linear combinations of which every element of the space can be canonically expressed.

**HAMILTON, WILLIAM ROWAN (1805-1865):** Irish mathematician, whose main achievement was in the subject of geometrical optics, for which he laid a theoretical foundation that came close to anticipating quantum theory. He became Professor of Astronomy at Trinity College, Dublin in 1827. He was also a Royal Astronomer of Ireland at the age of twenty-two.

**HAMILTONIAN SYSTEM:** Any mathematical function or operator governed by Hamilton's equations that is used to express the rate of change in time of the condition of a dynamic physical system. It specifies its total energy in terms of the sum of its kinetic energy and its potential energy in terms of the Lagrangian function. This is often used in mechanics and quantum systems.

**HAMILTONIAN CHAIN (HAMILTONIAN PATH):** A chain in a graph that uses each vertex exactly once.

**HAMILTONIAN CIRCUIT:** A circuit in a graph that uses each vertex exactly once.

**HAMILTONIAN FUNCTION:** A mathematical definition to express the rate of change in time of the condition of a dynamic physical system. It specifies its total energy in terms of the sum of its kinetic energy and its potential energy in terms of the Lagrangian function.

It is given by  $-L + q \frac{\partial L}{\partial \dot{q}}$ , where  $L$  is the Lagrangian and  $q$  ranges over the generalised coordinates. It was introduced in 1835 by Sir William Rowan Hamilton.

**HAMILTONIAN WALK:** A path in a graph.

**HAMILTON'S EQUATIONS OF MOTION (CANONICAL EQUATIONS OF MOTION):** A set of first-order, highly symmetrical equations describing the motion of a classical dynamical system.

They are given by:  $q = \frac{\partial H}{\partial p}$ ,  $\dot{p} = -\frac{\partial H}{\partial q}$ , where  $H$  is the Hamiltonian function,  $p = \frac{\partial L}{\partial \dot{q}}$ .

**HAMILTON'S PRINCIPLE OF LEAST ACTION (HAMILTON'S PRINCIPLE):** A variational form of Newton's second law for a discrete set of particles and Euler's laws of motion for a Green-elastic body under the influence of conservative forces. It states that a set of particles under the influence of conservative forces will move in the period  $t_0$  to  $t$  in such a way as to minimise the action from  $t_0$  to  $t$  of the given set.

**HAMMER:** 1. One of the bones in the ear. 2. A tool consisting of a heavy metal head for driving nails into wood or for striking things to break them or remove them.

**HAMMER MILL:** A machine for grinding cereals into meal for the preparation of animal feed.

**HAMMETT CONSTANT:** A constant which measures the influence of a substituent on the stability of a benzoic acid by comparing the logarithmic ratios of the dissociation constants of the substituted benzoic acid to that of benzoic acid. Species which withdraw electrons from aromatic rings and hence stabilise the benzoate anion formed, increase the acidity of benzoic acid giving a positive ratio and vice versa.

**HAMMING CODES:** Named after Richard Hamming, who invented it in 1950, it is a class of efficient error-correcting codes in telecommunication that can detect up to two simultaneous bit errors and is capable of correcting single-bit errors that can occur when computer data is moved or stored.

**HAMMOND PRINCIPLE (HAMMOND POSTULATE; LEFFLER'S ASSUMPTION):** The hypothesis that, when a transition state leading to an unstable reaction intermediate (or product) has nearly the same energy as that intermediate, the two are interconverted with only a small reorganisation of molecular structure.

**HAMSTRING MUSCLES:** A group of muscles at the back of the thigh. The upper ends of the hamstring muscles are attached by tendons to the pelvis. The lower ends are attached by tendons called hamstrings to the tibia and fibula. The hamstring muscles bend the knee and swing the leg backwards from the thigh.

**HAND:** 1. The appendage at the end of the arm of humans and other primates which is used for grasping, gripping and manipulating objects. It is made up of the wrist, palm and fingers. The bones of the hand are the carpals (wrist bones), metacarpals (in the palm) and phalanges (finger bones). Movements of the hand are achieved mainly by tendons that attach the muscles of the forearm to the bones of the hand. These tendons are surrounded by sheaths containing synovial fluid, which reduces friction. Other movements are controlled by short muscles in the palm of the hand; some of these muscles make up the prominent areas along the sides of the hand from the bases of the thumb and little fingers to the wrist. Blood is supplied to the hand by two arteries, the radial artery on the thumb side of the wrist and the ulna artery on the little finger side and is returned by veins that are prominent on the back of the hand. 2. A unit of measurement of the height of a horse. One hand is 10.16 cm and the measurement is taken from the ground to the withers of the horse.

**HAND FORK:** A small hand-held implement or tool with a handle and two or more metal sharp tines or prongs that is used in gardening for digging the ground for planting, aerating, weeding and mixing additives into the soil. It is especially useful in tight places where bigger tools cannot reach.

**HAND HOE (HOE):** A garden tool for breaking up the surface of the soil or to cut off weeds with a small sharp blade.



**HAND LENS:** A hand-held magnifying glass used to produce a magnified image of an object. It may be steel or plastic mounted with the lens usually mounted in a frame with a handle. Hand lenses are used in various fields including scientific research; it is a very useful tool that geologists use to examine rocks, sediments, soils, sand, minerals, and other materials with tiny features.

**HAND MATING:** A method of breeding, in which the male and female are kept apart except during mating. In hand mating, the animals are often guided to mate successfully.

**HAND SPAN (GREAT SPAN):** The distance from the tip of the thumb to the tip of the little finger of the fully stretched hand of an adult, which measures 22.9 centimetres.

**HANDEDNESS:** 1. In anatomy, it is the side of the body and in particular the hand, that is most used in motor tasks. Most people are right-handed and few are truly ambidextrous (either-handed). In the brain, the paths for sensory and motor information are crossed, so the right side of the body is controlled by the left cerebral hemisphere and vice versa. The left hemisphere is usually dominant and also contains centres for speech and calculation. The nondominant side deals with aspects of visual and spatial relationships, while other functions are represented on both sides. In some left-handed people, the right hemisphere is dominant. Suppression of left-handedness may lead to speech disorder. 2. In chemistry, handedness is known as **chirality**. It is having different left-handed and right-handed forms, not mirror symmetric. In the case of molecules, the existence of distinct right- and left-handed forms, known as **enantiomers**, is due to the molecule containing an asymmetric carbon atom. Chirality is an important property of life-supporting carbon compounds. Only right-handed carbohydrates and left-handed amino acids are metabolically active in Earth's organisms. Enzymes, which regulate a variety of biological processes, have the ability to recognise the desired chirality of a substrate. This enhances the versatility of carbon, though chirality is not limited to carbon compounds and occurs in most macromolecules.

**HANDLE:** 1. In computer science, it is the name used as a screen name in online chat without giving away your real identity. 2. In mathematics, it is a piece of a surface constructed by identifying two disjoint disks on another surface at the ends of a cylinder. This produces a torus from a sphere and a coffee cup from a soup bowl.

**HANDOFF (HANDOVER):** In telecommunications, it is the process such that when mobile users move from one cell to another while a call is in progress, their connections are passed from cell to cell.

**HANDSHAKE:** In telecommunications, it is the process of one computer establishing a connection with another computer or device. It involves transmitting signals back and forth over a communications network to verify the connection, the speed or the authorisation of the computer trying to connect to it. An example of handshaking is when a modem connects to another modem; the tones heard after the dialing is the handshake. Hardware handshake uses dedicated wires such as the request-to-send (RTS) and clear-to-send (CTS) lines in an RS-232 serial transmission. Software handshake sends codes such as "synchronise" (SYN) and "acknowledge" (ACK) in a TCP/IP transmission.

**HANDTOP COMPUTER (ULTRA PERSONAL COMPUTER):** A type of personal computer small enough to fit into the palm of a user's hand and operate with the other. Its functions are similar to the laptop computer and can be used in place of the laptop or desktop computers.

**HANDWRITING RECOGNITION:** The ability of a computer system to accept handwritten input from sources such as paper documents, photographs, touch-screens and other devices and convert it into digital data that can be processed and displayed or stored as ASCII characters on the computer screen. It is used for identification and also on devices such as PDA (personal digital assistant) and tablet PCs where a stylus is used to handwrite on a screen, which is converted by the computer into digital text.

**HANGING VALLEY:** A tributary valley that joins a larger glacial trough at a higher level than the trough floor. A river or stream flowing along the hanging valley often forming a waterfall as it enters the trough.

**HANKEL MATRIX:** A square matrix in which all the elements are the same along any diagonal that slopes from northeast to southwest, named after Hermann Hankel, (1839-1873) a German mathematician. An example is shown in the diagram.

$$\begin{bmatrix} a & b & c & d & e \\ b & c & d & e & f \\ c & d & e & f & g \\ d & e & f & g & h \\ e & f & g & h & i \end{bmatrix}$$

**HANTAVIRUS:** A single-stranded spherical virus in the Bunyaviridae family that causes various haemorrhagic diseases with renal complications and acute respiratory illnesses in humans. The virus was first identified in 1976 in Korea and named after Hantaan River. Its carriers are some rodents; and the virus is transmitted to humans through contact with faeces and urine of infected animals. Early symptoms include fever and chills, headaches and muscle aches, vomiting, diarrhoea or abdominal pain; which progresses in a few days time to coughing with secretions, shortness of breath, fluid accumulating within the lungs, low blood pressure, etc.

**HANTZ-WILDMAN SYSTEM:** A naming method for heterocyclic compounds introduced for naming 5- and 6-membered nitrogen monocyclic compounds and subsequently extended to other ring sizes and heteroatoms.

**HAPAXANTHIC (SEMELPAROUS; MONOCARPIC):** Referring to a plant that flowers only once during its lifetime. After fruiting, the leaves senesce and the plant dies completely. Examples include some species of bamboo, annual plants including all grain crops and most domestic vegetables, long lived plants such as century plant (agave) and *Corypha utan*, commonly known as Gebang Palm or Cabbage Palm, a giant palm tree reaching up to 20 metres high, found in some parts of Asia such as Malaysia, Indonesia and the Philippines. The term is also used for an animal or fish that reproduces only once during its lifetime and dies, for example, Pacific salmon, squid, mayflies, Atlantic eel, some species of butterflies, cicadas, some molluscs such as squid and octopus and many arachnids.

**HAPLOBIOTIC:** Referring to a life cycle in which either the sporophyte or gametophyte generation is lacking. For example, in the green alga *Chlamydomonas* the plant is haploid and the diploid condition is represented only by the zygospore, which gives rise to haploid zoospores immediately on germination. In diatoms the plant is diploid and arises directly by the fusion of gametes formed by the sporophyte generation.

**HAPLOCHEILIC:** A term used to describe a gymnosperm stomatal complex in which the subsidiary cells are not derived from the same initial as the guard cells, as occurs in cycads.

**HAPLOID:** A nucleus, cell or organism having a single set of unpaired chromosomes. Male honeybees are haploid because they have developed from eggs that have not been fertilised. Humans are diploid because they have 2 complete haploid sets.

**HAPLOID NUMBER:** The number of chromosomes in a gamete or germ cells, which is half the number of chromosomes found in the other cells of an organism. It is denoted by  $N$ . In humans it is 23 since the full set of chromosomes ( $2N$  or the diploid) is 46.

**HAPLONT:** An organism such as some algae and fungi that is haploid at one stage of its life cycle.

**HAPLOSTEMONOUS:** 1. Plants that have their stamen arranged in a single whorl. 2. Stamens having the same number as the true petals and aligned with them.

**HAPLOTYPE:** 1. A combination of linked genes or other genetic markers at different loci on the chromosome that are inherited together as a unit because during the meiosis there is little or no recombination with the corresponding region on the homologous chromosome and hence shuffling of alleles between the homologous

regions is rare. A haplotype block refers to the whole stretch of DNA-containing haplotype; or the entire set of genes occurring on a single chromosome or haploid set of chromosomes; hence an individual has two haplotypes for each chromosome derived each from mother and father. 2. The set of phenotypic features associated with either the paternal or maternal alleles inherited by an individual; or the antigenic constitution of individual resulting from the inheritance of a particular haploid combination of hisocompatibility alleles at the HLA locus.

**HAPTEN:** A foreign molecule that can bind to an antibody to evoke an immune response when attached to a large carrier such as a protein. Examples of haptens are dinitrophenols, phosphorylcholine and dextran.

**HAPTERON:** 1. Also known as **holdfast**, it is a structure found at the base of many algae in flowing or tidal water that serves to attach the plant to a support. It is often dissected into many finger-like processes or rhizoids. 2. The outer wall of the spore in species of *Equisetum* (horsetails). It differentiates into an elongated X-shaped structure, the arms of which are tightly coiled around the spore.

**HAPTICITY:** A group of atoms or electrons in a ligand that are used in forming a coordinate complex with a metal.

**HAPTO:** The greek letter  $\eta$  (**eta**) used as an affix giving a topological indication of the bonding between a  $\pi$ -electron ligand and the central atom in a coordination entity. A right superscript numerical index indicates the number of ligating atoms in the  $\pi$ -electron system of the ligand which bind to the central atom. Examples are  $[\text{Fe}(\text{CO})_3(\text{C}_6\text{H}_5\text{SO})]$ , tricarbonyl( $\eta^2$ -dihydrothiophene 1-oxide- $\kappa\text{O}$ )iron and  $[\text{Cr}(\text{CO})_4(\text{C}_4\text{H}_9)]$ , tetracarbonyl( $\eta^4$ -2-methylene-1,3 propanediyl) chromium.

**HAPTONASTY:** A non-growth nastic movement in which a plant part moves in response to a tactile stimulus. An example is the rapid and progressive collapse of *Mimosa pudica* (sensitive plant) leaflets, often throughout the whole plant, after those of one particular branch are touched.

**HAPTONEMA:** A threadlike structure that arises between the two flagella in algae belonging to the Haptophyta. It is not involved in cell movement, but may be used for temporary anchorage. The length varies and in some cases it is coiled when the cell is motile. Internally it consists of three concentric membranes surrounding a ring of fibres and a central space.

**HAPTOTROPISM (THIGMOTROPISM):** A tropism in response to an external contact stimulus. The movements are most clearly exhibited by the tendrils of certain species.

**HARD ACID:** A Lewis acid with an acceptor centre of low polarisability. Metal ions with the highest affinities for hard bases are hard acids. Other things being approximately equal, complexes of hard acids and bases or soft acids and bases have an added stabilisation (sometimes called 'HSAB' rule). For example, a metal in a low oxidation state, such as Ni(0), is a soft acid that is stabilised when surrounded by soft bases, such as CO in  $\text{Ni}(\text{CO})_4$ . But in a high oxidation state, such as Ni(V), the metal is a hard acid that is stabilised by hard bases, such as oxide ions in  $\text{NiO}_4^{3-}$ . Examples of hard acids are  $\text{H}^+$ ,  $\text{Na}^+$ ,  $\text{Ca}^{2+}$ ,  $\text{Mn}_2^+$ ,  $\text{Al}^{3+}$ ,  $\text{N}^{3+}$ ,  $\text{Cl}^{3+}$ ,  $\text{Gd}^{3+}$ ,  $\text{Cr}^{3+}$ ,  $\text{Co}^{3+}$ ,  $\text{Fe}^{3+}$ ,  $\text{Be}^{2+}$  and  $\text{Mg}^{2+}$ .

**HARD BASE:** A Lewis base with a donor centre that contains small, relatively nonpolarisable donor atoms for example, an oxygen atom of low polarisability. Examples of hard bases are  $\text{NH}_3$ ,  $\text{RNH}_2$ ,  $\text{N}_2\text{H}_4$ ,  $\text{H}_2\text{O}$ ,  $\text{OH}^-$ ,  $\text{F}^-$ ,  $\text{CH}_3\text{CO}_2^-$ ,  $\text{SO}_4^{2-}$ ,  $\text{CO}_3^{2-}$ ,  $\text{NO}_3^-$ ,  $\text{PO}_4^{3-}$ ,  $\text{ClO}_4^-$  and  $\text{ROH}$ . The converse applies to soft bases: they contain larger, relatively polarisable donor atoms. Examples are  $\text{R}_2\text{S}$ ,  $\text{RSH}$ ,  $\text{I}^-$ ,  $\text{SCN}^-$ ,  $\text{S}_2\text{O}_3^{2-}$ ,  $\text{R}_3\text{P}$ ,  $(\text{RO})_3\text{P}$ ,  $\text{CN}^-$ ,  $\text{RNC}$ ,  $\text{CO}$ ,  $\text{C}_2\text{H}_4$ ,  $\text{C}_6\text{H}_6$ ,  $\text{H}^-$  and  $\text{R}^-$ .

**HARD COPY:** It refers to a printed document, which may be a text file, photograph, drawing or any other type of printable file. For example, instead of e-mailing a business memo, it may be sent out as a hard copy or an actual physical paper containing the memo. Printing a hard copy is done to create a physical backup of a document.

**HARD DISKDRIVE (HARD DISK, HARD DRIVE):** The primary computer storage device, which spins, reads and writes one or more fixed disk platters. It usually consists of a rigid aluminium or glass disc coated with a magnetic material.

**HARD FERROMAGNETIC MATERIALS:** A form of iron such as alnico and ferrite that contains little carbon, has high relative permeability and is easily magnetised and demagnetised and has a small hysteresis loss.

**HARD PAGE BREAK (PAGE BREAK):** A code inserted by a software program such as Word processor that tells the printing device where to end the current page and begin the next page. After a page break is inserted into a document, a symbol indicating the page break is displayed.

**HARD RADIATION:** Ionising radiation of high penetrating power able to penetrate a given thickness of material, such as a lead shield. They are usually gamma rays or x-rays and of short wavelength.

**HARD SEED:** Any seed with a tough impervious outer coat that will not allow the entry of water. Imbibition and germination, therefore, cannot occur until the seed coat is ruptured, either by scarification or microbial action. The sweet pea (*Lathyrus odoratus*) has hard seeds.

**HARD TOKEN (AUTHENTICATION TOKEN):** A hardware security device that is used to authorise a user. An example is a security card that gives a user access to different areas of building or allows a user to log in to a computer system. A username and password or a fingerprint scan may be required along with the hard token to gain access to a secure system. A **hard tag** is another type of hard token that is used as a security device on merchandise such as clothing and electronics to deter theft.

**HARD WATER (HARDNESS OF WATER):** Water that contains dissolved calcium or magnesium salts, principally as bicarbonates, chlorides and sulfates and sometimes iron that forms scum with soap or water that does not lather easily with soap and forms scales in kettles, when boiled. It forms scum with soap. There are two types: **temporary hardness** that is caused by dissolved calcium hydrogen carbonate  $\{\text{Ca}(\text{HCO}_3)_2\}$  and magnesium carbonate; and **permanent hardness** that is caused by the presence of chlorides, nitrates and sulfates of calcium and magnesium. Temporary hardness is removed by boiling the water. **Permanent hardness** cannot be removed by boiling the water, but by distillation, addition of sodium carbonate, calcium hydroxide or ion exchanger.

**HARD-SECTORED DISK:** A floppy disk that is sold already formatted so that disk formatting is not necessary.

**HARD-SEGMENT PHASE DOMAIN:** Phase domain of microscopic or smaller size, usually in a block, graft or segmented copolymer, comprising essentially those segments of the polymer that are rigid and capable of forming strong intermolecular interactions. Hard-segment phase domains are typically of 2 - 15 nm linear size.

**HARDCODED:** In computer science, pertaining to software instructions that are statically encoded and not intended to be altered.

**HARDENING:** 1. A metallurgical and metalworking process to increase the hardness of a metal. **Surface hardening** is the treatment of steel by heat or mechanical means to increase the hardness of the outer surface while the core remains relatively soft.

**Work hardening** is the increase in hardness of a metal induced, deliberately or accidentally, by hammering, rolling, drawing or other physical processes. 2. Also known as hardening of oils, it is the conversion of liquid oils to solid products by the addition of hydrogen (hydrogenation) or the process in which unsaturated oils react with hydrogen to produce saturated compounds, which are more suitable for the production of vegetable oils and margarines. 3. The process by which an individual plant becomes tolerant to the effects of freezing during a period of weeks to months. Seedlings raised in greenhouses are often hardened off in this manner before planting.

**HARDINESS:** A plant's ability to withstand certain environmental conditions such as cold, heat, drought, flooding, or wind.



**HARDNESS:** 1. A measure of the resistance of a surface or a mineral to scratching, abrasion or indentation on the Mohs scale representing the hardness of a mineral. Diamond is the hardest element, with a hardness of 10, while talc is the softest with Mohs scale hardness of 1. 2. The physical property of materials that governs their use. Hardness is dependent on ductility, elasticity, plasticity, strain, strength, toughness, viscoelasticity and viscosity. 3. The degree to which water contains mineral salts.

**HARDWARE (COMPUTER HARDWARE):** The physical parts of a computer and related devices. Examples of internal hardware devices include motherboards, hard drives and RAM. Examples of external hardware devices include monitors, keyboards, mice, printers and scanners.

**HARDWARE CONFIGURATION:** In computer science, an interactive tool that can be used to define hardware configurations to the operating system and the channel subsystem.

**HARDWIRED (HARD-WIRED):** 1. A function or device that is physically built into the computer instead of programmed into the software. These devices or functions cannot be removed from the computer and usually allow hardware to control the operation instead of software. 2. A fixed placement such as text and graphics that never change their location on a page.

**HARDWOOD:** Any close-grained wood from deciduous trees such as Mahogany.

**HARDY-WEINBERG EQUILIBRIUM (HARDY-WEINBERG LAW; HARDY-WEINBERG PRINCIPLE, HWP):** The balance in the relative number of alleles that in the absence of forces change is maintained within a large population over a period of time. Deviation from Hardy-Weinberg principle indicates evolution of species. Examples of disturbing influences include non-random mating, mutations, selection, limited population size, random genetic drift and gene flow.

**HARDY-WEINBERG THEOREM:** An axiom maintaining that the sexual shuffling of genes alone cannot alter the overall genetic makeup of a population.

**HARE:** An animal of the genus *Lepus*. It is like the rabbit but usually longer, with long ears, short tail and long back legs which enable it to run very fast. The female is highly productive, bearing four to eight litters a year. Unlike rabbit, the young are born furred and with open eyes. They weigh about 1 to 5 kg and of a length of about 30 to 60 cm. They feed mainly on herbs, tree bark and vegetables.

**HARE'S APPARATUS:** A specified kind of hydrometer used to compare the relative densities of two liquids or to find the density of one if the other is known.

**HARELIP:** A congenital disease with a cleft defect in the upper lip due to impaired facial development in the embryo and often associated with cleft palate. It may be corrected by plastic surgery.

**HARLEQUIN GLORYBOWER (*Clerodendrum trichotomum*):** A hardy deciduous shrub in the family Lamiaceae that can grow up to 6 metres in height and is native to China and Japan. Both the flowers and leaves are fragrant. The dark green leaves are opposite, toothed to entire and ovate. When crushed, the leaves produce peanut butter scent. The inflorescence have petals which are white, held within green sepals which turn red as the fruits ripen. The white drupes change to bright-blue to dark-blue on maturity. The plant is cultivated for its scented flowers and ornamental berries. The leaves are edible and is boiled to remove the fragrance.

**HARMATTAN:** A dry and dusty north-east wind that blows between the end of November and the middle of March over West Africa, from the Sahara desert, where it picks up fine dust particles, which it transports hundreds of kilometres out over the Atlantic Ocean depositing them along its route. In some countries in West Africa, the heavy amount of dust in the air can severely limit visibility and block the sun for several days. It lowers the humidity and creates hot days and cool nights in the regions in the north of West Africa.

The interaction of the Harmattan with monsoon winds can cause tornadoes.

**HARMONIC:** 1. In mathematics, able to be expressed in terms of sine and cosine functions. 2. Of or relating to numbers of which the reciprocals form an arithmetic progression. 3. Of a function, having two dimensional Laplacian equal to zero, occurring as the real or imaginary part of an analytic function. 4. Of a projective geometry, having a cross ratio equal to -1. 5. A component of a periodic quantity, such as a musical tone, with a frequency that is an integral multiple of the frequency of the given oscillation. In other words, it is one of the extra frequencies produced by a musical instrument such as a stringed instrument or an oscillation having frequencies that are integral multiples of the primary or fundamental frequency. The fundamental frequency of a sinusoidal oscillation is usually called the **first harmonic**; the **second harmonic** has twice the fundamental frequency and in music is called the first overtone; the **third harmonic** has three times the fundamental frequency and called the second overtone. For open pipes and strings, the fundamental frequencies and the first four overtones are respectively  $\frac{v}{2l}, \frac{v}{l}, \frac{3v}{2l}, \frac{2v}{l}, \frac{5v}{l}, \dots, f_n$ .

This means that fundamental frequencies and its overtones are  $f_0, 2f_0, 3f_0, 4f_0, 5f_0, \dots, (n+1)f_0$ . So that the overtones that are the whole number multiple of the fundamental frequency,  $f_0$  are all harmonics. Overtones may or may not be harmonics.

**HARMONIC ANALYSIS:** The representation of functions or signals as the superposition of basic waves by means of linear operations on characteristic sets of functions. Also, the study of such a process.

**HARMONIC CONJUGATE (FOURTH HARMONIC POINT):** The fourth point, collinear with a pair of points, A and B and another collinear point, X, that can be found by constructing the complete quadrangle corresponding to a trapezoid ABCD, for which X is the point of intersection of the extensions of AB and CD. If Y is the sixth point of the complete quadrangle and the diagonals of the trapezoid intersect at Z, then the intersection, H, of YZ with the line XAB is the fourth harmonic of X with respect to A and B. These four collinear points are then a harmonic set as their cross ratio is a harmonic ratio in which A, B and X, H are conjugate pairs. Iterating the process leads to a harmonic net of rationality. Dually, a complete quadrilateral produces an harmonic pencil of four lines.

**HARMONIC DIVISION:** A line segment AB is divided harmonically by two points C and D, when it is divided externally and internally in the same ratio.

**HARMONIC FUNCTION:** A function that is a solution to Laplace's equation. Harmonic functions are important in the areas of applied mathematics, engineering and mathematical physics. They are used, for example, to solve problems involving steady state temperatures, two-dimensional electrostatics and ideal fluid flow.

**HARMONIC MEAN:** The harmonic mean of two numbers  $a$  and  $b$  is  $\frac{2ab}{a+b}$  or the reciprocal of the arithmetic mean of the reciprocals.

**HARMONIC MOTION (SIMPLE HARMONIC MOTION, SHM):** A periodic motion that is neither driven nor damped. An object in simple harmonic motion experiences a net force which obeys Hooke's law (the force is directly proportional to the displacement from the equilibrium position and acts in the opposite direction of the displacement).

**HARMONIC OSCILLATOR:** A system that oscillates with simple harmonic motion or when displaced from its equilibrium position, experiences a restoring force, proportional to the displacement. The harmonic oscillator is exactly solvable in both classical mechanics and quantum mechanics. An example of this in classical mechanics is a simple pendulum whose atoms vibrate about their mean positions in molecules or crystal lattices when at low temperatures.

**HARMONIC PENCIL:** The configuration of four lines, passing through a single point, such that any line that is not parallel to one of the four cuts the four lines at points which are harmonic conjugates.

**HARMONIC POINTS:** A cross-ratio of four points  $P, Q, R$  and  $S$  lying on a straight line segment which divide it such that

$\left(\frac{PR}{QR}\right) \left(\frac{PS}{QS}\right) = -1$ . The negative is possible only if there are positive and

negative directions along the line, for example,  $QR = -RQ$ .

**HARMONIC RATIO:** A cross-ratio of four points whose value is equal to -1. Consider four points  $P, Q, R$  and  $S$ ,  $\frac{PQ}{QR} = -\frac{PS}{SR} = \frac{PS}{RS}$ ;

therefore,  $PQ(PS - PR) = (PR - PQ) \Rightarrow PQ \cdot PS - PQ \cdot PR$

$= PS \cdot PR - PS \cdot PQ \Rightarrow \frac{1}{PQ} + \frac{1}{PS} = \frac{2}{PR}$ , then  $PQ, PR$  and  $PS$  are in harmonic progression. If  $Q$  is in the middle of  $PR$ , then  $S$  is at infinity.

**HARMONIC SEQUENCE (HARMONIC PROGRESSION):** A sequence of the form:  $\frac{1}{a}, \frac{1}{a+d}, \frac{1}{a+2d}, \dots, \frac{1}{a+(n-1)d}$ , the

terms being the reciprocals of those in an arithmetic sequence. There is no simple expression for the sum of a harmonic progression. The harmonic mean of two terms,  $a_s$  and  $a_{s+2}$  is given by

$$2a_s a_{s+2} / (a_s + a_{s+2}) = a_{s+1}.$$

**HARMONIC SERIES:** A series or progression in which the reciprocals of the terms have a constant difference between them. An example is  $1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \dots + \frac{1}{n-1} + \frac{1}{n} \dots$

**HARPOON MECHANISM:** A mechanism explaining the reaction between such metal atoms as sodium or potassium and such halogen molecules as bromine. As the alkali atom approaches the bromine molecule its valence electron moves to the bromine molecule, thus providing a "harpoon" effect.

**HARROW:** An implement with sharp metal teeth for cultivating the surface of the soil and used to break up furrows or clods and lumps left by ploughing and to reduce the soil to fine consistency, tilt and to cover the seeds after sowing.

**HARTREE (HARTREE ENERGY):** Atomic unit of energy, with the symbol  $E_h$ . It is approximately equal to ( $\approx$ ) 4.359 7482 ( $26 \times 10^{18}$  J). It is approximately the electric potential energy of the hydrogen atom in its ground state and, by the virial theorem, approximately twice its ionisation energy. The relationships are not exact because of the finite mass of the nucleus of the hydrogen atom and relativistic corrections.

**HARTREE-FOCK METHOD:** An approximation method used in computational physics and chemistry to determine state wave and ground-state energy of many electron quantum systems.

**HARVARD CLASSIFICATION:** A system of classifying stars according to the characteristic spectral absorption lines and bands of the chemical elements present. It was introduced at Harvard College Observatory in 1890 by E. C. Pickering, which culminated in the Henry Draper Catalogue. At first, stars were ordered according to the strength of their hydrogen absorption lines (the Balmer series), from A (strongest) to P. Eventually several letters were merged or dropped and the remaining spectral types were rearranged into the sequence O, B, A, F, G, K, M, in order of decreasing surface temperature. Early in the 20th century, spectral types R and N were applied to carbon-rich versions of types G, K and M (now known as carbon stars) and type S (M stars with heavy-metal lines) was subsequently added. The Harvard system has since been superseded by the Morgan-Keenan classification system.

**HARVEST MOON:** The full moon that appears closest in time to the autumnal equinox, in late September or early October in the Northern Hemisphere. For several nights the full moon rises at about the same time (around sunset) and may be bright enough for harvesting to

continue into the night. In the Southern Hemisphere this occurs around the spring equinox.

**HARVESTING:** 1. Gathering crops from their places of growth and moving them to locations for processing, consumption or storage. The major factor determining harvesting is the maturity of the crop, but factors such as weather, availability of harvest equipment, pickers, packing and storage facilities and transport are important considerations. 2. The collection of cells from cell cultures or of organs from donors for the purpose of transplantation.

**HARVESTMAN:** An order of arachnids belonging to the order Opiliones (formerly Phalangida) with over 6,400 species. They look like spiders, but are distinctly different, with small rounded bodies and very long thin legs. They are mainly nocturnal but there are some diurnal species.

**HASHTAG:** A way of tracking topics on a social media such as Twitter by using the number symbol # to label keywords in a tweet. A **tweet** is a short message sent through the microblogging platform, Twitter. For example, you can tag the word "tech" in a tweet by typing "#tech" and have it easily found and sorted. All hashtags begin with a hash character, which is the symbol (#).

**HASSIUM (UNNILOCTIUM):** A synthetic element, with the symbol Hs, atomic number 108, which is a radioactive transactinide element. It is the heaviest member of the Group 8 elements. It has a half-life of approximately 9.7 seconds.

**HASTATE:** Having a tapering triangular shape like that of a spear. A hastate leaf is shaped like a spear, with three lobes, one pointing forwards and two pointing sideways on either side of the petiole. Examples are the leaves of the copse buckwheat or bindweed (*Polygonum dumetorum*).



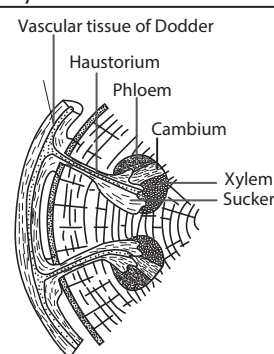
Hastate leaf

**HATCH-SLACK PATHWAY:** An alternative form of carbon dioxide ( $\text{CO}_2$ ) fixation found in C4 plants. In such plants the first product of  $\text{CO}_2$  fixation is not the three-carbon phosphoglyceric acid but the four-carbon oxaloacetate.

**HATCHERY:** A place for hatching eggs. **Hatchery waste** is surplus chicks or embryos produced in a hatchery.

**HATCHLING:** Newly hatched fish, amphibian, reptile, or bird.

**HAUSTORIUM:** A specialised structure produced by a parasite, especially the hyphal extensions formed by obligate fungal parasites that protrudes and penetrates the cells of the host to absorb nutrients. On penetration, the fungus increases the surface area in contact with host plasma membrane releasing enzymes that break down the cell wall, enabling greater potential movement of organic carbon from host to fungus.



Transverse section of host stem showing haustorium

**HARVESIAN GLANDS (CANALS):**

Narrow tubes within compact bone tissue containing blood vessels, nerves and lymphatics forming part of the Haversian system. Each canal is surrounded by a series of rings of bone known as lamellae.

**HAVERSIAN SYSTEM:** One of many structural units of vertebrate bone, consisting of concentric layers of mineralised bone matrix surrounding lacunae, which contain osteocytes and a central canal, which contains blood vessels and nerves.

**HAUSDORFF DIMENSION:** Named after the German mathematician Felix Hausdorff (1868- 1942), it serves as a measure of the local size of a set of numbers, taking into account the distance between each of its members. In other words, it is a further dimensional number associated with a given set of numbers, where the distances between all members of that set are defined and where the dimension is drawn from the real numbers,  $R$ , to which two elements have been added,  $+\infty$  and  $-\infty$ . Applying its mathematical formalisms provides that the Hausdorff dimension of a single point is zero, of a line is 1 and of a square is 2, of a cube is 3.

**HAUSDORFF DISTANCE:** The distance between two sets  $A$  and  $B$  in a metric space, defined as the supremum of the distances from points in  $A$  to the set  $B$  and from points in  $B$  to the set  $A$ . When restricted to compact convex subsets of Euclidean space, this distance becomes a complete metric, the Hausdorff metric and provides a useful tool in the study of many economic and isoperimetric problems.

**HAUSDORFF MAXIMALITY THEOREM:** The theorem that every non-empty partially ordered set contains a maximal totally ordered subset.

**HAUSDORFF SPACE:** A topological space in which distinct points have disjoint open neighbourhoods.

**HAVERSINE (HAVERSED SINE):** Half the value of the versed sine or versine (denoted by versin) of an angle that is equal to  $1 - \cos(\theta)$ .

It is defined by  $\text{hav}(z) \equiv \frac{1}{2} \text{vers}(z) = \frac{1}{2}(1 - \cos z) = \sin^2\left(\frac{1}{2}z\right)$ , where

$\text{versin}(z)$  is the versine,  $\cos z$  is the cosine and  $\sin z$  is the sine. The haversine formula is used to calculate the shortest distance between two points on the surface of a sphere such as the Earth using the longitude and latitude of each point.

**HAWKING PROCESS:** The theoretical emission of particles from just outside the event horizon of a black hole, as a result of quantum-mechanical effects, proposed in 1974 by Stephen W. Hawking, British theoretical physicist and mathematician. The gravitational field of the black hole causes production of particle-antiparticle pairs in the vicinity of the event horizon.

**HAWKING RADIATION:** A theoretical electromagnetic radiation with a faint glow from a black hole, predicted by Stephen Hawking as a result of quantum-mechanical effects. The radiation is due to the black hole capturing one of a particle-antiparticle pair created spontaneously near to the event horizon.

**HAWKING, STEPHEN WILLIAM (1942-):** British cosmologist and theoretical physicist and a professor of mathematics at Oxford University from 1979. He has worked primarily in the field of general relativity and particularly on the physics of black holes. He worked with Roger Penrose, who had shown how a singularity results from a black hole; and postulated that the original big bang must have come from a black hole. He also showed how black holes can emit particles by the Hawking process.

**HAWTHORN:** A small tree or shrub, of the genus *Crataegus* in the rose family, native to the northern temperate zone with spiny shoots, used for hedges around grazing areas.

**HAY:** Stems of grass and other herbaceous plants, which have been cut and dried. It is used as an ingredient in compost or as mulch or cattle feed. Hay supplies organic matter and some nutrients to the soil.

**HAY FEVER (ALLERGIC RHINITIS):** An allergic illness rather like a bad cold caused by allergic reaction to pollen, which is breathed in from the air. It is more common when airborne plant pollens are at their highest levels in the spring and summer.

**HAYLAGE:** A feed that is halfway between hay and silage, made from forages dried to 40 to 55 percent moisture.

**HAZARD:** A set of inherent properties of a substance, mixture of substances or a process involving substances that, under production, usage or disposal conditions, make it capable of causing adverse

effects to organisms or the environment, depending on the degree of exposure. It is a source of danger.

**HAZARD ANALYSIS AND CRITICAL CONTROL POINT (HACCP):** A food safety process used to identify and control hazards that are most likely to cause food-borne illness.

**HAZARD RATE FUNCTION (FORCE OF MORTALITY; FAILURE RATE):** The probability of failure of a component or the death of an organism, within the next unit of time,  $\Delta t$ , given survival to time  $t$ . It is defined mathematically as  $\Delta t \cdot h(t) = P[X \leq (t + \Delta t) : X > t]$ .

**HAZARDOUS SUBSTANCES:** Any of a wide variety of hazardous waste and chemical contaminants that are dangerous to humans or other organisms, property or the environment in general. They may be radioactive, flammable, explosive, corrosive, asphyxiants, bio-hazardous or toxic. Hazardous substances are often subject to regulation.

**HAZARDOUS WASTE (TOXIC WASTE):** Waste material that can cause death or injury to living organisms. It is usually the product of industry or commerce, but comes also from residential use, agriculture, the military, medical facilities, radioactive sources and light industry, such as dry cleaning establishments.

**HAZCHEM CODE (EMERGENCY ACTION CODE, EAC):** A code displayed in some countries such as Australia, Malaysia, New Zealand and the United Kingdom, when hazardous chemicals are transported or stored in bulk.

**HAZE:** A state of reduced visibility (1 - 2 km) resulting from the increased light scatter due to the presence of fine dust or aerosol particles ( $\text{H}_2\text{SO}_4$ ,  $\text{NH}_4\text{HSO}_4$ , products of the ozone-terpene reactions, etc.).

**HAZE FACTOR:** A unit of visibility, which is the ratio of the brightness of a mist or fog through which an object is viewed to that of the object.

**HAZE HORIZON:** The top of a haze layer which is confined by a low-level temperature inversion so that it gives the appearance of the horizon which it may obscure.

**HDMI (HIGH-DEFINITION MULTIMEDIA INTERFACE):** A digital interface for transmitting audio and video data in a single cable. It is supported by most HDTVs and related components, such as DVD and Blu-ray players, cable boxes and video game systems.

**HDTV (HIGH DEFINITION TELEVISION):** A high-quality digital TV broadcasting format where the broadcast transmits widescreen pictures with more detail and quality than found in a standard analogue television or other digital television formats. It offers a significantly greater number of scanning lines and therefore produces much sharper images than that provided by conventional systems. American TV (NTSC) has 525 lines per image and European TV (PAL) has 625 lines per image, while high definition systems may have more than 1000. It is developed to replace older video formats often referred to as SDTV (standard definition television). Types of HDTV displays include direct-view, plasma, rear screen and front screen projection.

**HEAD:** 1. A part of the body which contains the eyes, ears, nose, mouth and the brain. 2. An electromagnetic component in a tape recorder, video recorder and computer input or output device that can read, erase or write signals off or onto tapes and disks. 3. In botany, also known as **capitulum**, it is a special type of inflorescence which is short and densely clustered to form a single flower-like structure. It is characteristic of plants of the family *Compositae*. Examples are dandelion, sunflowers, broccoli and cauliflower.

**HEADAGE:** The number of animals of a specified type, such as cattle, to supplement producer returns earned through sales at market



Head  
(capitulum)



prices. It is used as a basis for calculating subsidy payments made to individual producers.

**HEADER:** 1. The unit of information that precedes a data object and contains control information for the user agents. 2. One or more lines of text that appears at the top of each page of a document.

**HEADFLY:** A parasitic insect, *Hydrotaea irritans*, which mainly affects sheep and can transmit summer mastitis.

**HEADING DATE:** The average date by which a certain percentage of a crop has formed seedheads. This is used by farmers to make decisions on which variety of a crop is more suitable for the environment in which they are planning to grow it.

**HEADLAND:** Area at the end or edge of a field where a tractor turns when ploughing. It could be cultivated or left as pasture to serve as access to the field.

**HEADPHONES:** A pair of left and right speakers that are positioned in or over the ears. They contain transducers that convert an audio signal into sound waves and used for listening to music or any audio signal. When a microphone is included for telephone or video calls, it is called a **headset**.

**HEALTH:** The absence of disease and a state of well being in the body and mind.

**HEALTH PHYSICS:** The branch of environmental and occupational safety health sciences concerned with the protection of medical, scientific and industrial workers from the hazards of ionising radiation and other dangers associated with atomic physics. The main activities in this field include establishing the permissible dose of radiation, the disposal of radioactive waste and the shielding of dangerous equipment.

**HEARING (AUDITION):** One of the traditional five senses, it is the use of a specialised sense organ such as the ear in higher animals to collect sound waves, convert them to signals to pass to the brain and use them to obtain information from outside.

**HEARING AID:** A small electronic component fitted near the ear, which detects and amplifies sound for people with weak hearing.

**HEART:** A muscular pumping organ found in all animals with a circulatory system, which maintains blood circulation. It is usually equipped with valves to prevent backward flow or a hollow muscular organ that rhythmically contracts to pump or force blood around the body of an animal with a circulatory system. The average human heart weighs about 250 to 300 grams in females and 300 to 350 grams in males and beats at about 72 beats per minute for a healthy adult.

**HEART ATTACK (CORONARY THROMBOSIS; MYOCARDIAL INFARCTION):** A sudden

onset of gripping central chest pain, often accompanied by sweating and vomiting, when the flow of blood to the heart is blocked due to death of a portion of the heart muscle following obstruction of a coronary artery by thrombosis (formation of a blood clot). This is most often caused by a build-up of fat, cholesterol and other substances, which form a plaque in the coronary arteries, obstructing the circulation of blood. Some other symptoms include nausea, indigestion, heartburn or abdominal pain, shortness of breath, fatigue,

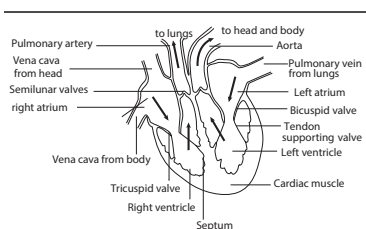


Fig. 1: Vertical section through a mammalian heart

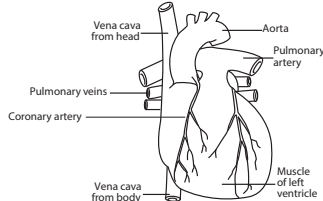


Fig. II: External appearance of a human heart

light-headedness or sudden dizziness. A heart attack is a major cause of death and disability worldwide and death can occur within the first two hours, if immediate medical help is not sought.

**HEART BEAT:** The regular contraction and relaxation of the heart and the accompanying sounds, which is expressed as beats per minute (bpm). It varies in individuals as the body's need for oxygen changes, such as during exercise or sleep. The average human heart beats at about 72 beats per minute for a healthy adult. The measurement of heart beat is used by medical professionals to assist in the diagnosis and tracking of medical conditions such as hypertension or high blood pressure (b.p.).

**HEART DISEASE:** Any disorder that affects the heart. Examples include hypertension (high blood pressure); coronary artery diseases such as arrhythmia, an irregular heartbeat; and angina pectoris, the reduced blood flow to heart muscle cells, resulting in pain in the chest, shortness of breath, nausea, vomiting, sweating and weakness.

**HEART FAILURE (CONGESTIVE HEART FAILURE):** A condition arising when the heart's pumping mechanism is defective and cannot supply adequate blood to the tissues and other organs such as the brain, liver and kidneys. Its causes include hypertension (high blood pressure); failure of the right or left or both ventricles; defective heart valves; congenital heart defects (heart problems that develop before birth) such as a narrowed valves; or various types of heart disease. Some of the signs and symptoms include shortness of breath; asthma due to the heart; fatigue and weakness; oedema or swelling in the legs, ankles and feet; rapid or irregular heartbeat; and enlargement of the heart.

**HEART ROT:** A disease of trees, root crops and celery, caused by boron deficiency. A dry rot spreads from the crown downwards and attacks the roots. The growing point is killed and replaced by a mass of small deformed leaves.

**HEARTWOOD (DURAMEN):** The inner layers of wood in a growing tree which have ceased to conduct and contain no living cells and provide structural support. The reserve materials have been removed or converted into heartwood substance. They are generally darker in colour than sapwood.

**HEAT (THERMAL ENERGY):** The measure of the total internal energy of body. It is a form of internal energy possessed by a substance by virtue of the kinetic energy in the motion or vibration of its molecules or atoms. It is energy in transit, it always flows from a substance at a higher temperature to the substance at a lower temperature, raising the temperature of the latter and lowering that of the former substance, provided the volumes of the bodies remain constant.

**HEAT BALANCE:** 1. The calculation used in fluidisation roasting so that the addition or removal of heat can be controlled to maintain the optimum temperature in the reacting vessel. 2. The equilibrium that exists on the average between the radiations received from the Sun by the Earth and its atmosphere and that emitted by the Earth and its atmosphere.

**HEAT BARRIER:** The speed above which friction with an atmosphere may damage a spacecraft. The heat barrier is of particular importance during re-entry. Special systems are needed to cool the spacecraft when it goes beyond the heat barrier speed.

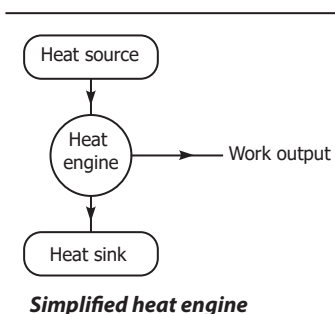
**HEAT CAPACITY (THERMAL CAPACITY):** The amount of heat required to raise the temperature of a substance by 1°C or 1 K or the ratio of heat supplied to an object or specimen to its consequent rise in temperature. In other words, it is heat brought to a system to increase its temperature divided by that temperature increase. Its

SI unit is joule per kelvin (JK<sup>-1</sup> or J/K) and expressed as  $C = \frac{Q}{\Delta\theta}$ , where  $Q$  is the heat transferred and  $\Delta\theta$  is the rise in temperature. The **specific heat capacity** is the heat energy needed to raise the temperature of 1kg of a substance by 1 °C/1 K. Its unit is Jkg<sup>-1</sup>K<sup>-1</sup>.

**HEAT DEATH OF THE UNIVERSE:** The condition of the universe when entropy is maximised and all large-scale samples of matter are

at a uniform temperature. In this condition no energy is available for doing work and the universe is finally unwound.

**HEAT ENGINE:** A device for converting heat into mechanical work by exploiting the temperature gradient between a hot source and a cold sink. It is one of the most important practical applications of the principles of thermodynamics. All heat engines operate on a cycle. As a result of a difference in temperature, heat is transferred from a high temperature source in the direction of decreasing temperature, a cold sink. In the process, some of the heat increases the temperature of a working substance such as water to be converted into steam to perform work. The unused heat is transmitted into a low temperature sink (cold sink). For example, in a thermal electrical power plant, heat from fuel is used to boil water, which is converted into high-pressure steam. Since the steam has much higher pressure than the water, heat is converted into mechanical energy in the form of pressure, which is used to turn a turbine, which turns a shaft. Thus, pressure of the steam is converted into the rotational energy of the turbine and the shaft. The shaft turns an electrical generator, converting rotational (mechanical) energy into electrical energy. Heat engine's efficiency is fundamentally limited by Carnot's theorem. Examples of heat engines are various engines used in motor vehicles and thermal electrical power generating plants.



**HEAT EQUATION (DIFFUSION EQUATION):** The partial differential equation defined by  $\nabla^2 u = c^2 \frac{\partial u}{\partial t}$ , where  $\nabla^2$  is the Laplacian in one, two or three dimensions, generally solved by using Fourier series. The solution of the heat equation gives the distribution of temperature in a region as a function of space and time when the temperature at the boundaries, the initial distribution of temperature and the physical properties of the medium are specified.

**HEAT EXCHANGE:** A process based on heat transfer, where heat energy contained in one material is transferred to another material of lower temperature.

**HEAT EXCHANGER:** A device for transferring heat from a hot to a cold fluid without permitting the fluids to contact each other. Examples of heat exchangers are boilers, evaporators, superheaters, condensers and coolers. Heat exchangers are used extensively in fossil fuel and nuclear power plants, gas turbines, heating and air conditioning, refrigeration and the chemical industry. In nuclear reactors, this allows the nuclear reactions to occur in a place that is sealed off from the rest of the environment. Reactions increase temperature in the core and this thermal energy is transferred to water and the steam that is produced turns the turbines.

**HEAT FLUX:** Heat transferred through a cross section perpendicular to the flow in a small time interval divided by that time interval and the cross sectional area. Its symbol is  $J_g$ .

**HEAT ISLAND (URBAN HEAT ISLAND, UHI):** An area within an urban area characterised by ambient temperatures higher than those of the surrounding area due to human activities, for example, materials like asphalt and concrete absorb energy from the Sun, and have surface radiative properties than the surrounding rural areas. The temperature difference usually is higher at night than during the day, and is most apparent when winds are weak.

**HEAT OF ATOMISATION/ATOMIZATION:** The energy required to dissociate one mole of a given compound into its constituent atoms at standard conditions. It is the heat energy required to change one mole of atom in the solid state to one mole of the atom in its gaseous state.

**HEAT OF COMBUSTION:** The heat energy released when one mole of a substance is completely burnt in excess oxygen or oxidised at

constant volume under standard conditions. It may be expressed as energy/mole of fuel (J/mol), energy/mass of fuel or energy/volume of fuel. It is the heat energy released when one mole of a substance is completely burnt in excess of oxygen. Its value determines the calorific values of fuel and the enthalpy of formation of substances from their oxidation products.

**HEAT OF CONDENSATION:** The amount of heat energy required to transform a given quantity of a substance into a gas at a given pressure or the quantity of heat that must be removed from one gram of a vapour at its condensation point in order for the vapour to condense at that constant temperature.

**HEAT OF CRYSTALLISATION/CRYSTALLIZATION:** The quantity of heat energy that must be removed from one gram of a liquid at its freezing point in order for the liquid to freeze at that constant temperature or the increase in enthalpy when 1 mole of a substance is transformed into its crystalline state at constant pressure.

**HEAT OF FORMATION:** The energy liberated or absorbed when one mole of a compound is formed from its constituent elements at standard conditions. In other words, it is the enthalpy change when one mole of a compound is formed from its elements in their standard state.

**HEAT OF FUSION:** The amount of heat required to melt one gram of solid at its melting point into a liquid at constant pressure and temperature, expressed in J/g. An example of heat of fusion is the melting of ice to water. Applications of heat of fusion are found in many areas of the manufacturing industry such as coin making, glassblowing, forging metal objects and transforming blow moulded plastics into household products.

**HEAT OF NEUTRALISATION/NEUTRALIZATION:** The amount of heat evolved when 1 mole of hydrogen ion ( $H^+$ ) from an acid reacts with 1 mole of hydroxide  $OH^-$  from alkali to form one mole of water. This can be represented by:  $H^+ + OH^- \rightarrow H_2O$ . Its value determines the strength of acid and its basicity.

**HEAT OF REACTION (ENERGY OF REACTION):** The amount of energy liberated or absorbed as a result of the complete chemical reaction of molar amounts of the reactants. If heat is absorbed, it is positive and the reaction is endothermic; if heat is liberated, it is negative and the reaction is exothermic.

**HEAT OF SOLUTION (INTEGRAL HEAT OF SOLUTION; TOTAL HEAT OF SOLUTION):** The energy liberated or absorbed when one mole of a given substance is completely dissolved in a large volume of solvent or the enthalpy of a solution minus the sum of the enthalpies of its components.

**HEAT OF VAPOURISATION (HEAT OF VAPORIZATION):** The amount of heat required to vapourise one gram of a liquid at its boiling point without an increase in temperature usually expressed in J/kg. It is the heat energy released when one mole of a substance in liquid state is changed into gas.

**HEAT PUMP:** A device such as a refrigerator for transferring heat from a substance or space of a lower temperature source to a higher temperature region by doing work. It consists of a compressor, a condenser, a throttle or expansion valve, an evaporator and a working fluid or refrigerant.

**HEAT RADIATION (RADIANT HEAT; THERMAL RADIATION):** Energy in the form of electromagnetic waves emitted by a solid, liquid or gas as a result of its temperature. An example is infrared radiation emitted by a household radiator or electric heater. Thermal radiation is generated when heat from the movement of charges in the material is converted to electromagnetic radiation. Sunshine or solar radiation, is thermal radiation from the extremely hot gases of the Sun, which heats the Earth. The Earth also emits thermal radiation, but at a much lower intensity because it is cooler. The balance between heating by incoming solar thermal radiation and cooling by the Earth's outgoing thermal radiation is the primary process that determines the Earth's overall temperature.

**HEAT SENSITIVE PIT (PIT ORGAN):** An organ located on each side of the head, between the nostril and the eye of some snakes, especially the vipers. These organs are very sensitive to small changes in temperature and are therefore used in detecting preys such as birds and rodents that maintain constant body temperatures.

**HEAT SHIELD:** Any barrier or any heat-protecting coating or system that prevents the heating of a space or an object by absorbing, reflecting or dissipating external heat. An example is the protective structure on a spacecraft or missile that dissipates heat on atmospheric re-entry by melting and vapourising.

**HEAT SINK:** A metal plate specially designed to conduct and radiate heat from an electrical component. Most heat sinks also have fins, which are thin slices of metal that are connected to the base of the heat sink. These additional pieces of metal further dissipate the heat by spreading it over a much larger area. A fan is often used to cool the air surrounding the heat sink, which prevents the heat sink from getting too hot. An example is the heat sink made out of metal such as a zinc or copper alloy and is attached to the processor with a thermal material that draws the heat away from the processor towards the heat sink.

**HEAT STORAGE:** Any means of storing heat for release later. For example, the ocean acts as heat storage medium, this results in smaller daily and annual variations in temperature over the sea.

**HEAT TRANSFER (THERMAL TRANSFER):** The transfer of energy from one body or system to another by the process of conduction, convection, radiation and evaporation. When an object is at a different temperature from another body or its surroundings, heat flows so that the body and the surroundings reach the same temperature, at which point they are in thermal equilibrium. Such spontaneous heat transfer always occurs from a region of high temperature to another region of lower temperature, as described by the second law of thermodynamics. Heat is transferred in three ways: conduction, convection and radiation. Heat conduction occurs mostly in solids as hot, rapidly moving or vibrating atoms and molecules interact with neighbouring atoms and molecules, transferring some of their energy (heat) to these neighbouring particles. Heat transfer by convection is the transfer of heat from one place to another by the movement of fluids, a process that is essentially the transfer of heat via mass transfer. This occurs mostly in liquids and gases. Heat or thermal radiation is energy emitted by matter as electromagnetic waves, due to the pool of thermal energy in all matter with a temperature above absolute zero. It propagates without the presence of matter through the vacuum of space. An example is the heat from the Sun, which reaches the Earth, providing heat energy.

**HEAT TREATMENT:** A process of controlled heating and cooling of a material without melting after fabrication to relieve internal stresses and improve their physical properties such as hardening, strengthening, softening, stress relief, etc. Heat treatment methods include annealing, normalising, stress relief anneals, process anneals, hardening, tempering, etc.

**HEAT-SHOCK PROTEIN (HSP):** Any protein that is synthesised by living cells in response to elevated temperatures or other stress. It is found in all living organisms, including bacteria and humans.

**HEATER CELL:** A type of muscle cell, found in certain fishes that are specialised for generation of heat. It lacks muscle fibril but retain the sarcoplasmic reticulum.

**HEATH (HEATHLAND):** An open area that is exposed to strong winds and with infertile acidic soils, where low shrubs such as heather and gorse grow and which are treeless as a result of grazing by animals.

**HEATING DEGREE DAYS (HDD):** A measure of how cold the temperature is on a given day or over a period of days, calculated against a standard reference temperature of 18° Celsius. For example, a day with a mean temperature of 16 °C has 2 heating degree days (HDD). This is used to assess the climate and the heating and cooling needs for different locations, or demand for energy needed to cool a building.

**HEATSTROKE (SUNSTROKE):** A rise in body temperature and failure of sweating in hot climates, often following exertion. Delirium, coma and convulsions may develop suddenly and rapid cooling should be effected.

**HEAVES:** A disease of horses, in which spores from mouldy hay, block the animals' lungs; causing difficult breathing, wheezing, flatulency and strange coughing.

**HEAVISIDE-KENNELLY LAYER (E-LAYER; E REGION, HEAVISIDE LAYER):** Named after Arthur Edwin Kennelly and Oliver Heaviside, it is a layer of ionised gas occurring roughly 90-150 km above the ground that contains a high concentration of free electrons. It reflects medium-frequency radio waves and because of this reflection radio waves can be propagated beyond the horizon. Propagation is affected by time of day. During the daytime the solar wind presses this layer closer to the Earth, thereby limiting how far it can reflect radio waves. On the night side of the Earth, the solar wind drags the ionosphere further away, thereby greatly increasing the range which radio waves can travel by reflection, called **sky wave**. The extent of the effect is further influenced by the season and the amount of sunspot activity.

**HEAVISIDE-LORENTZ UNITS (LORENTZ-HEAVISIDE UNITS):** A system of units for electric and magnetic quantities based upon centimetre-gram-second (cgs) electrostatic and electromagnetic units. They are like metric units rationalised forms of Gaussian units.

They differ by factors of  $\sqrt{4\pi}$  in the definitions of the electric and magnetic fields and of electric charge.

**HEAVY ATOM EFFECT:** The enhancement of the rate of a spin-forbidden process by the presence of an atom of high atomic number, which is either part of or external to, the excited molecular entity. Mechanistically, it responds to a spin-orbit coupling enhancement produced by a heavy atom.

**HEAVY ATOM ISOTOPE EFFECT:** An isotope effect due to isotopes other than those of hydrogen.

**HEAVY CHEMICALS:** Chemicals that are used extensively in industries and are produced in large quantities.

**HEAVY HYDROGEN (DEUTERIUM):** A stable isotope of hydrogen with atomic weight 2.0144, symbol  $^2\text{H}$  or D. It is a gas ( $\text{D}_2$ ) at room temperature. Its natural abundance in the oceans of Earth is approximately one atom in 6,400 of hydrogen ( $\sim 156.25$  ppm). Deuterium thus accounts for approximately 0.0156% of all naturally occurring hydrogen in the oceans on Earth. It is prepared from heavy water,  $\text{D}_2\text{O}$ , either by electrolysis or by reaction of  $\text{D}_2\text{O}$  with metals such as zinc, iron, calcium and uranium. It is also prepared directly by the fractional distillation of liquid hydrogen. In the uncombined state it finds uses as a research tool. Liquid deuterium is used in bubble chambers to study the reactions of elementary particles with the deuterium nucleus, the deuteron. Deuterons are frequently accelerated in cyclotrons to study their reactions with other nuclei and also to produce radioactive nuclides. Deuterium gas is used in the direct synthesis of organic compounds for tracer studies.

**HEAVY ISOTOPES:** Forms of atoms such as  $^{15}\text{N}$ ,  $^{13}\text{C}$  that contain greater numbers of neutron.

**HEAVY METAL:** An ill-defined subset of elements that exhibit metallic properties and include mainly the transition metals, some metalloids, the lanthanoids and actinoids. Many definitions have been proposed, some based on density, some based on atomic number or atomic weight and some on chemical properties or toxicity. A common definition is based on the density of the metal; a heavy metal must have a density of  $5000\text{kg/m}^3$ . Examples of heavy metals are lead, mercury, iron, zinc and cadmium but can include very light elements such as beryllium and aluminium. Heavy metals are a source of environmental pollution (heavy metal pollution) and some are very dangerous to human health. Living organisms require some



amounts of heavy metals; for example, humans require iron, cobalt, copper, manganese, molybdenum and zinc. However, excessive levels in organisms can be damaging to them. Heavy metals such as mercury, plutonium and lead have no known vital or beneficial effect on organisms and their accumulation over time in the bodies of animals are carcinogenic or toxic, affecting, among others, the central nervous system, the kidneys, liver, skin, bones or teeth causing serious illnesses.

**HEAVY SOIL:** A soil such as clay or silt that retains more moisture and has a high content of fine separates than those with larger particles, such as sand and decomposed granite. Most plants do not thrive very well in heavy soils, as they prefer good drainage soils. Heavy soil is also difficult to cultivate.

**HEAVY WATER:** Water that contains heavy hydrogen (two atoms of deuterium, which is a heavy isotope of hydrogen), instead of ordinary hydrogen and one atom of oxygen (O), with the chemical formula  $D_2O$  and with the symbol D or  $^2H$ . It is used as a moderator to slow down the fast neutrons so that they can react with the fuel in the reactor in some nuclear reactors. Water containing tritium ( $T_2O$ ) may also be called heavy water. Water from most natural sources contains about 0.015% deuterium oxide, which can be purified by distillation, electrolysis or chemical processing.

**HEAVY-FERMION SYSTEM:** A metallic compound with low-temperature specific heat, in which the electrons have a very high effective mass (they act as if they had masses several times the normal mass of the electron). Examples of a heavy-fermion system are  $CeCu_6$  and the superconductors  $LiV_2O_4$ ,  $UBe_{13}$ ,  $CeAl_3$  and  $CeCu_2Si_2$ .

**HEAVY-METAL POLLUTION:** Environmental pollution by metals such as lead, cadmium and mercury with a high relative atomic mass. It can arise from many sources but mainly from the purification of metals such as the smelting of copper and the preparation of nuclear fuels. Although some heavy metals such as iron, cobalt, copper, manganese, molybdenum and zinc are actually necessary for humans in minute amounts others are carcinogenic or toxic, affecting, among others, the central nervous system the kidneys or liver or skin, bones or teeth.

**HEAVY-METAL TOLERANCE:** The ability of certain plants to grow in areas where the concentration of heavy metals, such as copper, lead and zinc, is so high as to prevent the growth of most plants.

**HECT-:** A prefix used to denote six. For example, a hectad is a set of six elements and a hectic equation has degree six.

**HECTARE:** The SI unit of area, which is equal to 10,000 square metres. Its symbol is ha.

**HECTO:** A prefix with the symbol *h*, used in the metric system to denote 100 times or  $10^2$  of the physical units. For example, one hectogram is equal to 100 grams.

**HEDGE (HEDGEROW):** A row of bushes planted to provide a barrier around a field or garden or for ornamental or decorative purposes. Hedges may also serve in erosion control.

**HEDGEHOG:** Any spiny insectivorous small mammal (*Erinaceus albiventris*) of the subfamily Erinaceinae, which is in order Erinaceomorpha. There are seventeen species of hedgehog in five genera. It is native to Europe, Asia and Africa and introduced to New Zealand. It has a long snout and can roll into a ball. The body is 3-30 cm, with a tail length of 3-5 cm; and weights 400-100 g.

**HEDGING:** In climate change, it is balancing the risks of acting too slowly against acting too quickly, and it depends on society's attitude towards risks.

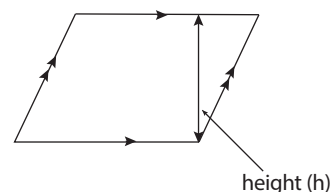
**HEDRON:** A suffix denoting geometrical solids having a specified number of plane faces; for example, enneahedron and eikosihedron are polyhedra having nine and twenty faces respectively.

**HEEL IN:** Placing plants in trenches until they are needed for permanent planting.

**HEFT:** A group of mountain sheep grazing the same area in which they were born, although not kept in by fences.

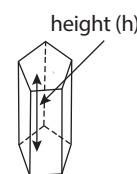
**HEIFER:** A young cow which has not yet given birth.

**HEIGHT:** 1. The vertical measurement from head to foot or from base to top. 2. The vertical distance that something such as a mountain is above the sea or ground levels or another chosen reference point. 3. In mathematics, the length of the altitude of a polygon in a given orientation. 4. The maximum of the absolute values of the coefficients of a given polynomial or linear form. 5. The angular distance of a celestial body above the horizon.



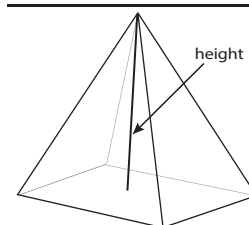
**HEIGHT OF A PARALLELOGRAM:** The length of the shortest line segment between a base of a parallelogram and the line containing the opposite side (or between two opposite sides). The height is perpendicular to the base.

**HEIGHT OF A PRISM/CYLINDER:** The length of the shortest line segment from a base of a prism or cylinder to the plane containing the opposite base. The height is perpendicular to the bases.



Prism height (h)

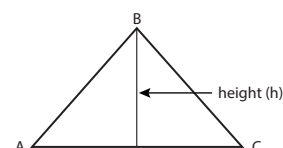
**HEIGHT OF A PYRAMID/CONE:** The length of the shortest line segment from the apex of a pyramid or cone to the plane containing the base. The height is perpendicular to the base.



Height of a pyramid/cone

**HEIGHT OF A RECTANGLE:** The length of a side perpendicular to its base.

**HEIGHT OF A TRIANGLE:** The length of the shortest segment from a vertex of a triangle to the line containing the opposite side. The height is perpendicular to the base.



Height of a triangle

**HEIMLICH MANOEUVRE:** A first-aid procedure used to relieve choking due to a blockage in the larynx (windpipe). The manoeuvre consists of the following steps: (i) stand behind the patient if the person is sitting or standing and wrap your arms around the waist; if the person is lying on his or her back, straddle the person facing the head. (ii) Make a fist with the hand and place it between the navel and the ribcage and apply a sudden strong pressure on the abdomen. This is repeated until the blockage is dislodged.

**HEINE-BOREL COVERING THEOREM:** Named after German mathematician Heinrich Eduard Heine (1821-1881) and French mathematician Emile Borel (1871-1956), it is the theorem that a subset of Euclidean space is closed and bounded if and only if it is compact.

**HEINE'S THEOREM:** Named after German mathematician Heinrich Eduard Heine (1821-1881), it is the result that if  $M$  and  $N$  are metric spaces,  $A$  is a compact subset of  $M$  and  $f$  is a continuous function from  $A$  to  $N$ , then  $f$  is uniformly continuous on  $A$ .

**HEISENBERG UNCERTAINTY PRINCIPLE:** It states that the precise position and momentum of an electron cannot be determined simultaneously: the more precisely one property is measured, the less precisely the other can be measured or the more you know the position of a particle, the less you can know about its velocity and the more you know about the velocity of a particle, the less you can know about its instantaneous position. It is expressed as

$\Delta x \Delta p \geq \frac{h}{4\pi}$ , where  $\Delta x$  is the uncertainty in the  $x$ -coordinate of the particle,  $\Delta p$  is the uncertainty in the  $x$ -component of the particle's momentum and  $h$  is Planck's constant. If a measurement of precision  $\Delta x$  and a simultaneous measurement of momentum is made with precision

$\Delta p_x$ , then the product of the two uncertainties can never be smaller than  $\frac{h}{2}$ , where  $\hbar = \frac{h}{2\pi}$ . Hence,  $\Delta x \Delta p_x = \frac{\hbar}{2} = \frac{h}{4\pi}$ .

**HELICAL THICKENING (SPIRAL THICKENING):** A type of secondary wall patterning in tracheary elements in which the secondary cell wall is laid down in a helical pattern. Spiral thickening, like annular thickening, permits further extension of the cell. It is, therefore, found in tissues such as protoxylem, which develop in young organs that are still elongating.

**HELICITY:** The chirality of a helical, propeller or screw-shaped molecular entity. A right-handed helix is described as P (or plus), a left-handed one as M (or minus).

**HELICASES:** Enzymes that unwind the DNA double helix. There are DNA and RNA helicases. DNA helicases are essential during DNA replication because they separate double-stranded DNA into single strands allowing each strand to be copied.

**Helicobacter pylori:** Spiral-shaped swimming bacterium that causes gastritis (stomach inflammation), stomach ulcers or stomach cancer when it infects the stomach lining. The infection may be acquired from consuming contaminated food and water or through contact with an infected person.

**HELICOID:** 1. In botany, coiled like a spiral or helix as present in some cymose inflorescence. 2. In mathematics, any geometric solid or surface shaped like a screw thread.

**HELICONIA (FALSE BIRD-OF-PARADISE; LOBSTER-CLAWS; WILD PLANTAINS):** Any of the medium to large rhizomatous herbaceous herbs in the Heliconiaceae family, native to the tropical Central and South America and some islands in the South Pacific and grown throughout the tropical and sub tropical regions of the world. Each erect shoot is composed of a stem and leaves, usually terminating with an inflorescence. Stem lengths range from less than about 0.5-5 metres. It has long, oblong simple leaves. The leaves may be arranged alternately or opposite on non woody petioles. The flowers, pollinated by bats, are borne on long, erect or drooping panicles, made up of brightly coloured waxy bracts. The colour of the flowers are shades of yellow, green and orange. It is grown as an ornamental plant. The leaves are used as food wrappers in some regions.

**HELICOPTER:** A powered aircraft that achieves both lift and propulsion by means of one or more power-driven horizontal rotors on top of the fuselage, enabling it to take off and land vertically, move in any direction or remain stationary in the air.

**HELIOCENTRIC UNIVERSE:** The theory that the solar system is centred on the Sun, with Earth and other planets revolving around it. It was first postulated by Polish astronomer, Nicolaus Copernicus.

**HELIOCHROMISM:** A term, which is applied to photochromic compounds possessing a high efficiency for colouring with near-ultraviolet radiation and a low efficiency for bleaching with visible radiation, but a moderate efficiency for thermal fades at ambient temperature. These compounds are activated by unfiltered sunlight and deactivated under diffuse daylight conditions, therefore being suitable for sun-lens applications.

**HELIOGRAPHIC COORDINATES:** The latitude and longitude of a feature on the Sun's surface. **Heliographic latitude** is an object's angular distance north or south of the solar equator. **Heliographic longitude** can be measured east or west of the central solar meridian or given in terms of the **Carrington rotation number**.

**HELIOSEISMOLOGY:** The study of the propagation of wave oscillations, particularly acoustic pressure waves, in the Sun, which are pressure waves and not seismic waves. Large numbers of distinct,

resonating sound waves can be detected by the Doppler shift they cause in spectral lines emitted at the Sun's surface. The periods of these waves depend on their propagation speeds and the depths of their resonant cavities similar to the lengths of organ pipes that produce different notes.

**HELIOSPHERE:** The region of space, surrounding the Sun, which is inflated by the solar wind and within which the Sun exerts a magnetic influence. Despite its name, it is almost certainly greatly elongated due to the movement of the Sun with respect to the interstellar medium. It is bounded by the heliopause. Between the heliopause and the bow shock is presumed to be a layer of interstellar gas known as the heliosheath.

**HELIOSTAT:** A movable flat mirror used to reflect sunlight into a stationary solar telescope. If the telescope is aligned with Earth's axis, only one mirror is needed. A heliostat is similar to a coelostat, but has a simpler design and (unlike the coelostat) makes the image of the Sun rotate as the Sun moves along the sky.

**HELITE:** A type of inorganic molecules containing a double helix of bipyridyl-derived molecules formed around a chain of up to five copper(I) ions.

**HELIUM:** A colourless, odourless, inert gaseous element occurring in natural gas and with radioactive ores with the symbol He, atomic number 2, relative atomic mass of 4.0026 and a boiling point -268.9 °C (4 K). It is non-reactive and forms no compounds. It is the second most abundant element in the universe and has the lowest boiling point of all elements as well as the second lightest element. It is used for inflating balloons and a dilutant for oxygen in deep-sea breathing.

**HELIUM STAR:** An O star or B star in which the absorption lines of helium are abnormally strong and those of hydrogen are absent or weak.

**HELIUM-CADMIUM LASER (CADMIUM-HELIUM LASER):** A continuous wave laser emitting mainly at 325.0 and 441.6 nm from singly ionised cadmium.

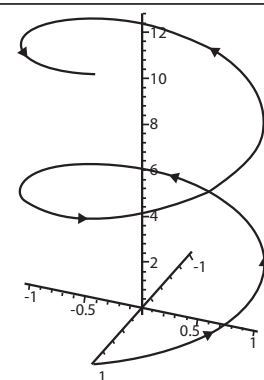
**HELIUM-NEON LASER (HENE LASER):** A small gas laser, in which the medium is a mixture of helium and neon in the mole ratio 1:5 respectively. It emits mainly at 632.8 nm, 1152.3 nm and 3391.3 nm from excited neutral Ne atoms.

**HELIX:** 1. A three dimensional curve in the form of a spring, corkscrew or screw thread generated by a line that encircles a cylinder or cone at a constant cycle. 2. The rim of the external ear. 3. The molecular conformation of a spiral nature, generated by regularly repeating rotations around the backbone bonds of a macromolecule.

**HELIX RESIDUE:** In a polymer, it is the smallest set of one or more successive configurational base units that generates the whole chain through helical symmetry.

**HELIX-TURN-HELIX (HELIX-LOOP-HELIX):** A structural motif characteristic of certain proteins that regulate gene expression and capable of binding DNA. It consists of two alpha-helices ( $\alpha$  helices) connected by short non-helical segment called a turn.

**HELLY'S THEOREM:** Named after Austrian mathematician and physicist, Eduard Helly (1884-1943), it is the basic result in discrete geometry, which states that if a finite family of bounded closed subsets of an  $n$ -dimensional vector space is such that every subcollection with at least  $n + 1$  members has a common point, then the entire family has a common point. The example of the sides of a triangle shows that the sub-collections must have  $n + 1$  members and that  $n$  are insufficient.



**HELMET (HOOD):** In botany, it refers to having a hollow or arched covering.

**HELMHOLTZ COILS:** A pair of flat circular coils with equal numbers of turns and radius placed a distance apart that is equal to the radius and connected in series to have a common current. If the same current is flowing in both coils then the value of the magnetic field strength is approximately uniform between the coils. Coils of this type are used to create fields and, in some cases, are employed to counter the effect of the Earth's magnetic field.

**HELMHOLTZ ENERGY (HELMHOLTZ FUNCTION):** Internal energy minus the product of thermodynamic temperature and entropy. Its symbol is  $A$ .

**HELMHOLTZ FREE ENERGY:** A thermodynamic quantity, which measures the useful work obtainable from a closed thermodynamic system at a constant temperature and volume. It is denoted by  $A = U - TS$ , where  $U$  is the internal energy,  $T$  the thermodynamic temperature and  $S$  the entropy. Developed by Hermann von Helmholtz, it represents the maximum work done aside expansion in a reversible isothermal process.

**HELMHOLTZ, HERMANN LUDWIG FERDINARD VON (1821-94):** German physiologist and physicist and also a physician, mathematician, philosopher and lecturer on popular science. In 1850, he measured the speed of a nerve impulse and in 1851 invented the ophthalmoscope. He also extended Thomas Young's theory of colour vision and explained the mechanism of lens accommodation in the eye. In physics, he discovered the conservation of energy and in 1847 formulated it mathematically and also introduced the concept of free energy.

**HELMINTH:** Any of the various parasitic worms, including the flukes, tapeworms and nematodes. **Helminthiasis** is the diseased condition resulting from an infestation with helminths. **Helminthology** is the study of parasitic worms.

**HELOPHYTE:** A marsh plant with its perennating buds situated in the mud at the bottom of a lake or pond. Examples are bulrushes (*Typha*) and water plantains (*Alisma*).

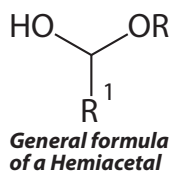
**HELOTIALES (CUP FUNGI):** An order of the Discomycetes containing some 1750 species in about 225 genera. Its members are similar to the Pezizales except that the asci lack an operculum. Most are saprobes but some are plant parasites. An example is *Sclerotinia*, which causes brown rot of stored fruit.

**HELPER T CELLS (T-HELPER CELL):** A type of lymphocyte that stimulates the production of antibodies by activating B cells when an antigen is present.

**HEMES (HEME DERIVATIVES):** Complexes consisting of an iron ion coordinated to a porphyrin acting as a tetradentate ligand and to one or two axial ligands.

**HEMI:** A prefix denoting half.

**HEMIACETAL:** The product formed by the condensation of an aldehyde with an alcohol; it contains one oxygen atom linked to a central carbon in a hydroxyl fashion and one oxygen atom linked to the same central carbon by an ether linkage. The general formula of a hemiacetal is as shown in the diagram, where  $R$  and  $R^1$  are organic residues.



**HEMIAMINALS:** Functional group or type of chemical compound that has a hydroxyl group and an amine attached to the same carbon atom:  $\text{-C(OH)(NR}_2\text{)-}$ .  $R$  can be hydrogen or an alkyl group. Hemiaminals are intermediates in imine formation from an amine and a carbonyl by alkylimino-de-oxo-bisubstitution.

**HEMIASCOMYCETES:** A class of the Ascomycotina containing ascomycetes that do not develop ascocarps. It contains about 330 species in some 60 genera. Such fungi are unicellular or produce a poorly developed mycelium.

**HEMICELLULOSE:** Any polysaccharide found in the cell walls of plants, made soluble by dilute alkali or which go into solution quite readily in hot dilute mineral acids with the formation of simple sugars. The branched chain of this molecule binds to cellulose microfibrils together with pectins, forming a network of cross-linked fibre. There are two distinct hemicelluloses in plants: the acidic and the neutral. Acidic hemicelluloses contain a larger number of uronic acids than neutral hemicelluloses. Hemicelluloses are partially fermented by the microorganisms of the colon, producing some volatile fatty acids. Hemicelluloses are insoluble in water but soluble in alkaline solutions. Hemicelluloses constitute about one-fourth of perennial plants and about one-third of annual plants.

**HEMICHORDATA:** A phylum of soft-bodied, worm-shaped marine invertebrates that comprises two classes, the Enteropneusta and the Pterobranchia.

**HEMICRYPTOPHYTE:** A plant life-form in Raunkiaer's system of classification. They are herbaceous perennial and produce perennating or aerial buds at the surface of the soil.

**HEMICYCLE:** A semicircle or semicircular structure.

**HEMIHEDRAL FORM:** The form of a crystal in which only half the number of faces required for the symmetry are present.

**HEMIHYDRATE:** A compound containing one water molecule for every two molecules of the compound.

**HEMIKETALS:** Hemiacetals having the structure  $\text{R}_2\text{C(OH)}$  or  $(\text{R} \neq \text{H})$ , derived from ketones by formal addition of an alcohol to the carbonyl group. This term, once abandoned, has been reinstated as a subclass of hemiacetals.

**HEMIMETABOLISM (HEMIMETABOLY; HEMIMETAMORPHOSIS INCOMPLETE; METAMORPHOSIS):** A type of development in certain insects, in which there is incomplete or partial metamorphosis, with three distinct stages: the egg, nymph and the adult stage or imago; the nymphs resemble adults but are smaller and have different body proportions. They go through a series of molts, each time looking more like an adult, until it reaches full size, without going through a pupal stage. Examples of insect groups showing incomplete metamorphosis include: Hemiptera (scale insects, aphids, whitefly, cicadas, leafhoppers and true bugs) orthoptera (grasshoppers and crickets), Mantodea (praying mantises), Blattaria (cockroaches), Dermaptera (earwigs) and Odonata (dragonflies and damselflies).

**HEMIMETABOLOUS:** Development of certain insects in which there is incomplete or partial metamorphosis, with three distinct stages: the egg, nymph and the adult stage or imago, which does not include any pupal stage. The nymph usually resembles the adult stage but lacks wings and functional reproductive organs.

**HEMIPARASITE (SEMIPARASITE; WATER PARASITES):** A parasitic plant, that lacks a fully developed root system and forms connection with plant hosts from which it absorbs water and nutrients, mainly water and minerals. Examples are witchweed that damages sugarcane, corn and other grass-like mistletoe, which attaches itself to broadleaf trees such as ash, maple, walnut, birch and some conifers. Mistletoe roots bore into the host's branches in order to draw out water and nutrients.

**HEMIPENIS:** One-half of paired erectile copulatory organs found in the males of the order Squamata (lizards and snakes).

**HEMIPTERA:** An order of insects commonly called the true bugs, comprising species of cicadas, aphids, planthoppers, leafhoppers, shield bugs, etc. They are about 1 millimetre to around 15 centimetres in size.

**HEMISPHERE:** 1. In anatomy, **cerebral hemisphere** is either of the lateral halves of the cerebrum or cerebellum. 2. In mathematics, half of a sphere, bounded by a plane containing its centre. 3. Any half of the Earth, which may be either south or north of the equator or east or west of the prime meridian.

**HEMISPHEROID:** One half of a spheroid.



**HEMITROPOUS (HEMIANATROPOUS) OVULE:** An ovule that is half converted such that the micropyle is at right angle to the funicle and the funicle attached near the middle. The embryo sac remains straight.

**HEMIZYGOUS:** Having only one of a pair of genes, this may be due to a result of deletion or chromosome loss. In male mammals it usually applies to genes on the male's X chromosome.

**HEMLOCK:** Any of 14 species of coniferous evergreen trees that make up the genus *Tsuga*, of the family Pinaceae (pine family), native to North America and Asia. The poison hemlock and water hemlock are herbs of the family Umbelliferae (carrot family) of the division Magnoliophyta. A true hemlock is a tall, pyramidal tree with purplish or reddish-brown bark, slender horizontal or drooping branches and short, blunt leaves that grow from woody cushion-like structures on the twigs. Some hemlocks are important timber trees and many are popular ornamentals.

**HEMLOCK POISONING:** Poisoning of young cattle by eating fresh shoots of the hemlock. Certain animals such as sheep and goats are believed to be resistant to the poison.

**HEMP:** An annual herbaceous plant, *Cannabis sativa*, of the Cannabaceae family. It is grown primarily for its fibre, which is strong and durable and used for cord and for artificial sponges; coarse fabrics as sacking (burlap) and canvas. Its flowers and leaves are used to produce the narcotic addictive drugs, bhang, hashish and marijuana. The seed yields oil, called oil of hemp, used in the manufacture of soap and oil paints.

**HENDECA-:** A prefix denoting eleven, for example, **hendecagon** is a polygon having 11 angles and 11 sides.

**HENRY:** The SI unit of inductance, with the symbol H, which is equal to the inductance of a closed circuit with a magnetic flux of 1 weber per ampere of current. It is the reaction of an electric current against the magnetic field that surrounds it. An induced electromotive force of one volt is produced when the current is varied at the rate of one ampere per second. It is expressed as  $H = VA^{-1} s = m^2 kg s A^{-2}$ .

**HENRY, JOSEPH (1797-1878):** US physicist, who became professor of natural philosophy at Princeton University in 1832. He discovered several important principles of electricity and in 1829 made an electric motor; used insulated windings to produce a powerful electromagnet; and a year later discovered electromagnetic induction independently of Faraday. In 1832, he discovered self-induction. He invented the electric relay in 1835 and also helped Samuel F.B. Morse to develop the telegraph.

**HENRY'S LAW:** A law named after William Henry (1774-1836), the English chemist who first reported the relationship, which states that at a constant temperature the mass of gas dissolved in a liquid is proportional to the partial pressure of the gas provided no chemical reaction takes place between the liquid and the gas.

**HENSEN'S NODE (PRIMITIVE NODE; PRIMITIVE KNOT):** A group of cells that acts as an organiser for gastrulation in the early embryo of birds and mammals. It is located immediately anterior to the primitive streak, where the outer layer of cells will begin to migrate inwards and is analogous to the dorsal lip of the blastopore in the blastula of an amphibian embryo.

**HEPARAN SULFATE:** Any of a group of glycosaminoglycans found in all animal tissues consisting of alternating residues of glucosamine and hexuronic acid, usually attached to protein, forming a proteoglycan. It regulates a wide variety of biological activities, including developmental processes, angiogenesis, blood coagulation and tumour metastasis. They accumulate in a group of inherited metabolic disorders called mucopolysaccharidoses.

**HEPARIN:** A glycosaminoglycan with anticoagulant properties comprising complex carbohydrate molecules called **mucopolysaccharides**, occurring in vertebrate tissues, especially the lungs and blood vessels. It is a mixture of organic compounds

used as a short-term anticoagulant to prevent thrombosis during and after surgery and for initial treatment of various heart, lung and circulatory disorders in which there is increased risk of blood clotting.

**HEPATIC:** A term describing part of the body related to the liver and its functions.

**HEPATIC ARTERY:** An artery that distributes blood to the liver, pancreas and gallbladder and also to the stomach and duodenal portion of the small intestine.

**HEPATIC PORTAL VEINS:** The veins in the abdominal cavity that transport blood containing the absorbed products of digestion from the gastrointestinal tract and spleen directly to the liver.

**HEPATICAEE:** A class of plants within the Bryophyta comprising the liverworts, which occur in moist environments.

**HEPATITIS:** An inflammation of the liver that causes physical weakness and jaundice. There are seven known types of viral hepatitis labelled A-G with different routes of transmission and different prognoses. The commonest single cause, accounting for up to 40 percent of cases worldwide, is the hepatitis A virus (HAV), which colonises the digestive tract, causing an acute illness. Much more serious is infection with the hepatitis B virus (HBV). It can lead to chronic inflammation of the liver and, in some cases, to liver cancer. Hepatitis C can also become chronic. HDV, common in the Mediterranean region, can only replicate in the presence of HBV. HEV is endemic in some tropical countries. Vaccines are available against HAV and HBV.

**HEPATOCYTE:** A specialised epithelial cell of the main tissue of the liver that is the most abundant type of cell in the liver. They are involved with the various functions of the liver, including protein synthesis, protein storage, transformation of carbohydrates, synthesis of cholesterol, bile salts and phospholipids and detoxification, modification and excretion of exogenous and endogenous substances.

**HEPATOPHYTA:** A phylum comprising the liverworts which are plants that lack vascular tissue and possess rudimentary root-like organs.

**HEPATOTOXIN:** Any chemical such as alcohol capable of causing damage to the liver.

**HEPTA-:** A prefix that denotes seven. For example, a heptagon is a seven-sided plane figure.

**HEPTAD:** A group or sequence of seven.

**HEPTADECANOIC ACID (MARGARIC ACID; ARGARIC ACID; n-MARGARIC ACID; n-HEPTADECANOIC ACID; HEPTADECYLIC ACID; n-HEPTADECYLIC ACID):** A white crystalline carboxylic acid with a linear chain of carbon atoms with the chemical formula  $C_{16}H_{33}COOH$ , density 0.853 g/cm<sup>3</sup>, melting point 61.3 °C (334 K) and boiling point 227 °C (500 K). It is soluble in ether and alcohol and insoluble in water. It is present as a trace component of the fat and milkfat of ruminants. It does not occur in any natural animal or vegetable fat at concentrations over half a percent.

**HEPTAGON (SEPTAGON):** A two-dimensional geometric figure formed of seven sides and seven angles. In a regular heptagon, in which all sides and all angles are equal, the sides meet at an angle of  $5\pi/7$  radians (128.5714286 degrees).

**HEPTAHYDRATE:** A hydrate, for example, magnesium sulfate,  $MgSO_4 \cdot 7H_2O$ , whose solid contains seven molecules of water of crystallisation per molecule or per unit cell.

**HEPTANE (n-HEPTANE):** A straight-chain alkane hydrocarbon with the molecular formula  $C_7H_{16}$ , density 679.5 mg mL<sup>-1</sup>, melting point -91 to -90 °C (182.2-183.0 K) and boiling point 98-99 °C (371.2-371.8 K). It is present in petrol and it is either a liquid or grease. It is the seventh member of the methane series. Heptane is commercially available as mixed isomers for use in paints and coatings, as a rubber cement solvent, as an outdoor stove fuel, as pure n-heptane for research and development and pharmaceutical manufacturing and as a minor component of gasoline.

**HEPTAVALENT (SEPTIVALENT):** Element such as manganese in potassium permanganate having a valence of 7 or an element that is in oxidation state of +7.

**HERB:** 1. Any plant, in which the aerial parts wither at the end of the growing season. 2. Any plant, usually a flowering plant, used medicinally in perfumes or as a spice. Herbs have a variety of tastes, which may be sweet, bitter, aromatic or pungent.

**HERBACEOUS (HERBACEOUS PLANT):** An annual, perennial or biennial plant with soft succulent non-woody stems whose aerial parts die after the end of the growing season. In annuals, the whole plant dies after the growing season, when they have flowered and fruited and grow again from seed. The aerial parts of biennials and perennials such as stems and leaves die at the end of the growing season, but parts of the plant such as bulbs, corms and tubers survive under or close to the ground from season to season. Examples of herbaceous biennials include carrot, parsnip and common ragwort. Herbaceous perennials include potato, hosta, mint, most ferns and most grasses.

**HERBAGE:** The green plants, especially grass, eaten by grazing animals.

**HERBALISM:** The study or practice of the medicinal and therapeutic use of plants and their derivatives.

**HERBARIUM:** A collection of dried, pressed plants or pickled fruits and fleshy organs used as an aid to the identification of unknown plants and also in the classification of plants.

**HERBICIDE:** A chemical substance or cultured biological organism that kills unwanted plants such as weeds or inhibits their growth. Examples are contact herbicides such as sulfuric acid, diquat, paraquat, which are effective on only the plant organs with which they are in contact; and translocated herbicides such as amitrole, picloram, 2,4-D, which are effective against roots or other organs, to which they are transported from aboveground treated surfaces.

**HERBICIDE-TOLERANT (HERBICIDE-RESISTANT):** Inherent ability of a plant or weed to survive and reproduce after it has been treated with herbicide. This resistance may be naturally occurring or induced by such techniques as genetic engineering or selection of variants produced by tissue culture or mutagenesis.

**HERBIG-HARO OBJECT:** A region in interstellar space named after G. Herbig and G. Haro, who discovered them in the early 1950 that is small, bright, containing gas and electromagnetic radiation by a jet of gas, ranging in sizes of 300 to 1000 astronomical units and can vary in intensity over periods of only a few years.

**HERBIVORE:** An animal which subsists mainly on plants, vegetation or their products including seeds, fruits and nectar. They are more numerous than other animals and their food is the most abundant. Examples are sheep, rabbits, cattle, cows and horses. Some animals adapted to herbivorous diet have the four-chambered stomach of ruminants, the incisor teeth of rodents and the specialised grinding molars of cattle, sheep and goats.

**HERCULES:** The Northern Hemisphere constellation located between Lyra and Corona Borealis and the fifth largest constellation. It is a red giant and perhaps the largest visible star in the sky. It contains the globular star cluster M13, which is only slightly visible to the naked eye but can be viewed with even a small telescope. Hercules reaches its highest point in the evening sky in late July.

**HEREDITARY:** 1. In biology, capable of being passed genetically from one generation to the next. 2. In mathematics, of a set, containing all those elements that have a given relation to any element of the set; closed under that relation. For example, the positive integers are hereditary with respect to  $\geq$  on the integers. 3. Of a property, transferred by a given relation, so that if  $x$  has the property and  $R_{xy}$ , then  $y$  also has the property. For example, evenness is hereditary with respect to addition of 2 but not 3. 4. Of a topological property, possessed by any subspace of a topological space that has the property.

**HEREDITARY EFFECTS OF RADIATION:** In radiation protection, they are those effects present in the offspring of those exposed to probabilistic/stochastic levels of ionising radiation.

**HEREDITY:** The complete set of inherited characteristics of an organism or the functional units of heritable material such as hair colour from one generation to the next in living organisms that are found within all living cells. Each member of the parental generation transmits only half its genes to the offspring and different offspring of the same parents receive different combinations of genes.

**HERITABILITY:** A fundamental concept in genetics that is the measure of the degree to which the variance of phenotypic variation in a population is attributable to genetic variation among individuals. It is given by a value between 0 and 1 and effectively measures the extent to which offspring resemble their parents relative to the population mean.

**HERMANN-MAUGUIN SYSTEM (INTERNATIONAL NOTATION):** A notation used to describe the symmetry of point groups, used in crystallography. It is named after the German crystallographer Carl Hermann and the French mineralogist, Charles-Victor Mauguin.

**HERMAPHRODITE:** 1. An animal or organism such as the earthworm, pulmonate snails, opisthobranch snails and slugs that have both male and female reproductive organs. The term is also used for a plant whose flowers contain both stamens and carpels. Such plants are referred to as monocious. Most plants are hermaphrodites and also some fish. 2. Also called **intersex**, it refers to a human that has biologically several variations in sex characteristics that does not apply to the typical definitions of a female or male. Such characteristics include chromosomes, gonads, sex hormones, or genitals.

**HERMITE INTERPOLATION:** A method of interpolating data points as a polynomial function; named after Charles Hermite, French mathematician.

**HERMITE, CHARLES (1822- 1901):** French mathematician, who developed the theory of functions, used elliptic functions to solve the general quintic in one variable and proved that  $e$ , the base of natural logarithms, is transcendental.

**HERMITE'S POLYNOMIALS:** The polynomials  $H_n(x)$ , which may be defined by  $e^{2nx-x^2} = \sum_{n=0}^{\infty} \frac{H_n(x)t^n}{n!}$ . The polynomial  $H_n(x)$  is a solution of the differential equation  $y'' - 2xy' + 2ny = 0$  for the integer  $n$ . The first six Hermite polynomials are  $H_0(x) \equiv 1$ ,  $H_1(x) \equiv 2x$ ,  $H_2(x) \equiv 4x^2 - 2$ ,  $H_3(x) \equiv 8x^3 - 12x$ ,  $H_4(x) \equiv 16x^4 - 48x^2 + 12$  and  $H_5(x) \equiv 32x^5 - 160x^3 + 120x$ .

**HERMITIAN:** 1. Denoting or relating to a square matrix over the complex numbers, whose transpose is equal to the matrix of the complex conjugates of its entries, so that it is its own Hermitian conjugate. For example, every real symmetric matrix is Hermitian. 2. An operator that is equal to its own Hermitian conjugate.

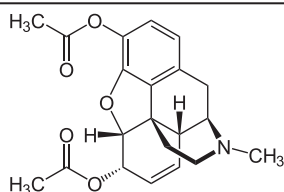
**HERMITIAN CONJUGATE (ADJOINT OF A MATRIX OR):** Often denoted  $A^*$ ,  $A'$  or  $A^\dagger$ , it is the matrix that is the transpose of the matrix of the complex conjugates of the entries of a given matrix. Alternatively, it is also known as **dual** and is the operator  $A^*$  that is conjugate to a given linear operator  $A$  between normed spaces  $X$  and  $Y$ .

**HERMITIAN VECTOR SPACE (UNITARY SPACE):** A complex vector space on which is defined an inner product.

**HERNIA:** A protrusion of an anatomical part such as a section of the intestine through an opening, tear or weakness in the abdominal wall musculature. Common types of hernia are protrusion of an intestinal loop through the umbilicus (umbilical hernia) or inguinal canal of the groin (inguinal hernia) or protrusion of part of the stomach or oesophagus into the chest cavity through the opening (hiatus) of the oesophagus into the diaphragm (hiatus hernia).

**HEROIN (DIACETYLMORPHINE; MORPHINE DIACETATE; (5 $\alpha$ , 6 $\alpha$ )-7,8-DIDEHYDRO-4,5-EPOXY-17-METHYLMORPHINAN-3,6-DIOL DIACETATE):** A highly addictive

alkaloid with the chemical formula  $C_{17}H_{17}NO(C_2H_3O_2)_2$ , made from sap collected from a type of poppy plant or derived from morphine. It is a narcotic drug that causes an ecstatic, warm, glowing sensation, followed by relaxation and contentment, if injected. It is about three times more potent than morphine and because of its increased lipid solubility it rapidly enters the brain. In many parts of the world, it is used as an analgesic for relief of pain. It is illegal in most countries.



Chemical structure  
of Heroin

**HERON OF ALEXANDRIA:** First century Greek mathematician and inventor, whose major geometric work, *Metrica*, was lost and found in 1896. He invented a jet-propelled rotary steam engine and a method of calculating square roots similar to that used by modern computers.

**HERON'S FORMULA (HERO'S FORMULA):** An important formula in plane geometry that allows the area of any triangle to be calculated without knowing the altitude (perpendicular height) of any of its sides. Let  $a$ ,  $b$  and  $c$  be the side lengths of a triangle and  $A$  its area, it is expressed as  $A = \sqrt{s(s-a)(s-b)(s-c)}$  where  $s = \frac{a+b+c}{2}$ .

**HERON'S METHOD (HERO'S METHOD):** A method for determining the square root of a number, which can be derived from Newton's method.

**HERONIAN MEAN:** Named after Hero of Alexandria, the Heronian mean of two numbers  $a$  and  $b$  is given by  $\frac{1}{3}[a + \sqrt{ab} + b]$ . It is always at least equal to the geometric mean and at most equal to the arithmetic mean. It is used in finding the volume of a frustum of a pyramid or cone. The volume is equal to the product of the height of the frustum and the Heronian mean of the areas of the opposing parallel faces.

**HERONIAN TRIANGLE:** A triangle named after Hero of Alexandria, with integer sides and integer area.

**HERPES:** Any of several infectious diseases caused by herpes simplex viruses (HSV) of the herpes group. There are two types of the virus: HSV type 1, also known as **human herpesvirus 1 (HHV-1)** and HSV type 2 (HSV-2). HSV type 1 is the causative agent of common inflammation, the cold sore and fever blisters in the mouth and around it. HSV type 2 is responsible for genital herpes, a highly contagious, sexually transmitted disease characterised by painful blisters in the genital area. The virus can also be transmitted from mother to an infant through the birth canal.

**HERPES ZOSTER (SHINGLES):** A viral infectious disease that affects the sensory nerves, characterised by pain and eruption of blisters along the course of the affected nerves. It is caused by the varicella-zoster virus that also causes chickenpox.

**HERPESVIRUS:** One of a family of viruses of complex DNA-containing viruses causing diseases that tend to recur in man and most vertebrates.

**HERPETOLOGY:** The branch of zoology concerned with the study of reptiles and amphibians, which are sometimes collectively known as herps or herptiles.

**HESSENBERG MATRIX (HESSENBERG FORM):** A matrix that has only zeros more than one row above. **Lower Hessenberg matrix** has zero entries more than one row above the first main diagonal. An **upper Hessenberg matrix** has zero entries more than one row below the first main diagonal. The two kinds are shown below:

$$\begin{vmatrix} 4 & 1 & 0 & 0 \\ 8 & 5 & 2 & 0 \\ 2 & 9 & 6 & 3 \\ 4 & 3 & 1 & 7 \end{vmatrix}$$

Lower Hessenberg Matrix

$$\begin{vmatrix} 1 & 4 & 2 & 3 \\ 3 & 4 & 1 & 7 \\ 0 & 2 & 3 & 4 \\ 0 & 0 & 1 & 3 \end{vmatrix}$$

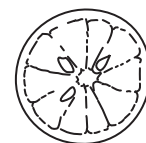
Upper Hessenberg Matrix

**HERTZ:** SI unit of frequency, which is equal to one cycle per second,  $\text{Hz} = \text{s}^{-1}$ . Its symbol is Hz. It is often combined with metric prefixes to denote multiple units such as the kilohertz (1,000 Hz), megahertz (1,000,000 Hz) and gigahertz (1,000,000,000 Hz).

**HERTZ, HEINRICH RUDOLPH (1857):** German physicist, who was born in Hamburg and educated at the University of Berlin. He was a Professor of Physics at the Technical School in Karlsruhe from 1885 to 1889 and also a Professor of Physics at the University in Bonn. Hertz clarified and expanded the electromagnetic theory of light that had been put forth by the British physicist, James Clerk Maxwell in 1873. He proved that electricity can be transmitted in electromagnetic waves, which travel at the speed of light and which possess many other properties of light. His experiments with these electromagnetic waves led to the development of the wireless telegraph and the radio. The unit of frequency that is measured in cycles per second was renamed the hertz, which is commonly abbreviated Hz.

**HERTZSPRUNG-RUSSELL DIAGRAM (H-R DIAGRAM):** A graphical representation of the absolute magnitude of stars (a measure of their temperatures), plotted against their colours, named after the Danish astronomer Ejnar Hertzsprung (1873-1967) and the U.S. astronomer Henry Norris Russell (1877-1957), who worked independently of each other.

**HESPERIDIUM:** A fruit such as an orange or other citrus fruit consisting of a thick leathery rind, juicy pulp and soft segmented pulp.



Hesperidium

**HESS'S LAW (THE LAW OF CONSTANT HEAT SUMMATION):** It states that the total energy change resulting from a chemical reaction is dependent only on the initial and final states and is independent of the reaction route.

**HESSIAN (HESSIAN MATRIX):** The Hessian matrix of a multivariate function is the second partial derivatives of a given function. For

example, the Hessian of  $f(x, y) = x^2 - y^2$  is given by:  $\begin{bmatrix} 2 & 0 \\ 0 & -2 \end{bmatrix}$ .

**HETERO ATOM:** An odd atom in the ring of a heterocyclic compound that is not carbon or hydrogen to indicate that non-carbon atoms have replaced carbon in the backbone of the molecular structure. For instance, nitrogen is the hetero atom in pyridine. Some other hetero atoms include oxygen, sulfur, phosphorus, boron, chlorine, bromine or iodine.

**HETEROBARIC LEAF ANATOMY:** An anatomical arrangement in which the smaller veinlets of a leaf, form a network of fine translucent lines on the leaf surface.

**HETEROBLASTIC DEVELOPMENT:** A progressive change in the form and size of successive organs. In many grasses, for example, successive leaf blades are progressively longer and only after a maximum length is achieved will the plant begin to flower.

**HETEROCHROMATIN:** Highly condensed regions of chromosomes lying on a continuum between the two extremes of constitutive and facultative heterochromatin, which are not usually transcriptionally active.

**HETEROCHRONY:** Any evolutionary phenomenon that involves changes in the relative rates or timing of development of different cell lines in the body. Such changes lead to significant alteration in form. Supposing that a substance  $A$  can be converted into another substance  $D$  either directly, when the heat of reaction is  $w$  joule or indirectly by way of substances  $B$  and  $C$  when the heat of reaction of

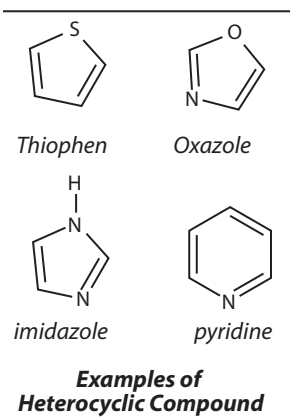


the three stages are  $x$ ,  $y$  and  $z$  joule respectively, thus:

(i)  $A \rightarrow D$ ;  $\Delta H = -WJ$ , (ii)  $A \rightarrow B$ ;  $\Delta H = -xJ$ , (iii)  $A \rightarrow C$ ;  $\Delta H = -yJ$ ,

(iv)  $A \rightarrow D$ ;  $\Delta H = -zJ$ , then according to Hess's Law, the heat of reaction in going from  $A$  to  $D$  is the same whether the change is achieved directly or through a series of separate stages and  $w = x + y + z$ .

**HETEROCYCLIC COMPOUND:** Any of a class of cyclic organic compounds with at least one of the carbon atoms in the ring system replaced by a non-carbon atom. In other words, it is an organic compound having a ring system made up of more than one kind of atoms. The most common heterocyclic elements include oxygen, nitrogen and sulfur. Examples of other heterocyclic elements are thiophen, oxazole, imidazole and pyridine.



**HETEROCYST:** A specialised nitrogen-fixing cells found in nitrogen-fixing cyanobacteria *Anabaena sperica*, during nitrogen starvation. They are enlarged cells with thick cell walls and lack chlorophyll, which gives them a colourless appearance.

**HETERODONT:** A term describing animals that possess teeth of more than one type. For example, such members of the synapsida including mammals possess incisors, canines, premolars and molars, each adapted for a particular function.

**HETERODUPLEX DNA:** Double-stranded DNA in which the two strands are derived from different DNA molecules. It is formed during genetic recombination and can be produced in vitro in DNA hybridisation.

**HETERODYNE:** A method of radio reception in which two new frequencies are produced by superimposing a locally generated radio wave on an incoming wave. For example, in the heterodyne wavemeter, a variable frequency local oscillator is adjusted to give a predetermined beat frequency with the incoming wave, enabling the frequency of the incoming wave to be determined.

**HETEROEICIOUS:** Parasitic organisms such as rust fungi that pass different phases of their life cycle on different host plants.

**HETEROFERMENTATION:** Any fermentation where there is more than one main end product. Lactic acid bacteria, for example, ferment glucose with lactic acid, ethanol/acetic acid and carbon dioxide ( $\text{CO}_2$ ) as by-products.

**HETEROGAMETIC SEX:** The sex of some species such as humans and many other mammals, which is determined by the possession of two dissimilar sex chromosomes, XY. It produces reproductive cells of two kinds, with half containing an X chromosome and the other half a Y chromosome.

**HETEROGAMY:** 1. Alternation of sexual and parthenogenic generations, in which parthenogenesis alternates with bisexual reproduction in the life cycle as seen in certain aphids. 2. The state or condition in which conjugating gametes are dissimilar in structure and size as well as in function.

**HETEROGENEOUS:** A system made up of multiple components in different phases. A heterogeneous mixture is a mixture of two or more compounds. A reaction between a solid substance and a liquid or a gaseous substance can be considered as a heterogeneous system or reaction.

**HETEROGENEOUS EQUILIBRIUM:** An equilibrium state in which the reacting species are not in the same phase of matter, for example, the reacting species are solid, liquid and gas.

**HETEROGENEOUS MIXTURE:** A mixture in which the individual components remain physically separated and the composition can easily be identified, as there are two or more phases present.

**HETEROGENEOUS NUCLEAR RNA (hnRNA):** An assortment of RNA molecules assembled around newly transcribed RNA in the cell nucleus. The hnRNA associates with protein from heterogeneous nuclear ribonucleoprotein to form hnRNP.

**HETEROGENEOUS REACTION:** A reaction where there is an interphase between the different components or reactants; or reactions that have reactants in two or more phases. Examples are those between a gas and a solid, a gas and a liquid or two different solids. Practical applications include in catalytic converters, fuel cells and chemical vapour deposition among others. **Heterogeneous mixture** is a substance such as toothpaste, milk, perfume and steel in which components are not evenly mixed.

**HETEROGENESIS (ALTERNATION OF GENERATIONS; METAGENESIS):** The occurrence in one life cycle of an organism of two or more different forms, usually an alternation of a sexual with an asexual form. The sexual phase, called the gametophyte, produces gametes or sex cells; the asexual phase or sporophyte, produces spores asexually. The two phases or generations, are often morphologically and sometimes chromosomally, distinct. It occurs more commonly in algae, fungi, mosses, ferns and seed plants; many invertebrates, for example, protozoans, jellyfish and flatworms also have an alternation of sexual and asexual generations, but the alternation of haploid and diploid generations is unknown.

**HETEROGRAFT (HETEROLOGOUS GRAFT; HETEROPLASTIC GRAFT; XENOGRAFT):** An organ, tissue, skin or cells that is transplanted from one species to another. An example is an organ or tissue that is obtained from a baboon and transplanted into a human body.

**HETEROKARYOSIS:** The presence in the same cell of two or more genetically different nuclei within the same mycelium of a fungus, which results from different strains without the fusion of their nuclei. It can also be induced *in vitro* to study the interaction between the cellular components from different species. A **heterokaryon** is a cell that contains at least two genetically different nuclei. **Heterokaryotic** refers to having two or more genetically different types of nuclei within the same mycelium.

**HETEROKONT:** Referring to an organism having two different types of flagella, for example, as in the motile stages of algae in the Xanthophyta.

**HETEROLACTIC FERMENTATION (MIXED LACTIC FERMENTATION):** A form of anaerobic respiration found in certain microorganisms, such as *Lactobacillus brevis*, in which the end products are a molecule each of lactic acid, ethanol and carbon dioxide.

**HETEROLYSIS (HETEROLYTIC):** The cleavage of a covalent bond so that both bonding electrons remain with one of the two fragments between which the bond is broken. Heterolytic bond fission is a feature of many bimolecular reactions in solution such as electrophilic substitution and nucleophilic substitution. It results in the formation of nucleophile and electrophile.

**HETEROLYTIC BOND-DISSOCIATION ENERGY:** The energy required to break a given bond of some specific compound by heterolysis. For the dissociation of a neutral molecule  $\text{AB}$  in the gas phase into  $\text{A}^+$  and  $\text{B}^-$  the heterolytic bond dissociation energy  $D(\text{A}^+ \text{B}^-)$  is the sum of the bond dissociation energy,  $D(\text{A-B})$  and the adiabatic ionisation energy of the radical  $\text{A}$  minus the electron affinity of the radical  $\text{B}$ .

**HETEROLYTIC DISSOCIATIVE ADSORPTION:** The formal nature of the cleavage of a single bond. If in the bond of the adsorptive  $\text{A:B}$ ,  $\text{A}$  or  $\text{B}$  retains the electron pair, the adsorption is heterolytic dissociative adsorption.

**HETEROLYTIC FISSION (HETEROLYTIC CLEAVAGE):** The breaking of covalent bond made of two electrons to give two fragments, with one fragment retaining both electrons. An example

is  $\text{HCl} \rightarrow \text{H}^+ + \text{Cl}^-$ . The fragments produced include an electrophile and a nucleophile.

**HETEROMEROUS:** Referring to lichens in which the algal cells are restricted to a layer situated between an upper plectenchymatous cortex and a lower loosely woven medulla of fungal tissue. It is the more usual tissue arrangement.

**HETEROMORPHIC (ANTITHETIC):** A term used to describe a life cycle in which the alternating generations are morphologically and physiologically distinct, as in the bryophytes and pteridophytes.

**HETEROMORPHOSIS (HYPERMETAMORPHOSIS):** Regeneration of an organ or part that differs from the original structure at the site. For example, the tiny insects of the order Strepsitera have active first instar lava, which is succeeded by several inactive, often legless, grublike stages.

**HETERONUCLEAR MOLECULE:** A molecule made up of two different atoms. It could be diatomic ( $\text{HCl}$ ) or polyatomic ( $\text{NH}_3$ ). Heteronuclear diatomic molecules are usually polar.

**HETEROPHYLLY:** Possession of two or more leaf types, often differing widely in morphology and function. For example, certain species of *Lycopodium* and *Selaginella* have two rows of expanded lateral leaves and one or two rows of smaller adaxial or abaxial leaves. Many aquatic and semiaquatic plants have dissected submerged leaves and entire floating or aerial leaves. The submerged leaves are adapted to reduce resistance to water flow while the floating leaves have a broad lamina to maintain buoyancy. **Heterophyllous** refers to having dissimilar leaves on the same plant.

**HETEROPLASMY:** The presence within a cell or organism of mitochondria with different genetic constitution.

**HETEROPOLYMER:** A polymer containing more than one type of monomer units that are different from one another.

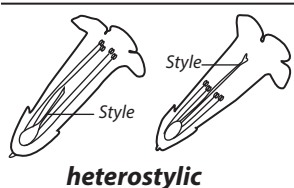
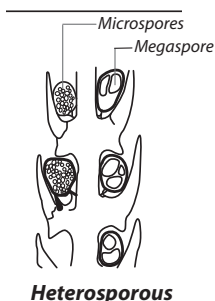
**HETEROPOLYSACCHARIDE (HETEROGLYCAN):** The class name for polysaccharides composed of two or more different kinds of monomeric units (monosaccharides).

**HETEROSCEDASTIC:** 1. Of a number of distributions, having different variances. 2. Of a bivariate or multivariate distribution, not having any variable with constant variance for all values of the other or others. 3. Of a random variable in a multivariate distribution, having different variances for different values of the other variables.

**HETEROSIS (OUTBREEDING ENHANCEMENT; HYBRID VIGOUR):** An increased function in physical capacities such as vigorous growth, greater yield or increased disease resistance than either parent that sometimes occur in the hybrid produced by mating two genetically different parents. The effect is thought to be due to an accumulation of dominant alleles, each having additive effects and masking the effects of deleterious recessive alleles.

**HETEROSPORY:** The condition of producing more than one type of spore, differing in size and sex. The male is known as microspore and the female as megaspore. They finally develop into separate male and female gametophytes. **Heterosporous** refers to bearing two different kinds of spores having different size and sex.

**HETEROSTYLY:** The condition or state of flowering plants in which flowers of the same species have styles of different length so that the stigma is positioned below the anther in some flowers and above them in others ensuring pollinating insects are more likely to transfer pollen from the anther of one flower to the stigma of another to promote cross-pollination.



**HETEROTHALLIC:** Producing male and female gametangia in different structures or plants, as in some algae and fungi.

**HETEROTHERM:** An organism which is both poikilothermy and homeothermy. It sometimes maintains its body temperature at a certain level, such as when active, but at other times allows it to fluctuate with the environment. Examples are hummingbirds and certain rodents.

**HETEROTRICHY:** The condition occurring in certain filamentous algae in which the algal body is composed of both prostrate filaments and protruding upright filaments.

**HETEROTROPH (HETEROTROPHIC ORGANISMS):** Any living organism that uses organic carbon for growth and obtains its energy from organic substances produced by other organisms. Examples are all animals and fungi including herbivores, carnivores and animals that live on dead animal and plant material.

**HETEROTROPHIC NITRIFICATION:** The oxidation of ammonium to nitrite and nitrate by heterotrophic organisms.

**HETEROTROPHIC NUTRITION (HETEROTROPISM):** A type of nutrition occurring in animals, fungi, many prokaryotes and protists, called heterotrophs, which are unable to synthesise organic compounds to use as food and which derive energy from the intake and digestion of organic substances, comprising mainly plant or animal tissue.

**HETEROTROPHIC RESPIRATION:** The conversion of organic matter to  $\text{CO}_2$  by organisms in ecosystems other than plants.

**HETEROZYGOTE:** A nucleus, cell or organism possessing two different alleles, usually one normal and the other mutant (abnormal), for a particular gene. An example is a zygote having one dominant allele and one recessive allele for a particular gene.

**HETEROZYGOTE ADVANTAGE:** A selection model in which heterozygotes have the highest fitness than individuals homozygous for any one of the associated alleles.

**HETEROZYGOUS:** Having inherited a dominant and a recessive allele controlling a characteristic or feature. A dominant allele will override the traits of a recessive allele in a heterozygous pairing and the traits displayed by that organism is determined by the dominant allele. In some traits, however, alleles may be codominant and neither acts as dominant or recessive. An example is the human ABO blood system; persons with type AB blood have one allele for A and one for B.

**HEURISTIC:** Using or obtained by informal methods or reasoning from experience, often since no precise algorithm is known or is relevant. It can be used to control problem solving in human beings and machines and applied in computing to indicate the process by which a program attempts to improve its performance by learning from its own experience.

**HEUSLER ALLOYS:** Ferromagnetic metal alloys containing no ferromagnetic elements as a result of the double-exchange mechanism between neighbouring magnetic ions.

**HEWLETT-PACKARD (HP):** A multinational information technology (IT) company that manufactures computers, plotters, LaserJet and DeskJet printers and scanners. It was founded in 1939 by William Hewlett and David Packard.

**HEWLETT-PACKARD GRAPHICS LANGUAGE (HPGL; HP-GL):** An object-oriented language used by Hewlett-Packard plotters to represent images through the use of geometrical formulae. A plotter is a type of printer that interprets commands from a computer to make line drawings on paper and is capable of producing continuous lines directly from vector graphics files or commands.

**HEX:** Abbreviation for hexadecimal, a number system to the base 16.

**HEXA-:** A prefix denoting six. For example, a hexagon has six sides.

**HEXACHLOROBENZENE (PERCHLOROBENZENE; BENZENE HEXACHLORIDE; HCB; BHC):** A colourless crystalline compound with the chemical formula  $C_6Cl_{16}$ , density  $2.04 \text{ g/cm}^3$ , melting point  $23^\circ\text{C}$  ( $504 \text{ K}$ ) and boiling point  $323\text{--}326^\circ\text{C}$  ( $596\text{--}599 \text{ K}$ ). It is made by the chlorination of benzene with an iron(III) chloride catalyst or by treating hexachlorocyclohexane with chlorine in hexachloroethane. It is a fungicide used as a seed treatment, especially on wheat to control the fungal disease bunt and also in preserving wood. Long-term oral exposure to hexachlorobenzene in humans results in a liver disease with associated skin lesions. It has been banned worldwide under the Stockholm Convention on persistent organic pollutants.

**HEXACHLOROCYCLOHEXANE, HCH (1, 2, 3, 4, 5, 6, -HEXACHLOROCYCLOHEXANE):** A chemical that exists in eight isomers. An example is gamma-HCH ( $\gamma$ -HCH) or gammexene, commonly called lindane. It is a white solid that may evaporate into the air as a colourless vapour with a slightly musty odour. Some other forms of HCH are alpha ( $\alpha$ ), beta ( $\beta$ ), delta ( $\delta$ ) and epsilon ( $\epsilon$ ). Lindane is used as an insecticide and also used to treat head and body lice and scabies.

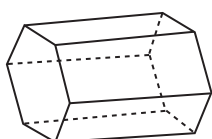
**HEXACHLOROPHENE [2,2'-METHYLENEBIS(3,4,6-TRICHLOROPHENOL); NABAC]:** A white, colourless, bactericide, with the molecular formula  $C_{13}H_6Cl_6O_2$ . It is produced by alkylation of trichlorophenol with formaldehyde. It is used in minute quantities in soaps and surgical disinfectants. It is also used in agriculture as a soil fungicide, plant bactericide and acaricide.

**HEXADECANE (CETANE):** A colourless liquid straight-chain alkane hydrocarbon with the chemical formula  $C_{16}H_{34}$ , density  $773 \text{ mg mL}^{-1}$ , melting point  $17\text{--}19^\circ\text{C}$  ( $290\text{--}292 \text{ K}$ ) and boiling point  $271\text{--}290^\circ\text{C}$  ( $544\text{--}564 \text{ K}$ ). It is used in standardising cetane ratings of diesel fuel.

**HEXADECIMAL:** Of or denoting a place value notation to base 16. **Hexadecimal number system** is the number system to the base 16. It is often used in computing since each hexadecimal digit is equivalent to a byte of four bits. In base-16 or the hexadecimal number system, each digit can have sixteen values instead of ten. The values of a hexadecimal digit can be: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B, C, D, E, F. Therefore, the number 12 (in the common base-10 format) would be represented as "C" in hexadecimal notation. The number 24 would be 18 ( $16 + 8$ ). 100 is 64 in hexadecimal ( $16 \times 6 + 4$ ) and 1000 is 3E8 ( $256 \times 3 + 16 \times 14 + 8$ ). To avoid confusion, an H is added to the symbol to denote that it is a hexadecimal number, for example, OH, 2H, AH or CH.

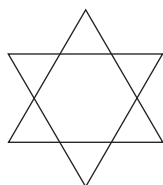
**HEXAGON:** A two-dimensional geometric figure formed by six sides. In a regular hexagon, all sides have equal length and all the six interior angles are equal,  $120^\circ$  each with an exterior angle of  $60^\circ$  each. A **hexagonal number** is a number of the form  $n(2n - 1)$ .

**HEXAGONAL PRISM:** In geometry, it is a prism with a hexagonal base. It is actually an octahedron composed of two hexagonal bases and six rectangular sides (faces). A regular hexagonal prism has regular hexagon bases and an irregular hexagonal prism has irregular hexagon bases with the sides of the bases having different lengths.



Hexagonal Prism

**HEXAGRAM (SOLOMON'S SEAL; STAR OF DAVID):** A star polygon or a plane figure formed by extending six sides of a regular hexagon to meet one another.



Hexagram

**HEXAHEDRITE:** One of the main types of iron meteorite. Hexahedrites have a nickel content of about 6% and are 92% composed of the nickel-iron mineral kamacite.

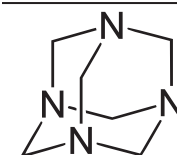
**HEXAEDRO-:** An affix used in names to denote eight atoms bound into a hexahedron such as a cube.

**HEXAEDRON:** A polyhedron having six faces with eight corners and 12 edges. A regular hexahedron, with all its faces perfect squares, is a cube. A hexahedron is a generalisation of a quadrilateral in three dimensions. There are seven topologically distinct convex hexahedra.

**HEXAHYDRATE:** A crystalline compound that has six moles of water per mole of compound.

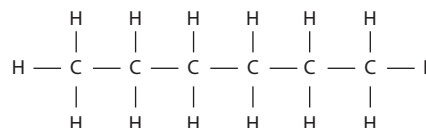
**HEXAMEROUS:** Having parts arranged in groups of six.

**HEXAMINE (HEXAMETHYLENETETRAMINE):** An organic compound made by condensing formaldehyde (methanol) with ammonia. Its molecular formula is  $C_6H_{12}N_4$ . It is used as a camping fuel and an antiseptic drug for treating urinary infections in medicine and also in the production of cyclonite. It can be nitrated to make the high explosive cyclonite.



Chemical structure of Hexamine

**HEXANE:** A colourless liquid alkane, with the molecular formula  $C_6H_{14}$ , density  $654.8 \text{ mg mL}^{-1}$ , melting point  $-96$  to  $-94^\circ\text{C}$  ( $177\text{--}179 \text{ K}$ ) and boiling point  $68\text{--}69^\circ\text{C}$  ( $341.6\text{--}342.2 \text{ K}$ ). It is derived from the fractional distillation of petroleum and used as a solvent and as a working fluid in low-temperature thermometers.



Structure of Hexane

**HEXANEDIOATE (ADIPATE):** A salt or ester of hexanedioic acid.

**HEXANEDIOIC ACID (ADIPIC ACID; HEXANE-1,6-DIOIC ACID):** A carboxylic acid with the chemical formula  $(CH_2)_4(COOH)_2$ , density  $1.36 \text{ g/cm}^3$ , melting point  $152.1^\circ\text{C}$  ( $425 \text{ K}$ ) and boiling point  $337.5^\circ\text{C}$  ( $611 \text{ K}$ ). It is formed from oxidation of various fats. It is slightly soluble in water and soluble in alcohol and acetone and used especially in the manufacture of nylon and polyurethane foams.

**HEXANOIC ACID (CAPROIC ACID):** A liquid fatty acid, with the chemical formula,  $CH_3(CH_2)_4COOH$ , density  $0.93 \text{ g/cm}^3$ , melting point  $-3.4^\circ\text{C}$  ( $270 \text{ K}$ ) and boiling point  $205^\circ\text{C}$  ( $478 \text{ K}$ ). Glycerides of the acid occur naturally in cow and goat milk and in some vegetable oils or it can be synthesised. It is used in the manufacture of pharmaceuticals and flavourings.

**HEXAPLOID:** Having six times the haploid number of chromosomes.

**HEXAPODA (INSECT):** A class of the phylum Uniramia comprising about a million known species of arthropods, distributed worldwide in all terrestrial habitats and includes the insects as well as three much smaller groups of wingless arthropods: Collembola, Protura and Diplura

**HEXAPRISMO-:** An affix used in names to denote twelve atoms bound into a hexagonal prism.

**HEXOSE (HEXOSE MONOSACCHARIDE):** A monosaccharide such as glucose and fructose that has six carbon atoms in its molecules. Its chemical formula is  $C_6H_{12}O_6$ . Some are widely distributed in nature, especially in ripe fruits.

**HEXYL GROUP (HEXYL RADICAL):** The organic group  $C_6H_{13}$ , derived from hexane, especially the isomeric form of this group,  $CH_3(CH_2)_4CH_2-$ .



**HIBERNAL:** Appearing in winter.

**HIBERNATION:** 1. A condition of dormancy and torpor found in cold-blooded vertebrates and invertebrates, including bats, hedgehogs, amphibians, reptiles and many fish, in which they pass the winter months as a way of surviving food scarcity and cold weather. Various physiological changes occur such as lowering of the body temperature and slowing of the pulse rate and the animal lives on its reserves of body fat. 2. In computer science, the term **hibernate** refers to an option that powers down the computer to save the current state of the system from the computer's RAM to the hard disk, then shuts down. Hibernation mode is similar to sleep mode, however, in sleep mode the power cannot be shut off. In sleep mode, when the computer is restarted, instead of going through the typical boot sequence, the previously saved state is automatically loaded into the RAM.

**HIBISCUS:** Any of about 250 species of shrubs, trees and herbaceous tropical plants with large bright flowers belonging to the family Malvaceae, cultivated for its ornamental flowers.

**HICCUP:** A brief involuntary contraction of the diaphragm that may follow dietary or alcoholic excess and rapid eating. It may also be a symptom of uraemia, mineral disorders and brain-stem disease.

**HIDDEN FILE:** In computing, a file that is not normally visible to users when examining the contents of the directory in which it resides. This is done to prevent unauthorised access and to minimise its being modified or deleted by mistake.

**HIDDEN DIMENSIONS:** Hypothetical additional dimensions of space-time, either a classical dimension in which particles can move or a quantum dimension that converts a force particle into a matter particle and vice versa.

**HIDDEN MATTER:** One of the most unexpected findings of 20th century physics is the so-called dark matter: the fact that the vast majority of the universe may be made up of matter that is invisible to us. According to physicists, only 5% of all the matter and energy in the universe is in a perceptible form, meaning the stuff that makes up stars, planets and living beings. The other 95% is composed of matter that is completely hidden from us.

**HIDDEN-VARIABLES THEORY:** A theory that denies that the specification of a physical system given by a state described by quantum mechanics is a complete specification. A successful hidden variables theory has never been constructed. There are important considerations preventing the construction of a simple hidden-variables theory, notably Bell's theorem.

**HIDE:** An animal skin, especially when removed to be used for leather.

**HIERARCHICAL CLASSIFICATION:** A system in which individuals are grouped into an ascending series of successively larger and broader categories, so that lower groups are always subordinate to and included in, those that are higher in the hierarchy. The order in which the 12 principal ranks are used is governed by the International Code of Botanical Nomenclature and, although all available ranks in the hierarchy need not be used, their sequential order must not be altered. In ascending order, the ranks are: form, variety, species, series, section, genus, tribe, family order, class, division and kingdom. Additional subcategories can also be designated (for example, subseries and subclass) thus further extending the total number of ranks to 24.

**HIERARCHICAL FILE SYSTEM (HFS):** The file system developed by Apple Inc and used for organising files on a Macintosh hard disk. When a hard disk is formatted for a Macintosh computer, the hierarchical file system is used to create a directory that can expand as new files and folders are added to the disk. Since HFS is a Macintosh format, Windows computers cannot recognise HFS-formatted drives. Windows hard drives are typically formatted using WIN32 or NTFS file systems.

**HIERARCHICAL INCLUSION:** The counting and quantity principle with the understanding that all numbers preceding a number can be or are systematically included in the value of another selected number.

**HIERARCHICAL NETWORK:** Several computers that are connected to a central host computer. These computers themselves are host to other computers or devices. In other words, it is a network in which processing and control functions are performed at various levels by computers specially suited for the functions performed; for example, in factory or laboratory automation.

**HIERARCHICAL STORAGE MANAGEMENT:** In computing, it is data storage software that automatically moves data between high-cost storage media such as hard disk drives and low-cost storage media such as optical disks and magnetic tapes.

**HIERARCHY:** A type of social organisation in animals, in which individuals are ranked according to their status or dominance relative to other group members, which affects their feeding and mating behaviour. Many vertebrate animals and some invertebrates live in such hierarchies.

**HIGGS BOSON (HIGGS PARTICLE):** Postulated elementary particle with a non-zero mass which exists in certain gauge theories, in particular in electroweak theory. If it existed, it would explain why particles have mass.

**HIGGS FIELD:** The symmetry-breaking field associated with a Higgs boson, named after Scottish physicist Peter Higgs. The Higgs field can either be an elementary scalar quantity or the field associated with a bound state of two fermions. The Higgs boson is the only Standard Model particle that has not been observed in particle physics experiments.

**HIGH FREQUENCY (DECAMETER BAND; DECAMETER WAVE):** A radio frequency in the range of 3-30 megahertz, with a wavelength in the range of 10-100 metres. Frequencies immediately below HF are denoted Medium-frequency (MF) and the next higher frequencies are known as Very high frequency (VHF). Shortwave (2.310 - 25.820 MHz) overlaps and is slightly lower than HF.

**HIGH LEVEL LANGUAGE (HIGHER-LEVEL LANGUAGE, HIGHER-ORDER LANGUAGE):** A programming language in computing designed to suit the requirements of the programmer. It allows one program without having to know about the microprocessor actions that actually carry out the program actions. High level languages are used to solve problems and are described as problem-oriented language. Examples are BASIC, COBOL and FORTRAN.

**HIGH MASS FRACTION:** A high value of the ratio of propellant mass to total motor mass. Values of 0.9 and greater characterise high-mass fraction motors.

**HIGH MOLECULAR WEIGHT RDX, HMX (OCTOGEN; OCTAHYDRO-1,3,5,7-TETRAZOCINE):** A colourless crystalline compound with the chemical compound  $C_4H_8N_8O_8$ , molar mass 296.155 g/mol, density 1.91 g/cm<sup>3</sup> (solid) and melting point 276-286 °C (549-559 K). It has a structure similar to that of cyclonite, but with an 8-membered ring. It has high molecular weight and it is one of the most powerful chemical explosives manufactured and used for military purposes and as a rocket propellant.

**HIGH MORTALITY RATE:** A high percentage of animals in a group which die.

**HIGH PERFORMANCE COMPUTING (SUPERCOMPUTING):** This involves the use of many computing resources, usually parallel processors to achieve high throughput, (effectively, efficiently and quickly) in very advanced applications. Such systems achieve more than 10<sup>12</sup> floating point operating per second. It is mainly used in science and engineering as well as military applications.

**HIGH RESOLUTION ELECTRON-LOSS SPECTROSCOPY:** A technique for obtaining information about molecules absorbed on a surface by inelastic scattering of electrons.

**HIGH SPEED CIRCUIT SWITCHED DATA (HSCSD):** A wireless data transmission service for mobile users introduced in 2000 by UK

mobile phone operator Orange, allowing users of laptop computers and personal digital assistants (PDAs) to transmit data at a speed of 38.4 kilobytes per second (kbps) and is four times faster than the standard data rates of the Global System for Mobile (GSM) communication.

**HIGH TEMPERATURE SHORT TIME METHOD (HTST):** The usual method of pasteurising milk, perishable beverages like fruit and vegetable juices, beer, etc, where it is heated to 72 °C for 15 seconds and then rapidly cooled, done prior to filling into containers in order to kill spoilage microorganisms, to make the products safer and extend their shelf life. Compared to other pasteurisation processes, it maintains colour and flavour, but some cheeses were found to have varying responses to the process.

**HIGH TENSION (HT):** A high potential difference, usually several hundred volts or more. Two factors considered in the classification are the possibility of causing a spark in air and the danger of electric shock by contact or proximity. Batteries formerly used in the anode circuits of radio devices using valves were usually called high-tension batteries to distinguish them from the batteries supplying the relating filaments.

**HIGH WATER MARK (HIGH-WATER MARK):** A line or mark left upon tide flats, beach or along shore objects indicating the elevation of the intrusion of high water. The mark may be a line of oil or scum on along shore objects or a more or less continuous deposit of fine shell or debris on the fore shore or berm. This mark is physical evidence of the general height reached by wave run up at recent high waters. It should not be confused with the mean high water line or mean higher high water line.

**HIGH-DEFINITION MULTIMEDIA INTERFACE (HDMI):** A digital interface for transmitting audio and video data in a single cable. It is supported by most HDTVs and related components, such as DVD and Blu-ray players, cable boxes and video game systems.

**HIGH-DEFINITION TELEVISION (HDTV):** A high-quality digital TV broadcasting format where the broadcast transmits widescreen pictures with more detail and quality than found in a standard analogue television or other digital television formats. It offers a significantly greater number of scanning lines and therefore produces much sharper images than that provided by conventional systems. American TV (NTSC) has 525 lines per image and European TV (PAL) has 625 lines per image, while high definition systems may have more than 1000. It is developed to replace older video formats often referred to as SDTV (standard definition television). Types of HDTV displays include direct-view, plasma, rear screen and front screen projection.

**HIGH-DENSITY LIPOPROTEIN (HDL):** A complex of lipids and proteins that is the form in which cholesterol is transported in the bloodstream from the tissues to the liver. HDL cholesterol helps remove the plaque-carrying LDL cholesterol from the arteries and helps prevent the formation of plaque, leading to high cholesterol level in the blood.

**HIGH-ENERGY PHOSPHATE BOND:** A phosphate linkage with a negative standard free energy of hydrolysis, i.e. one that hydrolyses spontaneously, releasing energy. Examples are the last two phosphate bonds of adenosine triphosphate (ATP). Phosphate-bond energy is the major method of transfer of chemical energy from one enzymic reaction to another. For example, the high-energy phosphate bond of phosphoenolpyruvate can be broken to supply energy for the formation of ATP from ADP and phosphate. The ATP bond energy may then be used to drive an energy-requiring reaction, such as the formation of glutamine from glutamic acid.

**HIGH-ENERGY RADIATION:** Penetrating particle or electromagnetic radiation with energies of more than a few thousand electron volts. It may include electrons, neutrons, protons, mesons, X-rays and gamma rays.

**HIGH-GAIN ANTENNA (HGA):** A type of antenna used on board spacecraft to provide high amplification of either transmitted or received radio signals. HGAs typically consist of parabolic dishes and must be accurately pointed to be useful.

**HIGH-LEVEL LANGUAGE:** In computing, a programming language that is designed to suit the requirements of programmers. It is independent of the internal machine code of any particular computer; and is considered high-level because it is closer to human languages as their programming style and context is easier to learn and implement. High-level languages are used to solve problems and are often described as problem-oriented languages; for example, **BASIC** is a simple high-level computer programming language that is easily learnt by first-time programmers; **COBOL** is used to write programs used in solving business problems; and **FORTRAN** is for programs used in solving scientific and mathematical problems. With the increasing popularity of windows-based systems, there are programming languages designed to facilitate the development of graphical user interface (GUI); for example, Visual Basic wraps the BASIC language in a graphical programming environment. In contrast, low-level languages, such as assembly languages are designed to operate and manipulate the computing hardware and components without any interpretation or translation. They are closer to machine language than to human language and reflect the machine codes of specific computers and are therefore described as machine-oriented languages.

**HIGH-PERFORMANCE LIQUID CHROMATOGRAPHY (HIGH-PRESSURE LIQUID CHROMATOGRAPHY; HPLC):** A sensitive technique used in separating the complex mixture of molecules encountered in chemical and biological systems, in order to understand better the role of individual molecules. It is a chromatographic method in which a mixture of molecules dissolved in a solution (mobile phase) is separated into its constituent parts and forced at high pressure through a tightly packed column of finely divided particles that present a very large area. Even though it gives good separation very rapidly, it is very expensive hence manufacturers refer to it as High Performance Liquid Chromatography to encourage buyers.

**HIGH-PRESSURE GRAPHITISATION:** A term which refers to the solid-state transformation of non-graphitic carbon into graphite by heat treatment under elevated pressure (e.g. 100 to 1000 MPa) so that a definitely higher degree of graphitisation is achieved at lower temperature and/or for a shorter heat treatment time than in heat treatment of the same non-graphitic material at atmospheric pressure.

**HIGH-PRESSURE MERCURY LAMP (ARC):** Radiation source containing mercury at a pressure of approximately 8 MPa (80 bar) or higher, which emits lines over a background continuum between about 200 and 1400 nm.

**HIGH-PRESSURE SYSTEM:** An area where the atmospheric pressure at the surface of earth is greater than its surrounding environment and from which winds blow outward. In the Northern Hemisphere these winds blow in a clockwise direction. High-pressure system produces dry and stable weather often with clear and gentle breeze.

**HIGH-SPEED STEEL:** Alloy of steel introduced in 1900 that remains at dull red heat and can, therefore, be used in cutting tools for high-speed lathes. A common type of high-speed steel contains 18% tungsten, 4% chromium, 1% vanadium and only 0.5-0.8% carbon.

**HIGH-TECHNOLOGY INDUSTRY (HIGH-TECH INDUSTRY):** Any industry that makes use of advanced technology. The largest high-tech group is the fast growing electronics industry, especially the manufacture of computers and telecommunications equipment.

**HIGH-TEMPERATURE SUPERCONDUCTIVITY:** Materials that have a superconducting transition temperature ( $T_c$ ) above (30 K). From 1960 to 1980, -243.2 °C (30 K) was thought to be the highest theoretically possible  $T_c$ . The first high  $T_c$  superconductor was discovered in 1986 by IBM researchers Karl Müller and Johannes Bednorz, for which they were awarded the Nobel Prize in Physics in 1987. Until Fe-based superconductors were discovered in 2008, the term high-temperature superconductor was used interchangeably with cuprate superconductor for compounds such as bismuth strontium calcium copper oxide (BSCCO) and yttrium barium copper oxide (YBCO).

**HIGH-YIELD VARIETY (HYV):** A crop that has been genetically enhanced to produce more than the natural varieties of the same

species and which may have increased percentage of usable plant parts or an increased resistance against crop diseases. They may also require a higher level of agricultural care, such as intensive disease control, higher fertiliser levels and controlled water supply.

**HIGHER ARITHMETIC (NUMBER THEORY):** A branch of mathematics concerned with the properties of integers, rational numbers, irrational numbers and real numbers; and also sometimes the imaginary and complex numbers. It is more advanced than basic arithmetic, algebra, geometry and trigonometry

**HIGHER DIMENSIONS:** Dimensions of space or time that exists in addition to the three space dimensions and one time dimensions. Higher dimensions can be very small as in kaluza theory or large as in the brane world. High-dimensional spaces occur in mathematics and the sciences for many reasons, frequently as configuration spaces such as in Lagrangian or Hamiltonian mechanics; these are abstract spaces, independent of the physical space we live in.

**HIGHER MATHEMATICS:** Mathematics of greater abstraction than arithmetic, algebra, geometry and trigonometry. Some of the fields studied include analysis, topology, linear algebra and number theory.

**HIGHER-ORDER TRANSITION (SMOOTH TRANSITION):** A general term used to describe a transition in which the first and second derivatives of the molar Gibbs energy or molar Helmholtz energy (or chemical potential) with respect to temperature and pressure are continuous, but derivatives of some higher order are discontinuous at the transition point.

**HIGHEST COMMON FACTOR, HCF (GREATEST COMMON DENOMINATOR; GREATEST COMMON DIVISOR, GCD; GREATEST COMMON FACTOR, GCF):** The largest integer that divides each number without a remainder in a sequence of integers. For example, the HCF of 8 and 12 is 4.

**HIGHEST OCCUPIED MOLECULAR ORBITAL (HOMO):** The highest energy that is occupied by one or more electrons. It is the preferred orbital from which an electron would be lost if the atom or molecule were to lose an electron.

**HIGHLIGHT:** In computing, to select an item in a menu, or within an application by pointing to the object with the mouse and clicking the left mouse button in order to copy, move, delete, mark or launch it. Keyboard can also be used to highlight an item. Highlighting an object or text, makes it different such as appearing in colour, having dots around the selection, or bold lines around it.

**HIGHLY CONSERVED SEQUENCE:** A DNA sequence that is very similar in several different kinds of organisms. Scientists regard these cross species similarities as evidence that a specific gene performs some basic function essential to many forms of life and that evolution has therefore conserved its structure by permitting few mutations to accumulate in it.

**HILBERT CUBE:** Named after David Hilbert, it is a topological space, which is the Cartesian product of countably infinite dimensions, where the lengths of the edges in each orthogonal direction form the sequences  $\{a_n\}_{n=1}^{\infty}$  with  $a_n \leq 1/n$  or similar order intervals.

**HILBERT MATRIX:** An  $n \times n$  matrix whose entries are unit fractions. Its  $(i,j)^{\text{th}}$  entry is given as  $H_{ij} = \frac{1}{i+j-1}$ . For example, the unit fractions of a  $3 \times 3$  Hilbert matrix is

$$H = \begin{bmatrix} 1 & \frac{1}{2} & \frac{1}{3} \\ \frac{1}{2} & \frac{1}{3} & \frac{1}{4} \\ \frac{1}{3} & \frac{1}{4} & \frac{1}{5} \end{bmatrix}$$

**HILBERT SPACE:** A linear vector space that can have an infinite number of dimensions. In quantum mechanics, the state of a system is represented by a vector in Hilbert space.

**HILBERT TRANSFORM:** A linear operator which takes a function,  $u(t)$  and produces a function,  $H(u)(t)$ , with the same domain. It is used in signal processing to derive the analytic representation of a signal  $u(t)$ . This means that real signal  $u(t)$  is extended into the complex plane such that it satisfies the Cauchy-Riemann equations. For example, the Hilbert transform leads to the harmonic conjugate of a given function in Fourier analysis (harmonic analysis). It is also an example of a singular integral operator and of a Fourier multiplier.

**HILBERT, DAVID (1862 - 1943):** German mathematician, regarded as the most famous mathematician of his time. He did important work in the geometrical axioms in foundations of geometry. He also made many important contributions in many areas of mathematics including number theory, mathematical logic, differential equations, the three-body problem and formulated the theory of Hilbert spaces and has many other mathematical terms named after him including Hilbert's programme, Hilbert curves, Hilbert transform, Hilbert matrix, Hilbert's basis theorem, etc.

**HILBERT'S BASIS THEOREM:** It states that if  $R$  is a commutative Noetherian ring, then the polynomial ring  $R[X_1, \dots, X_n]$  is also a Noetherian ring.

**HILBERT'S PROBLEMS:** Twenty three mathematical problems compiled by David Hilbert (1862- 1943) in 1901 that include the Riemann Hypothesis, Fermat's last theorem, the Gelfond-Schneider theorem and the continuum hypothesis that were originally unsolved. Many of the problems have remained unsolved.

**HILBERT'S PROGRAMME:** The set of ideas proposed by David Hilbert in 1920 as solutions to his foundations of mathematics, which include formalisation of mathematics and some others. Kurt Gödel showed that most of the goals of the programme were unattainable; however it led to the development of proof theory and computability theory.

**HILDEBRAND RULE:** The rule stating that if the density of the vapour phase above a liquid is constant, the entropy of vaporisation of a mole of liquid is constant. The law does not hold if molecular association takes place in the liquid or if the liquid is subject to quantum-mechanic effects.

**HILL:** A raised area of land, not as high as a mountain and usually bare or rocky that extends above the surrounding terrain.

**HILL DRAINAGE:** A small open channel about ten to sixty metres apart, dug to drain hilly grazing areas.

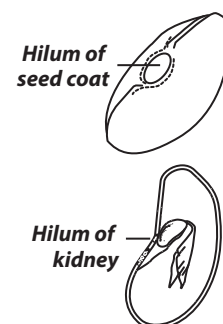
**HILL FARM:** A farm in mountainous country, with 95% or more of its land is classified as rough grazing, mainly for breeding ewe flocks.

**HILL FARM ALLOWANCE (HFA):** A support payment available for owners of hill farm to help with running costs.

**HILL GRAZING:** A grassland in hilly and mountainous areas used for sheep and cattle grazing.

**HILL REACTION:** 1. The release of molecular oxygen from isolated illuminated chloroplasts, when suitable electron acceptors such as ferricyanide are added to the surrounding water. 2. Production of substituted phenylacetic acids by the oxidation of the corresponding alkylbenzene by potassium permanganate in the presence of acetic acid.

**HILUM:** 1. A scar on a seed coat, marking the point where the seed (ovule) was attached to the ovary by the funiculus that distinguishes seeds from fruits. 2. Also known as **hilus**, it is a depression in the surface of organs such as kidney where blood vessels, nerve fibres, etc., enter. For example, the hilum of kidney is the medial depression for blood vessels and ureter to enter the kidney chamber.





**HINDBRAIN (RHOMBENCEPHALON):** One of the three primary divisions of the brain and the most posterior of a vertebrate embryo from which the metencephalon and myelencephalon develop. It controls and coordinates the fundamental physiological processes including respiration and circulation of blood. Hindbrain also refers to the corresponding part of the adult brain that includes the cerebellum, the medulla oblongata and in mammals the pons that controls autonomic functions and equilibrium.

**HINDGUT (EPIGASTER):** The posterior part of the alimentary canal of vertebrates and arthropods. It includes the distal third of the transverse colon and the splenic flexure, the descending colon, sigmoid colon and rectum.

**HINDQUARTERS:** 1. The back part of an animal, including the haunches and hind legs. 2. The posterior portion of a side of beef, lamb, veal or mutton, including a hind leg and one or two ribs.

**HINDU-ARABIC NUMBER SYSTEM:** A number system which uses the digits 0 – 9 and uses base 10.

**HINGE:** Soil which is left uncut by a plough when it has failed to cut a full furrow.

**HINGE JOINT:** A type of synovial joint such as the elbow which allows movement in one plane only and permits flexion and extension movements. Examples of hinge joints are the elbow, knee and ankle joints.

**HIP:** 1. The laterally projecting prominence of the pelvis or pelvic region from the waist to the thigh. **Hip joint** is the ball-and-socket joint between the upper end of the femur (thigh bone) and the pelvis. The rounded head of the femur fits into a deep hollow in the pelvis called the acetabulum. Both surfaces are covered by articular cartilage. 2. In botany, it is a succulent fruit which has the form of a berry and has its achenes surrounded by enlarged hypanthium (outer covering). They are formed from a flower which has many free carpels (aggregate). They belong to the family *Rosaceae* or the rose plant.

**HIPPOCAMPUS:** A part of the vertebrate brain consisting of two ridges, one over each of the lateral ventricles. In humans and other primates, the hippocampus is located inside the medial temporal lobe, beneath the cortical surface. It plays important roles in long-term memory and spatial navigation and also appears to be related to the expression of responses that generate emotions.

**HIPPOCRATIC OATH:** A moral code for ethical conduct and practice in medicine that is taken by all doctors on qualifying that lays down moral and ethical guidelines for the practice of medicine. It was laid down by the Greek physician Hippocrates (c.460 - c.370 BC), who is regarded as the father of medicine.

**HIPPOPOTAMUS:** Large semi-aquatic herbivorous ungulate mammal (*Hippopotamus amphibius*) belonging to the order Artiodactyla. It is native to sub-Saharan Africa, inhabiting rivers, lakes and mangrove swamps. Adults have an average weight of 1,500 kg for males and 1,300 kg for females. They have very delicate and sensitive skin and excrete reddish oil or spend a lot of time in the water to protect their skin from drying out. Their teeth are made of ivory, similar to the tusks of elephants.

**HIRSEL:** A piece of ground and flock looked after by one shepherd.

**HIRSUTISM:** The abnormal growth of masculine type hair in women, typically in the face, chest and back. It may be caused by genetic factors, hormone imbalance (e.g., high levels of male hormones called androgen), disease of the ovaries and use of some drugs. In some cases, especially if it is caused by high levels of androgen, it may be accompanied with deepening voice, balding, acne, decrease in breast size and enlargement of the clitoris.

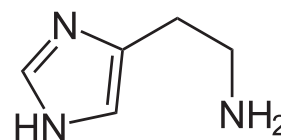
**HIRSUTULOUS (HIRELLATE, HIRELLOUS):** Covered with very small, coarse and stiff hairs.

**HIRUDINEA:** A class of freshwater, terrestrial and marine annelid worm that comprise the leeches. They have suckers at both anterior

and posterior ends but no bristles. Some are blood-sucking parasites of vertebrates and invertebrates but the majority are predators.

**HISPID:** Covered with bristles, rough hair of minute spines.

**HISTAMINE (2-(1H-IMIDAZOL-4-YL)ETHANAMINE):** A substance, with the molecular formula  $C_5H_9N_3$ , molar mass  $111.15 \text{ g mol}^{-1}$ , melting point  $83.5^\circ\text{C}$  (356.65 K) and boiling point  $209.5^\circ\text{C}$  (482.65 K). It is found in plant and animal tissue that is released during allergic reactions such as hay fever. It stimulates gastric secretion and causes dilation of capillaries, constriction of bronchial smooth muscle and decreased blood pressure.



Structural formula  
of Histamine

**HISTIDINE:** Colourless, soluble, crystalline essential amino acid, with the molecular formula  $C_6H_9N_3O_2$  which is the precursor of histamine. It is necessary for growth and repair of tissue, production of red and white blood cells. Nutritional sources of histidine include dairy products and meat.

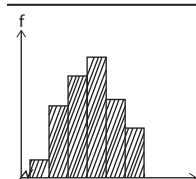
**HISTOCHEMISTRY:** The branch of science that deals with the distribution of the chemical constituents of tissues by means of their chemical reactions.

**HISTOCOMPACTIBILITY:** The degree to which genes influence acceptance or rejection of grafts or how tissues from one organism will be tolerated by the immune system of another organism.

**HISTOGEN:** Hanstein's term for a meristem in shoot or root tip that forms a definite tissue system in the primary plant body. The four types of histogens are calyptogens, dermatogens, periblem and plerome.

**HISTOGEN CONCEPT (HISTOGEN THEORY):** A theory in botany stating that the three tissue systems in the plant body, the epidermis, the cortex and the vascular system with associated ground tissue originate from three histogens.

**HISTOGRAM:** A graph in which no spaces exist in between the bars as in the case of ordinary bar chart, in which the lengths of its vertical or horizontal bars indicate quantities. The horizontal axis is made up of discrete units or class boundaries and the vertical axis represents the frequency. In other words, it is a bar chart used mainly to show the number of items in each group of a set. Together with the pie chart, they are the most common format for representing statistical data.



A Histogram on  
the cartesian  
plane

**HISTOLOGY:** The branch of anatomy concerned with the study of microscopic structure of animal and plant tissues by visual examination, usually with a microscope.

**HISTONES:** Any of a group of water soluble-protein within the nucleus of plant and animal chromosomes, which are involved in the coiling of chromosomes during cell division and have also been implicated in non specific suppression of gene activity.

**HISTOPLASMOSIS:** A fungal disease caused by inhaling spores of *Histoplasma capsulatum*, a fungus found in soil particularly that contaminated with droppings from poultry, wild birds and bats. It may cause acute respiratory infection, a rapidly disseminated form or a chronic type with fever, debility and specific organ involvement (especially of the lungs and resembling tuberculosis).

**HIV (HUMAN IMMUNODEFICIENCY VIRUS):** Either of two strains of a retrovirus HIV-1 or HVI-2 that destroys the immune system's helper T-cells, the loss of which causes AIDS. Retroviruses produce the enzyme reverse transcriptase, which allows transcription of the viral RNA genome into DNA which is inserted into the host cell chromosome. The retrovirus attacks and gradually destroys the immune system, leaving the host unprotected against infection.

**HIVE (BEEHIVE):** An enclosed natural structure occupied by some honey bee species of the subgenus *Apis* to live and raise their young or a man-made structure for raising domesticated bees.

**HIVE INTELLIGENCE:** A decentralised, multicomponent mind or appearance of mind, such as that displayed by social insects on Earth, notably ants, termites and some species of bees. The hive centres and functions around the queen, who alone produces new individuals each of which is a slave to the needs and demands of the community. The cohesion and complex cooperative behaviour of the hive is made possible by a tight communications, command and control system which, in the case of social insects, is based upon the rapid transmission of pheromones.

**HLA SYSTEM (HUMAN LEUCOCYTE ANTIGEN SYSTEM):** A series of gene groups, designated HLA-A, HLA-B, HLA-C and HLA-D that code for proteins on the surface of cells. They are responsible for the regulation of the human immune system by distinguishing the body's own proteins from proteins made by foreign invaders such as viruses and bacteria. The HLA system is used to assess tissue compatibility and people with the same HLA types are said to be histocompatible. Some autoimmune disorders such as psoriasis, type 1 diabetes mellitus, multiple sclerosis and rheumatoid arthritis are linked to specific HLA alleles.

**HNIW (2,4,6,8,10,12-HEXANITRO-2,4,6,8,10,12-HEXAAZAISSOWURTZITANE; CL-20):** A powerful explosive, with the molecular formula  $C_6H_6N_{12}O_{12}$ . It has a three-dimensional bridged structure containing six  $N-NO_2$ , which is extremely sensitive.

**HOARFROST:** A type of icing that occurs in clear air on a surface whose temperature falls below the frost point of the air in contact with it. The water vapour condenses into white crystalline coating of ice that is usually of a feathery nature. This type of clear air icing may occur on an aircraft parked on ground or in air when it enters into a region, where the air is warm and moist and after the aircraft has flown in a region of subzero temperature.

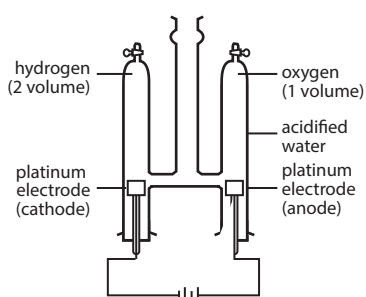
**HOARY:** Having grey or white and short fine hair; very old.

**HOCK:** 1. The hind leg joint of mammals, such as a horse or cow between the knee and the fetlock, which corresponds to the human ankle. 2. A cut of cured meat, especially ham, taken from the lower joint of the leg immediately above the foot, containing comparatively small amount of meat but has a good flavour and jelly properties and is often used in stocks and soups.

**HODGKIN'S DISEASE (HODGKIN LYMPHOMA; LYMPHADENOMA):** A rare form of cancer mainly affecting the lymphatic system, which is part of the immune system. This results in a weakened immune system with the individual susceptible to infection. The disease can spread to almost any organ or tissue in the body, including the liver, bone marrow and spleen. It is treated by radiotherapy and cytotoxic drugs. It is named after English physician, Thomas Hodgkin (1798-1866), who first recognised it.

**HOE (HAND HOE):** A long-handled garden tool used for breaking up the soil and removing wild plants or weeds.

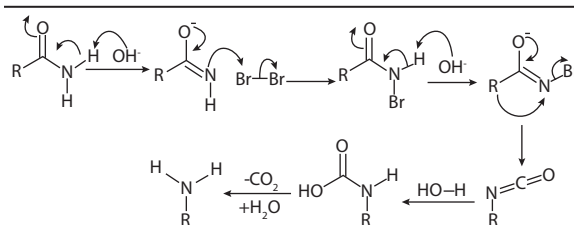
**HOFFMANN VOLTAMMETER:** An apparatus for collecting gases, invented by August Wilhelm von Hofmann (1818-1892). It is produced by the electrolysis of a liquid or an apparatus used in the laboratory to determine the volumetric composition of water. It consists of three joined upright cylinders, usually glass. The inner cylinder is open at the top to allow addition of water and an ionic compound to improve conductivity, such as a small amount of sulfuric acid. A platinum electrode is placed



**Hoffman voltammeter**

inside the bottom of each of the two side cylinders, connected to the positive and negative terminals of a source of electricity. When current is run through Hofmann's voltammeter, gaseous oxygen forms at the anode and gaseous hydrogen at the cathode. Each gas displaces water and collects at the top of the two outer tubes.

**HOFFMANN'S REACTION (HOFFMANN'S REARRANGEMENT; HOFFMANN DEGRADATION):** A reaction for making primary amines from amides using bromine or chlorine and sodium hydroxide,  $RC(=O)NH_2 \rightarrow RNH_2$ . It is named after August Wilhelm von Hofmann, who discovered it. It is applied to the following: aliphatic and aromatic amides are converted into aliphatic and aromatic amines, respectively; in the preparations of anthranilic acid from phthalimide; nicotinic acid is converted into 3-amino pyridine; the symmetrical structure of  $\alpha$ -phenyl propanamide does not change after Hoffmann reaction. The reaction mechanism is illustrated below:



**Hoffmann's Reaction**

**HOG:** Swine more than four months old or any member of the pig family, especially a castrated male.

**HOGG:** A young sheep before the first shearing.

**HOGGET:** A sheep roughly six to twelve months old.

**HOLANDRIC:** A trait that is inherited only by the male line, which is encoded by a gene or genes specific to the Y-chromosome and therefore occurring only in males.

**HOLD:** 1. To conceive after artificial insemination. 2. A scheduled or unscheduled pause in a testing or launching sequence or countdown of a missile or space vehicle. 3. To retain information in a computer storage device for further use after it has been initially utilised. 4. To maintain storage elements at equilibrium voltages in a charge storage tube by electron bombardment.

**HOLD-UP VOLUME:** The volume of eluent required to elute a component, the concentration of which in the stationary phase is negligible compared to that in the mobile phase.

**HÖLDER CONDITION:** In mathematics, a real or complex-valued function,  $f$  on  $d$ -dimensional Euclidean space satisfies a Hölder condition or is **Hölder continuous**, when there are nonnegative real constants  $C, \alpha$ , such that  $|f(x) - f(y)| \leq C|x - y|^\alpha$  for all  $x$  and  $y$  in the domain of  $f$ . More generally, the condition can be formulated for functions between any two metric spaces. The number  $\alpha$  is called the exponent of the Hölder condition. If  $\alpha = 1$ , then the function satisfies a Lipschitz condition. If  $\alpha = 0$ , then the function simply is bounded. The condition is named after Otto Hölder.

**HÖLDER MEAN (GENERALISED MEAN; POWER MEAN):** In mathematics, a family of functions for aggregating sets of numbers that include as special cases the Pythagorean means (arithmetic, geometric and harmonic means).

**HÖLDER'S INEQUALITY:** In mathematics, the generalisation of the Schwarz inequality: for real functions

$|\int f(x)g(x)dx| \leq (\int |f(x)|^p dx)^{1/p} \cdot (\int |g(x)|^q dx)^{1/q}$ , where  $1/p + 1/q = 1$ ,  $1 \leq p, q \leq \infty$  and all integrals taken with respect to the same measure.

**HOLDFAST (HAPTERON):** A structure found at the base of many algae in flowing or tidal water that serves to attach the plant to a support. It is often dissected into many finger-like processes or rhizoids.

**HOLDING (AGRICULTURAL HOLDING):** An economic unit engaged in agricultural production activities under a single management and comprising all the livestock kept and all the land used, wholly or partly, without regard to title, legal form or size.

**HOLDOVER:** A situation where a tenant farmer uses buildings and crops on a farm, after leaving the farm at the end of a tenancy. It may, for example, involve harvesting crops later in the year after the tenancy has expired.

**HOLES:** Missing electrons in the structure of a semiconductor or a vacant electron position in the lattice structure of a solid at a position where one could exist in an atom or atomic lattice that behaves like a mobile positive charge carrier with a negative rest energy. With a potential difference between the ends of a sample, electrons can move to fill the holes and in effect the holes move the other way to form an electric current.

**HOLISTIC MEDICINE:** A term used for all alternative therapies that treat a patient based on his/her overall health and lifestyle profile such as nutritional, physical, mental, spiritual and emotional wellbeing. Holistic medicine practitioners believe that the whole person is made up of interdependent parts and if one part is not working properly, all the other parts will be affected and therefore treats specific ailments not primarily as conditions to be alleviated but rather as symptoms of more fundamental disease. Holistic therapies tend to emphasise proper nutrition and avoidance of substances such as chemicals that pollute the body. Examples of alternative medicine systems that use the holistic approach include naturopathic medicine, homeopathy and traditional Chinese medicine.

**HOLLIDAY INTERMEDIATE:** An intermediate structure in genetic recombination named after Robin Holliday, who proposed it in 1964, in which two double stranded homologous DNA molecules are joined together by means of a reciprocal crossover involving one DNA strand of each molecule.

**HOLMIUM:** A soft, malleable, stable rare-earth, silvery metallic element, which is part of the Lanthanide series. Its chemical symbol is Ho, atomic number 67, atomic weight 164.930, melting point 1461 °C (1734 K), boiling point 2600 °C (2873 K), specific gravity 8.803 and valence 3. It occurs in gadolinite, monazite and other rare-earth minerals. The metal is paramagnetic, but as the temperature is lowered, it changes to antiferromagnetic and then to the ferromagnetic system. It has the highest magnetic strength of any element, and therefore used to create the strongest artificially generated magnetic fields, when placed within high-strength magnets as a magnetic pole piece. It is also used in nuclear control rods to absorb nuclear fission-bred neutrons. It is one of the colourants used for cubic zirconia and glass, providing yellow or red colouring.

**HOLOBLASTIC CLEAVAGE:** A type of cleavage in which there is complete division of the egg resulting in equal blastomeres. The holoblastic type of cleavage is commonly seen in smaller eggs containing moderate (mesolecithal) to sparse (isolecithal) yolk. Examples of eggs that divide holoblastically include those of amphibians, mammals, non-vertebrate chordates, echinoderms, most molluscs, annelids, flatworms and nematodes.

**Holocarpic:** A fungus in which the entire thallus is differentiated into a reproductive sporangium or fruiting body when matured.

**Holocene (Recent):** The most recent geological period belonging to the geologic time, rock series or sedimentary deposits of the more recent of the two epochs of the Quaternary period consisting roughly of the past 10,000 years since the end of the Pleistocene up to today.

**Holoenzyme:** A complex structure enzyme consisting of an apoenzyme combined with the appropriate cofactor or cofactors. Only in this state is an enzyme catalytically active.

**Hologram:** A sort of photographic image recorded on light-sensitive plates, paper or film that forms a three dimensional image when light is reflected from its surface.

**HOLOGRAPHIC HYPOTHESIS:** A key principle of quantum gravity postulating that it is possible to obtain information about the bulk of a system, such as a black hole, from the surface of that system.

**HOLOGRAPHIC OPTICAL ELEMENT (HOE):** An optical device consisting of a hologram, used to focus or deflect electromagnetic radiation. Holographic optical elements are relatively easy to make and can function as lenses, mirrors, diffraction gratings, beam splitters, etc, in place of more traditional optical components.

**HOLOGRAPHY:** A method of recording a three-dimensional image of an object using very pure light such as from a laser.

**Holometabolous:** Describing insect development such as in butterflies and bees, in which there is complete metamorphosis and the immature stage, called larvae, is markedly different from adults.

**Holomictic:** Referring to lakes, whose water at some point of time during the year have a uniform temperature and density from top to bottom, thus allowing the lake waters to mix completely.

**Holomorph:** A fungus of the phyla Ascomycota and Basidiomycota which consists of all sexual and asexual stages in its life cycle.

**Holomorphic function (analytic function; regular function):** A complex function that is single-valued and differentiable at every point of its domain and in consequence possessing derivatives of all orders and agreeing with its Taylor series locally. The function may not be differentiable everywhere, it is only analytic at points where the function is differentiable. For example,  $\exp z$  is holomorphic on the complex plane and  $\log z$  is holomorphic on  $\mathbb{C} \setminus (-\infty, 0)$ .

**Holophytic:** A term describing organisms such as certain protozoans capable of photosynthesis that feed like plants and do not require the ingestion of exogenous organic substances for nutrition.

**Holosericeous:** Having fine and silky hair.

**Holotype:** The single specimen or illustration designated as the type for naming a species or subspecies or used as the basis for naming a species or subspecies when no type has been selected.

**Holozoic nutrition:** The type of feeding where an animal ingests liquid or solid organic material. Humans and free living animals exhibit this type of nutrition. **Holozoic** describes the system of feeding like animals or heterotrophs.

**HOME KEY:** Key on a computer keyboard that controls cursor movement such as returning the cursor to the beginning of a line, document, page, screen or worksheet cell based on the position of the cursor.

**HOME PAGE:** The introductory document of a website. It typically serves as a table of contents to the site's other pages or provides links to other sites. For example, <http://www.apple.com/index.html> is the home page of Apple.com. When a basic URL such as "http://www.cnet.com," is typed in, it typically links to the home page of the Web site.

**HOME RANGE:** The geographic area to which an organism normally confines its activity. The size varies according to the size of the animal and its feeding habits. For example, the home range of African hunting dogs may extend over 4000 km<sup>2</sup> whereas that of the field mouse may be less than a hectare.

**HOME ROW:** In computing, the row on a typewriter or computer keyboard that contains a row of keys to which four fingers of each hand return as a base to rest when not in use. On a QWERTY keyboard they are A, S, D, and F for the left hand and J, K, L, and the semicolon for the right.

**HOMEBOX:** A nucleotide sequence containing about 180 base pairs, with identical base sequence in all genes that have it. It is very similar in many eukaryotic organisms that encode a series of amino acids. It is involved in the binding of regulatory proteins to the DNA molecule.

**HOMEOPENETIC INDUCTION:** The influence of a differentiated cell on an adjacent undifferentiated cell such that it brings about similar



differentiation in the adjacent cell, seen when vascular tissues are cut. The severed ends of the vascular bundles induce differentiation in adjacent pith cells and thus re-establish a connection.

**HOMEOMORPHISM (TOPOLOGICAL ISOMORPHISM; CONTINUOUS TRANSFORMATION):** A one-to-one transformation between points in two geometric figures or topological spaces that is continuous in both directions. A homeomorphism which also preserves distances is called an **isometry**. Two objects are homeomorphic if they can be deformed into each other by a continuous, invertible mapping.

**HOMEOPATHY (HOMOEOPATHY):** A system of alternative medicine based on the principle that symptoms of disease are part of the body's self-healing processes and treat ailments by administering minute doses of natural substances that larger amounts is capable of producing in healthy people the same symptoms as those of the disease being treated. It was developed by the German physician Samuel Hahnemann (1755-1843). The system is widely practised as an alternative to orthodox medicine.

**HOMEOSIS (HOMEOTIC TRANSFORMATION):** The formation of a normal structure or organ in place of another at an abnormal site, as caused by the expression of a mutated hox gene. An example is the formation of a leg from the head of a fruit fly instead of the expected antenna. This is caused by a loss of function mutation in antennapedia hox gene.

**HOMEOSTASIS:** Any self-regulating process by an organism, to maintain a state of equilibrium of the chemical composition of its blood and body fluids and other aspects of its internal environments. Physiologically, it is a complex system controlled by the hypothalamus and includes such processes as the body's self-regulation of hormone and acid-base levels, breathing and metabolic rates, blood vessel dilation, blood sugar level regulation, cell growth and body temperature regulation.

**HOMEOTHERM (HOMOEOTHERM):** An organism, such as a mammal that has a steady internal body temperature or maintains a constant narrow range of body temperature regardless of the ambient temperature. Homeothermic or warm-blooded animals mainly control their body temperature by using body chemical processes to compensate for heat loss or gain when external temperatures change. **Homeothermy** is the process of maintaining a constant body temperature in warm-blooded animals.

**HOMEOTIC GENES:** A class of genes including the Hox genes involved in developmental patterns and sequences. They control the early development and differentiation of embryonic tissues in eukaryotic organisms and direct the formation of particular body parts. These are proteins that bind to DNA and regulate the expression of a wide range of other tissues. Homeotic genes act by switching other genes on or off and code for transcription factors. For example, the Hox and ParaHox genes are important for segmentation. Alterations in these genes cause changes in patterns of body parts, sometimes causing abnormal effects such as development of an extra finger.

**HOMEOTIC MUTATION:** A mutation in homeobox genes resulting in homeotic transformation in animals. An example is the mutation in antennapedia hox gene that results in the formation of a leg from the head of a fruit fly instead of the expected antenna.

**HOMESTEAD:** A single building or collection of buildings on a large agricultural holding.

**HOMINID:** Any member of the primate family Hominidae, which includes humans and the great apes (orangutans, gorillas and chimpanzees).

**HOMININ:** A taxonomic tribe, Hominini, of hominids of the genus *Homo* and other genera that includes recent humans together with extinct ancestral and related forms, after the split from the tribe Panini (chimpanzees).

**HOMO:** 1. The genus of the primate family Hominidae that includes modern humans and various extinct species. The oldest homo fossils are those of *Homo habilis* and *Homo rudolfensis*, which appeared in Africa about 1.5 million years ago. Members are characterised by a relatively large cranium (braincase), limb structure adapted to erect

posture and a two-footed gait, well-developed and fully opposable thumbs, hands capable of power and precision grips and the ability to make precision tools. 2. An acronym for **Highest Occupied Molecular Orbital (HOMO)**, which is the highest-energy molecular orbital that is occupied by electrons. 3. A prefix consisting of lower case letters, homo, used to indicate a higher homologue of a compound. 4. **homo-** is an affix used to denote a ring expansion by inclusion of a methylene group into a ring-sector.

**HOMOAROMATICITY:** A special case of aromaticity in which conjugation is discontinued by a single  $sp^3$  hybridised carbon atom at one or several positions within the ring. P-orbital overlap apparently bridges these  $sp^3$  centres and features associated with aromaticity are manifest in the properties of the compound.

**HOMOBARIC LEAF ANATOMY:** An anatomical arrangement in which there are few or no internal barriers to lateral movement of air inside the leaf.

**HOMOCHAIN POLYMER:** A class of polymer in which the main chain is constructed from atoms of a single element. Homochain polymers are named by placing the name or symbol of the element in the main chain immediately before the expression '-chain polymer', for example, carbon-chain polymer or C-chain polymer; sulfur-chain polymer or S-chain polymer.

**HOMOCHROMOUS:** Composed of only a single colour as seen in composite flowers.

**HOMOCYCLIC COMPOUND:** A chemical compound that contains one or more closed rings, comprising carbon atoms only in its molecules. All atoms in the ring are the same. An example is benzene ( $C_6H_6$ ) and cyclohexane ( $C_6H_{12}$ ). If different atoms occur in the ring as in the case of pyridine ( $C_5H_5N$ ), the compound is said to be **heterocyclic**.

**HOMOCYSTEINE (2-AMINO-4-SULFANYLBUTANOIC ACID):** A sulfur-containing amino acid with the molecular formula  $C_4H_9NO_2S$  that is an intermediate in the synthesis of cysteine. Elevated concentrations in the blood are thought to increase the risk for heart disease by damaging the lining of blood vessels and enhancing blood clotting.

**HOMODESMOTIC REACTION:** A subclass of isodesmic reactions in which reactants and products contain equal numbers of carbon atoms in corresponding states of hybridisation. Moreover, there is matching of the carbon-hydrogen bonds in terms of the number of hydrogen atoms joined to the individual carbon atoms.

**HOMODETIC CYCLIC PEPTIDE:** A cyclic peptide in which the ring consists solely of amino-acid residues in eupeptide linkage.

**HOMODONT:** A term describing animals such as dolphins and reptiles, whose teeth are all of same size and shape and perform the same function. Most non mammalian vertebrates are homodont.

**HOMOEOLGY:** Similarity between chromosomes derived from different but related species. Depending on the degree of homoeology, pairing between homoeologous chromosomes may be seen at meiosis in segmental allopolyploids. Such polyploids display meiotic behaviour intermediate between autopolyploids and amphidiploids.

**HOMOFERMENTATION:** A type of fermentation having only one type of end product. An example is lactic acid bacteria which produce lactic acid as the major product of fermentation of glucose.

**HOMOGAMETIC SEX:** The sex that produces one type of gamete with respect to sex chromosome content. In humans and some other mammals, the homogametic sex is the female because they have two of the same kind of sex chromosome (XX). In birds and some reptiles, the homogametic sex is the male because they have two Z sex chromosomes (ZZ).

**HOMOGAMOUS:** Having flowers that do not differ sexually.

**HOMOGAMY (SIMULTANEOUS/SYNCHRONOUS HERMAPHRODISM):** The condition in some organisms such as a flower, in which the male and the female reproductive organs mature at the same time, thereby allowing self-fertilisation.

**HOMOGENEOUS:** 1. In general, composed of parts or elements that are all of the same kind; not heterogeneous. 2. In mathematics, it may refer to (i) a function of several variables that becomes multiplied

by some power of a constant when each variable is multiplied by that constant, for example,  $x^2y^3$  is a homogeneous expression of the fifth degree. (ii) Relating to a differential equation in which a linear combination of derivatives is set equal to zero. (iii) A homogeneous equation, for example, is one having all terms of the same degree. 3. Referring to a system such as solids and liquids in which all the components are in the same phase of matter. **Homogeneity** is the degree to which a property or a constituent is uniformly distributed throughout a quantity of material. A material may be homogeneous with respect to one analyte or property but heterogeneous with respect to another. The degree of heterogeneity (the opposite of homogeneity) is the determining factor of sampling error.

**HOMOGENEOUS CATALYST:** A catalyst that exists in the same phase of matter as the reactants. A homogeneous catalyst is molecularly dispersed in the reactants, which are most commonly in the liquid state.

**HOMOGENEOUS EQUILIBRIUM:** An equilibrium state in which all reacting substances are in the same phase of matter.

**HOMOGENEOUS MIXTURE:** A mixture in which the components are evenly mixed and are the same throughout the solution with the same colour.

**HOMOGENEOUS REACTION:** A reaction in which there is no interface between components or a reaction in which all the reactants and products are in the same physical state. For example, a reaction between two gases, two liquids or two solids is homogeneous.

**HOMOGENISATION (HOMOGENIZATION):** A process to delay the separation of fat from milk. **Homogenised milk** is one that has been emulsified of fat to prevent the separation of the cream layer. Milk, an unstable emulsion, contains fat globules that tend to coalesce. In homogenisation, the milk is heated to about 60 °C and passed at pressure through small openings. The fatty clusters are broken up by shearing as they pass through the holes, by the action of pressure and by impact with components of the homogeniser.

**HOMOGENEOUS COORDINATES (PROJECTIVE COORDINATES):** 1. The representation of a point with Cartesian coordinates  $(x, y)$  as  $(x', y', t)$ , where  $x' = tx$  and  $y' = ty$ . For a given plane, this is equivalent to selecting a point not in the plane and representing any point in it by the line joining it to the reference point. This allows any point of the plane to be represented in terms of a given set of basis points as a linear combination whose coefficients are the homogeneous coordinates of the point with reference to those basis points. 2. The representation of points in a two-dimensional algebraic geometry by equivalence classes of proportional triples from the ground field and similarly for higher dimensions. The coordinates are determined by the choice of a triangle of reference and a unit point.

**HOMOGONOUS:** Having all the flowers of a plant alike, in relation to the stamens and pistils. This is opposed to **heterogonous** which means having two or more complete flowers of different individuals on separate plants of the same species with differences in the relative length of the pistils and stamens.

**HOMOGRAFT (ALLOGENEIC TRANSPLANT; ALLOTRANSPLANTATION):** The transplantation of an organ or tissue from one individual to another of the same species with a different genotype. The transplant is called an allograft.

**HOMOIMEROUS:** A term used to describe lichens in which the algal and fungal cells are evenly distributed, as in *Collema*.

**HOMOIOOTHERMY:** The maintenance by an animal of its internal body temperature at a relatively constant temperature despite variation in environmental temperature by using metabolic processes. It occurs in birds and mammals, which are described as **endotherms**.

**HOMOKARYOTIC:** In fungi, having nuclei with the same genetic makeup within a mycelium. Such a fungus is called **homokaryon**.

**HOMOLACTIC FERMENTATION:** A type of lactic acid fermentation, in which lactic-acid bacteria convert all sugars into lactic acid.

**HOMOLECITHAL (ISOLECITHAL):** Of an egg, having small amount of yolk fairly uniformly distributed throughout the cytoplasm.

**HOMOLEPTIC COMPLEX:** A complex in which there is only one type of a ligand coordinated to the metal.

**HOMOLOGOUS (HOMEOLOGOUS):** 1. Having the same or allelic genes with genetic loci usually arranged in the same order, for example, homologous chromosomes. 2. Relating to a series of organic chemical compounds such as methylene group, each of which differs from the preceding by the addition of a constant component. 3. Referring to biological structures such as the wing of a bird and the fin of a fish that share the same origin but have different functions. 4. In mathematics, playing the same role in distinct but related figures or functions, such as the corresponding points of a figure and a projection of it.

**HOMOLOGOUS CHROMOSOMES (HOMOLOGUES):** A pair of chromosomes in a diploid cell having corresponding DNA sequences with one coming from each parent.

**HOMOLOGOUS RECOMBINATION (GENERAL RECOMBINATION):** Recombination that occurs between two similar or identical molecules of DNA during the formation of ova (eggs) and sperm. It is most widely used by cells to accurately repair harmful breaks that occur on both strands of DNA, known as double-strand breaks. Homologous recombination also produces new combinations of DNA sequences during meiosis, the process by which eukaryotes make gamete cells, like sperm and egg cells in animals.

**HOMOLOGOUS SERIES:** A series of organic compounds with the same general formula and possessing similar chemical properties due to the presence of the same functional group. They differ in formula by a fixed group of atoms; alkanes (paraffins), alkenes (olefins), ethers and alkynes (acetylenes) differ in mass by 14 atomic mass units. For example, the alkane homologous series begins with methane ( $\text{CH}_4$ ), ethane ( $\text{C}_2\text{H}_6$ ), propane ( $\text{C}_3\text{H}_8$ ), butane ( $\text{C}_4\text{H}_{10}$ ) and pentane ( $\text{C}_5\text{H}_{12}$ ), each member differing from the previous one by a  $\text{CH}_2$  group (or 14 atomic mass units). Successive members of such a series are called **homologues**.

**HOMOLOGOUS STRUCTURES:** Body parts in different organisms that have evolved from a shared ancestor and have similar bones and similar arrangements of muscles, blood vessels and nerves and undergo similar embryological development, but do not necessarily serve the same function. For example, the flipper of a whale, a bat's wing, a human arm and the forelimb of a horse look quite different from the outside, but the bones inside reveal that these limbs all have the same basic structure.

**HOMOLOGY:** 1. Similar characteristics in two different organisms that are a product of descent from a common ancestor rather than a product of a similar environment. 2. The correspondence between chemical compounds in a homologous series. 3. In mathematics, a general procedure to associate a sequence of Abelian groups or modules with a given mathematical object such as a topological space or a group.

**HOMOLYTIC DISSOCIATIVE ADSORPTION:** Relating to the formal nature of the cleavage of a single bond. If the electron pair in the bond of the adsorptive  $A : B$  is divided in the course of its dissociative adsorption, the adsorption is homolytic dissociative adsorption.

**HOMOLYTIC FISSION (HOMOLYTIC CLEAVAGE):** The breaking of covalent bond made up of two electrons to give two fragments, each with odd number of electrons. Such fragments are called free radicals; an example is  $\text{Cl}_2 \rightarrow \text{Cl}^\cdot + \text{Cl}^\cdot$ . For heterolytic fission, it is  $\text{HCl} \rightarrow \text{H}^+ + \text{Cl}^-$ . Such reactions can be induced by irradiation in the UV region, diffused sunlight or peroxide. High temperatures in the absence of oxygen (pyrolysis) can also induce homolytic elimination of carbon compounds.

**HOMOMORPHIC:** Having the same form.

**HOMOMORPHIC IMAGE:** The algebra that is the range of a homomorphism.

**HOMOMORPHISM (HOMOMORPHY):** 1. In biology, a correspondence in form or external appearance but not in type of structure or origin. 2. In botany, possession of perfect flowers of only one kind. 3. In zoology, resemblance between the young and the adult. 4. In mathematics, a mapping from one algebraic structure to another under which the structural properties of its domain are preserved in

its range in the sense that if  $\circ$  is the operation on the domain and  $\circ$  is the operation on the range, then  $\theta(x \circ y) = \theta(x) \circ \theta(y)$ .

**HOMOMORPHISM THEOREM (FIRST ISOMORPHISM THEOREM):** Any theorem stating that some specific algebraic structure,  $G$ , has the property that if  $\theta$  is a homomorphism, then  $G/\ker\theta$ , where  $\ker\theta$  is the kernel of the homomorphism, is isomorphic to the image of  $G$  under the homomorphism.

**HOMONID:** Any member of the family Hominidae of the order Primates of erect bipedal primate mammals including the closely-related species of humans and the extinct ancestors such as *Homo erectus*, *Homo habilis*, *Homo neanderthalensis* and *Australopithecus* species. *Homo sapiens*, the modern species of humans is the only species still in existence. **Hominoid** is a primate group that includes common ancestors of humans and apes.

**HOMONUCLEAR DIATOMIC MOLECULES:** A diatomic molecule containing atoms of the same elements, for example,  $\text{Cl}_2$ ,  $\text{H}_2$  and  $\text{I}_2$ .

**HOMOPLASMY:** The presence within a cell or organism of genetically identical mitochondria. This is a common condition for most groups of organisms, although there are some exceptions.

**HOMOPLASY:** The similarity of a particular character in two different yet often related groups of organisms that is not the result of common ancestry. Such similarity may be due to convergent, parallel evolution or an evolutionary reversal and therefore, potentially misleading when examining shared characters in constructing phylogenetic trees. Examples are wings in birds and bats which are convergent.

**HOMOPOLAR GENERATOR (UNIPOLAR GENERATOR; ACYCLIC GENERATOR; DISK DYNAMO, FARADAY DISC):** A simple direct current (dc) electric generator consisting of a metal disc rotating between two poles of a magnet. Contacts are made to the axle and the rim of the disc. At constant rotational speed, potential difference is created between the centre of the disc and the rim. The electrical polarity depends on the direction of rotation and the orientation of the field. It can also be used as a simple motor if a direct current is supplied.

**HOMOPOLYMER (ADDITION POLYMER):** A polymer that is formed by polymerisation of a single substance. It is also formed by addition polymerisation. An example is polyethylene, formed by polymerisation of ethylene.

**HOMOPOLYMERISATION:** Polymerisation in which a homopolymer is formed.

**HOMOPOLYSACCHARIDE (HOMOGLYCAN):** The class name for polysaccharides composed of only one kind of monosaccharide.

**HOMOSCEDASTIC:** In statistics, it can be defined as: (i) Of a number distributions, having equal variances. (ii) Of a bivariate or multivariate distribution, having one variable of which the variance is constant for all values of the other or others. (iii) Of a random variable in a multivariate distribution, having constant variance for all values of the other variables.

**HOMOSPOROUS:** In some plants, production of only one type of spore rather than differentiated types. The term is commonly used of certain nonseed-bearing vascular plants, such as the psilopsids and most of the ferns, to distinguish them from other vascular plants that produce megaspores and microspores.

**HOMOSTYLIC:** A condition in flowering plants in which flowers of the same species have styles of the same length.

**HOMOTHALLIC:** Referring to species of algae and fungi that has both male and female reproductive organs on one thallus and is therefore able to fertilise itself.

**HOMOTHETIC:** Of a pair of figures, related by an expansion or geometric contraction such that they lie in the same plane with corresponding sides parallel.

**HOMOTHETY:** A linear transformation that involves no rotation, the composition of a translation and a central dilation. It is of the form  $x' = kx$ ,  $y' = ky$  and it is stretching if  $k > 1$  and shrinks if  $0 < k < 1$ .

**HOMOTOPY:** A continuous transformation from one path in a topological space to another or, more generally, of one function to

another. A homotopy between a function and a constant is called null or inessential. Paths connected by a homotopy are called homotopic and are said to be in the same homotopy class. Properties left unchanged by such homotopies are known as homotopy invariants. Homotopy classes of paths can be composed to form the fundamental group or first homotopy group. Other maps can be used to form higher homotopy groups.

**HOMOZYGOTE:** A diploid organism that carries identical alleles at one or more genetic loci. If both of the two gametes or sex cells that fuse during fertilisation carry the same form of the gene for a specific trait, the organism is said to be **homozygous** for that trait. Individuals homozygous for a particular trait always produce offspring that resembles the main appearance when bred with a genetically similar individual.

**HONEY:** A viscous, sweet syrupy food, golden in colour, produced by honeybees from the nectar of flowers. It is stored in honeycombs and made in excess of their needs as food for a given season. Honey extracted for human consumption is usually heated to destroy fermentation-causing yeasts and then strained.

**HONEY FUNGUS:** A fungus that lives on trees and woody shrubs and attacks them, causing rhizomorphs.

**HONEY GUIDES:** The colourful spots or stripes on the petals of a flower that indicates to pollinating insects the position of the nectaries within the flower. An example is the orange spot on the lower lip of the flower of the common toadflax (*Linaria vulgaris*).

**HONEYCOMB:** A term used to describe a number of architectural forms in networks based on a hexagonal design. It is used in cellular systems design. An analogous term is **honeypots** that are decoy computers that monitor the logging activities of entities attempting to probe or attack a system.

**HOOF:** The foot of a mammal of the orders Perissodactyla and Artiodactyla, such as a horse, cow or deer, covered with horny material. The hoof is attached to the underlying soft tissues by lamellae which interdigitate with similar lamellae in the soft tissues.

**HOOF AND HORN MEAL:** A fertiliser made from animal hooves and horns. It is a great source of nitrogen (12%) for organic garden plants, especially corn, tomato and lettuce. It also has a small amount of phosphorus (2%) for healthy roots and stems.

**HOOKE, ROBERT (1635-1703):** English scientist, natural philosopher and architect who made contributions to various fields including architecture, astronomy, biology, chemistry, physics, surveying, map making, and the design and construction of scientific instruments. He studied comets, the motion of light, the rotation of Jupiter (in 1664 he suggested that Jupiter rotates on its axis.), gravity (In 1666 he suggested that the force of gravity could be measured by utilising the motion of a pendulum), and the properties of air. He was the first man to state that all matter expands when heated and that air is made up of particles separated from each other by relatively large distances. In 1662, at the age of 27, he was appointed as Curator of Experiments for the newly founded Royal Society. He is best known for the discovery of the law of elasticity (Hooke's law).

**HOOKE'S LAW:** A scientific law, named after the 17th century British physicist Robert Hooke, which states that when an elastic substance like rubber or spring is stretched or squashed the deformation produced is proportional to the applied force, provided the elastic limit is not exceeded. In other words, the stress applied to any solid is proportional to the strain it produces within the elastic limit for that solid or more simply, the load applied to an elastic material is proportional to the extension, provided the elastic limit is not exceeded. Mathematically, it is expressed as:  $F = ke$ , where  $F$  is the force,  $e$  is the extension in the elastic material, and  $k$  is the proportionality constant known as spring constant with unit  $\text{Nm}^{-1}$ .

**HOOKWORM:** A parasitic nematode worm that lives in the small intestines of its host, which may be human, dog or cat. Two species



of hookworm that affect man are *Ancylostoma duodenale* and *Necator americanus*. The most significant risk of hookworm infection is anaemia because the worms suck blood voraciously and damage the mucosa.

**HOOSE:** A popular name for a lungworm, a disease of sheep, cattle, goats and swine caused by the presence of various species of nematodes of the genera *Dictyocaulus*, *Metastrongylus* and *Protostrongylus* in the bronchial tubes or in the lungs. It is marked by cough, dyspnoea, anorexia and constipation.

**HOP:** A herbaceous perennial vine, classified in the division Magnoliophyta, class Magnoliopsida order Urticales, family Moraceae. It is a climbing plant that has a rough stem and heart-shaped leaves of three to seven lobes, which develop into cone-like catkin of achenes and bracts. The commercial hop (*Humulus lupulus*) has cone-like mature female flowers, called hops, borne on different plants from the male. The loose scales contain lupulin, a yellow resinous powder that is added to beer to impart a bitter flavour and is used medicinally as a tonic and soporific. Oil of hops is used for some perfumes and the hop stem is used for fibre.

**HOP BINE:** The new shoot of a hop plant, which has to be made to coil round climbing strings.

**HOP MILDEW (HOP POWDERY MILDEW):** A fungal disease of hops, caused by *Sphaerotheca macularis*.

**HOPE'S APPARATUS:** An apparatus consisting of a vessel containing water, a freezing mixture in a tray surrounding the vessel and thermometers inserted in the water at points above and below the freezing mixture. It is used to demonstrate the temperature at which water has its maximum density.

**HOPEFUL MONSTER:** A hypothetical new phenotype or biological monstrosity that arises due to instantaneous speciation, saltation or systemic mutation that radically alters an individual's developmental pattern. Such an individual could possess major innovation that equips it for an environment quite different from that of its immediate ancestor. The term was coined in 1933 by the German geneticist Richard B. Goldschmidt (1879-1958).

**HOPPER:** A container, usually funnel-shaped with a hole at the bottom, for holding seed or fertiliser granules. The seed drops from the hole onto the ground, taking it to drills.

**HORDEIN:** A storage protein found in the grains of barley.

**HORDEOLUM (STY):** 1. A structure for housing pigs usually with a small run next to it. 2. An acute infection of the secretory glands of the eyelids. This common infection results from blocked glands within the eyelid. When the gland is blocked, the oil produced by the gland occasionally backs up and extrudes through the wall of the gland, forming a lump (chalazion), which can be red, painful and nodular.

**HORIZON:** 1. The apparent line that separates Earth from sky or the limit to which one can see across the surface of the sea or a level plain where the sky seems to meet the Earth or sea. The true horizon is obscured by trees, buildings, mountains, etc. The resulting intersection of Earth and sky is called the **visible horizon**. 2. Also called **soil profile** or **soil horizon**, it is layers of soil lying one above the other, parallel to the soil surface whose physical characteristics differ from one another. They are formed through a number of geological, chemical, and biological processes over thousands of years. They are identified mainly by colour, structure, texture, particle size, as well as biological and chemical composition. Most soils have three main soil horizons or layers which are Horizon A, Horizon B, Horizon C, and sometimes Horizon O.

**HORIZON A (A HORIZON):** One of the main classifications of soil horizon, it is a weathered layer, usually dark-coloured, and found at the uppermost part of soil. It is made up of some small mineral materials, organic matter and also home to some organisms such as earthworms, fungi, and bacteria; plants grow in this layer.

**HORIZON B (SUBSOIL; ZONE OF ACCUMULATION; ZONE OF ILLUVIATION):** A layer of soil in a well-developed soil lying immediately below the A-horizon where rain water percolating through the soil has leached organic matter from the surface soils. It contains accumulation of soluble or suspended organic material, clay, iron or

aluminium. Well drained soils typically have a blocky structure and the brightest colour (reddish) due to the presence of oxidised metals.

**HORIZON C:** It consists mostly of weathered big rocks. This solid rock, gave rise to the horizons above it. Sand, silt and clay are the basic types of soil. Leaching may also bring some minerals from horizon B down to horizon C.

**HORIZON O:** Soil layer on the surface of the soil with a high percentage of organic matter. The "O" stands for organic. O horizons may be divided into three distinct categories: one of undecomposed leaves and other organic matter (Oi); underlain by a partially decomposed layer (Oe); and then a very dark layer of well decomposed humus (Oa).

**HORIZONTAL:** At right angles to the vertical; parallel to level ground or to the horizon.

**HORIZONTAL ASYMPTOTE:** A horizontal line that the graph of a function approaches as  $x$  tends to positive infinity or negative infinity. The horizontal line is parallel to the  $x$ -axis in such a way that the distance between curve and line approaches zero.

**HORIZONTAL COMPRESSION:** The squeezing of a graph or figure towards the  $y$ -axis. All distances on the coordinate plane are shortened by multiplying with a common factor less than 1.

**Horizontal stretching** is stretching a graph or figure away from the  $y$ -axis.

**HORIZONTAL ELUTION (HORIZONTAL DEVELOPMENT):** A mode of operation in which the paper or plate is in a horizontal position and the mobile-phase movement along the plane depends on capillary action.

**HORIZONTAL LINE TEST:** In mathematics, it is a test used to determine whether a function is injective (i.e., one-to-one). If a horizontal line can be drawn that crosses two points on the graph of the function, then the function is not one-to-one.

**HORIZONTAL MARKET SOFTWARE:** Software that can be used by several different types of industries. Examples are word processing and spreadsheet programs.

**HORIZONTAL POLARISATION:** One type of linear polarisation in which the electric waves vibrate or oscillate from left to right and their electromagnetic wave frequencies are parallel to and touch the earth. This is generated by having antennas vertical to the earth. The position of electric and magnetic field vectors are very important in radio wave communication. This is very important when choosing and installing an antenna as the direction of electric field vector specifies polarisation of the antenna.

**HORMOCYST:** A resting stage formed by certain filamentous blue-green algae from side branches of the filament.

**HORMOGONIUM:** A motile filament of cells produced during asexual reproduction in unicellular, filamentous cyanobacteria. Some contain heterocysts and akinetes that become detached from the parent organism and act as a means of vegetative propagation.

**HORMONE:** 1. A chemical secreted by the endocrine glands and deposited directly into the blood stream and transported to certain

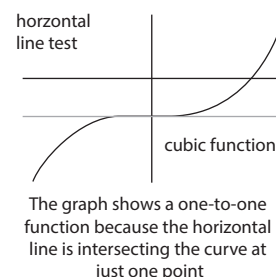


Fig 1:  
**A graph of a horizontal line test of the function  $y = x^3$**

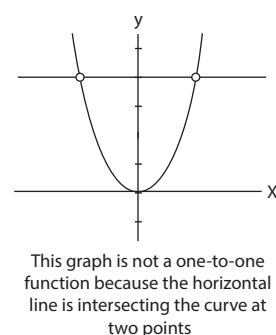


Fig 2:  
**A graph of a horizontal line test of the function  $y = x^2$**

organs, where they cause specific effect which are vital in regulating and coordinating body functions of a specific tissue. 2. The growth substance involved in many plant processes, including germination and tropism. An example in plants is auxin, produced at specific regions of the plant and translocated along living cells.

**HORMONE-REPLACEMENT THERAPY (HORMONE THERAPY, HT):** A system of medical treatment using oral oestrogen, progesterone and sometimes testosterone to help limit the effects of menopause and perimenopause in women.

**HORN:** 1. A hard pointed part that grows usually as one of a pair on the heads of cattle, sheep, goats and some wild animals. 2. An electronic device fixed on moving machines, especially cars to give warning.

**HORNBLLENDE:** A dark-coloured mineral, composed mainly of silicate of iron, calcium and magnesium with the general formula  $(Ca,Na)_{2-3}(Mg,Fe,Al)_5(Al,Si)_8O_{22}(OH,F)_2$  and Mohs scale hardness of 5-6. It is a major component of granite and many other metamorphic and igneous rocks. Metamorphic rocks with abundant hornblende are called **amphibolites**.

**HORNER'S METHOD:** Named after English mathematician, William G. Horner (1786-1837), it is a technique for finding real roots of algebraic equations that involves successive substitutions. The largest integer less than some root is identified and the variable is changed by subtracting this integer, so that the root of the new equation lies between 0 and 1. This process is then repeated to identify the interval of next smallest order in which the root lies and the root to any required degree of accuracy is the sum of the amounts subtracted in the successive changes of variable.

**HORNER'S RULE (SYNTHETIC DIVISION):** A rule for polynomial division in which an algorithm is used to simplify the process of evaluating a polynomial  $f(x)$  at a certain value  $x = x_0$  by dividing the polynomial into monomials or polynomials of the first degree. Each monomial involves a maximum of one multiplication and one addition processes. The result obtained from one monomial is added to the result obtained from the next monomial and so forth in an accumulative addition fashion.

**HORNWORT (HORNED LIVERWORT):** Any of a small group of bryophytes belonging to the class Anthocerotae. Hornworts have a very simple thallus, the cells of which contain a single chloroplast and a pyrenoid, the latter only occurring elsewhere in algae.

**HOROLOGY:** The measurement of time and of the construction of timepieces.

**HOROSCOPE (ASTROLOGICAL CHART):** Astrological chart showing the positions of the Sun, Moon and planets in relation to the signs of the zodiac at a specific time. It is used to analyse the character of individuals born at that time, providing information about the current state of their life and predicting their future.

**HORSE:** A mostly domesticated, large, strong, mammal belonging to the order Perissodactyla (odd-toed ungulates) and family Equidae with hooves. They are adapted to use speed to escape predators and they have a well-developed sense of balance and a strong fight-or-flight instinct. Horses are able to sleep both standing up and lying down. Horses have been used throughout human civilisation as transport animals and for sport activities such as racing.

**HORSE LATITUDES:** Two belts, characterised by low winds, about 30° north and south of the equator.

**HORSEFLY (CLEG):** A general name for many bloodsucking flies of the genus Tabanid.

**HORSEPOWER (HP):** A imperial unit of power defined as 550 foot pound force per second. It has now been replaced by the watt. 1 hp is equivalent to 745.699872 watts.

**HORSESHOE:** An u-shaped iron or synthetic material shoe nailed or glued to the hard part of a horse's hoof. It is used to protect the animal's feet from wear and tear.

**HORSETAIL:** A poisonous weed found in moist, rich soils of grassland. A member of Equisetum, family Equisetaceae (horse family), it is poisonous to livestock.

**HORSEWOOD (MAGGOT KILLER; *Clausena anisata*):** A small deciduous shrub or small tree belonging to the Rutaceae (Citrus). It occurs in savanna, thickets, disturbed areas, riverine forest and secondary forest from Guinea and Sierra Leone eastwards to Ethiopia, the Sudan and southward to the Cape Province, only avoiding the driest regions. It also occurs in tropical Asia and South-East Asia. The tree is about 10 metres tall with a smooth, thin, greyish-green bark becoming brownish and mottled when matured. The leaves are compound with absent stipules; the leaflets are ovate to narrowly elliptical, alternate or almost opposite. The apex of the leaflets is notched or obtuse with smooth leaf margins. The inflorescences are hairy, bisexual and regular with elliptical petals having cream to yellowish-white colours. The roots and leaves contain chemicals and are used in treating ailments like fever, pneumonia, whooping cough, hypertension, headache, venereal diseases, sinusitis, kidney ailments and epilepsy.

**HORTICULTURE:** The industry and science of plant cultivation, which is of particular importance in food production and ornamental plants. **Horticulturists** work and conduct research in the disciplines of plant propagation and cultivation, crop production, plant breeding and genetic engineering, plant biochemistry and plant physiology.

**HOSE-IN-HOSE:** Describing a flower that has an extra set of petals within the normal corolla. Examples include some primroses, certain Kurume azaleas and some forms of Canterbury bells.

**HOSIER SYSTEM:** A system of dairy cattle management, where the milking of cows is done in the field using a milking pail.

**HOST:** 1. The recipient of a transplanted or grafted embryo, tissue or organ. 2. In computer terminology, it is also known as the **server computer** and refers to the main computer that controls specific functions or files in a network that another computer can access by means of a network or modem. It can be a Web server, an e-mail server, an FTP server, etc. For example, a Web host is what provides the content of Web pages to the computers that access it. 3. An organism whose body provides nourishment and shelter for a parasite or parasitoid. A definitive host is one in which an animal parasite becomes sexually mature. An intermediate host is one in which the parasite passes the larvae or asexual stage of its life cycle. 4. An organism that lives in close association with an aquiline (resembling or connected with eagle). 5. A molecular entity that forms complexes with organic or inorganic guests or a chemical species that can accommodate guests within cavities of its crystal structure. Examples include cryptands and crowns, where there are ion-dipole attractions between heteroatoms and positive ions, hydrogen-bonded molecules that form clathrates, for example, hydroquinone and water and host molecules of inclusion compounds, for example, urea or thiourea. Van der Waals forces and hydrophobic interactions bind the guest to the host molecule in clathrates and inclusion compounds.

**HOST-PLANT RESISTANCE (VARIETAL RESISTANCE):** A particular plant's ability to resist injury caused by herbicides, pathogens, or insects. This may be genetically controlled innate or bred phenotypic or physiological property of the plant. Some plants use certain features such as hair covering on their leaves or plants with a thick leaf cuticle to deter organisms.

**HOST-VECTOR SYSTEM:** A compatible combination of host, for example, bacteria and vector such as plasmid that allows propagation of DNA.

**HOSTNAME:** In computing, it is a unique name that is assigned to a device connected to network and that is used to identify the device on various forms of electronic communication such as the World Wide Web, e-mail or Usenet.

**HOT ATOM:** An atom in an excited energy state or having kinetic energy above the ambient thermal level, usually as a result of nuclear processes.

**HOT CELL:** A heavily shielded enclosure for highly radioactive materials. It may be used for their handling or processing by remote means or for their storage.

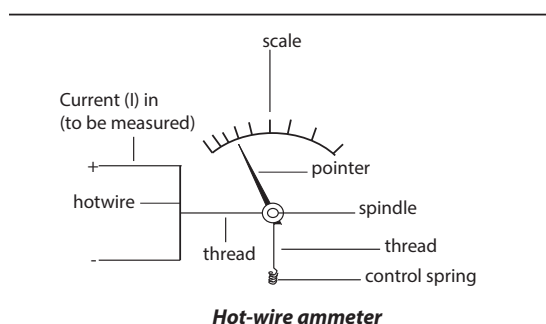
**HOT KEY (HOTKEY; SHORTCUT):** In computing, a key or a combination of keys on a keyboard that performs a specific task, usually faster than by using a mouse or other input device. For example, in operating systems such as Windows and Mac, Ctrl + C usually copies selected objects and Ctrl + V, pastes the objects.

**HOT SPOT (HOTSPOT):** 1. An area that lies above a strong current or plume of magma that rises upwards below the Earth's crust. The source of the heat is thought to be the decay of radioactive elements deep within the Earth that rises through the process of convection. The magma spreads horizontally in all directions and may break through where the crust is thin. Hot spots occur within, rather than on the edges of, lithospheric plates. However, the magma usually reaches the surface at plate margins. Hot spots can be found in places such as Hawaii, Iceland and Yellowstone National Park in Wyoming, USA. 2. Also known as **Wi-Fi hotspot**, it is a location often in public places such as airports, train stations, libraries, marinas and hotels equipped with wireless LAN (local area network), from where visitors with wireless-based devices such as laptops, smartphones or tablets can have access to the Internet. To gain access, a user must know the network name and the password. 3. In genetics, it is a part of a chromosome that has a tendency for mutation or recombination.

**HOT STATE REACTION:** A reaction proceeding from an ensemble of molecular entities possessing a higher average vibrational, rotational or translational energy than they would at thermal equilibrium with the surrounding medium.

**HOT SWAPPABLE DEVICE:** In computer terminology, it refers to a peripheral or component such as a pen drive that can be removed or added while a computer is running. Replacing a device while a computer is powered on is called **hot swapping**. Modern ports that support hot swapping include USB, FireWire and Thunderbolt. The term is also used most often to describe internal components such as server hard drive that can be replaced while the computer is running.

**HOT-WIRE INSTRUMENT:** An electrical measuring instrument such as an ammeter that depends for its operation on the expansion by heat of a wire carrying a current. The current to be measured is passed through a thin wire which causes its temperature to change. Some mechanical device is used to magnify the actual increase in the length of the wire and causes it to rotate a pointer over a circular scale.



**HOTHOUSE LAMB:** A lamb between 9 to 16 weeks of age old. It is normally born in autumn (the season between summer and winter) and raised indoors. Hothouse lambs are usually slaughtered at 18-27 kg.

**HOURLY:** 1. One of the 24 equal parts of a day, which is equal to 60 minutes or 3,600 seconds. In one hour, the Earth rotates through 15 degrees. The time of day at any point on Earth is expressed as the number of hours and minutes that have elapsed since midnight for the time zone in which the point is situated. The time zones are fixed intervals behind or ahead of Greenwich Mean Time. 2. The unit of right ascension, equivalent to 15 degrees of arc.

**HOURLY ANGLE:** The angle between the hour circle of a celestial object and the observer's meridian, measured anticlockwise along

the celestial equator. It is sometimes used as a coordinate in place of right ascension.

**HOURLY CIRCLE:** A great circle on the celestial sphere that passes through a celestial object and the north and south celestial poles. Everywhere on an hour circle the right ascension is the same.

**HOURLY GLASS:** An ancient instrument, which measures the passage of time. A quantity of fine, dry sand is contained in a bulb constricted at its centre to a narrow neck. The device is turned so that all the sand is in the upper chamber: the time taken for the sand to trickle into the lower chamber depends on the amount of sand and on the diameter of the neck.

**HOVER:** 1. In computing, also called **mouse hover**, it is to move the mouse pointer over an object or link on a webpage and hold it there to trigger an event such as making a program to respond. When you click the mouse button while the cursor is hovering over a link, it presses down on the link to activate it. When hovering is done, the cursor will change from a pointer to a small hand. In some cases the link text changes colour. 2. **Hover!** is a video game from Microsoft that came with Windows 95 and has been available for subsequent versions of Windows. 3. In poultry keeping, it is a canopy used on brooders to direct heat downwards to chicks.

**HOVERCRAFT:** A vehicle supported above the surface of land or water by an air cushion of high-pressure air, free from all contact with the surface beneath. It is used mainly as ferries.

**HOX GENES (HOMEOTIC GENE):** A class of developmental control genes that controls development of structures along the head to the tail axis of a wide range of animals and defined by a DNA sequence known as the homeobox.

**HOYLE, SIR FRED (1915-2001):** British astrophysicist and cosmologist, who did important work in nucleosynthesis and founded the steady-state theory of the universe. He studied mathematics and theoretical physics at Cambridge University, where in 1958, he became a professor of astronomy. His early work on stellar evolution led to his famous collaboration with Margaret and Geoffrey Burbidge and William Fowler on the synthesis of the elements beyond helium in stars. Hoyle successfully predicted the existence of a resonance in carbon-12 that was essential to helium burning in stars. He is best known for his proposal in 1948, with Herman Bondi (1919-2005) and Thomas Gold (1920-2004), of the steady-state theory of the universe. He also carried out theoretical work on the formation of elements in stars. Hoyle provided a mathematical theory of the model consistent with the general theory of relativity and served as the leading spokesman for the new theory, coining the term "Big Bang" for the competing model during a radio lecture.

**HSPA (HIGH SPEED PACKET ACCESS):** A third-generation (3G) mobile broadband communications technology used to increase data capacity and speeds. It is composed of protocols: **high speed-downlink packet access (HSDPA)** and **high speed-uplink packet access (HSUPA)**. The HSDPA data has transfer speeds of up to 14.4 Mbps per cell for downloads and HSUPA supports data rates up to 5.7 Mbps.

**HTML (HYPERTEXT MARKUP LANGUAGE):** The language that is used to create documents on the Web. Web pages must conform to the rules of HTML in order to be displayed correctly in a Web browser. The HTML syntax is based on a list of tags that describe the page's format and what is displayed on the Web page. HTML supports inclusion of audio, video and animation into an electronic document through helper software such as ActiveX, Java applets, Quick-Time.

**HTML EXTENSION:** In computing, any proprietary addition to the standard specification of HTML (hypertext markup language).

**HTTP (HYPERTEXT TRANSFER PROTOCOL):** The protocol used to transfer hypertext requests and information between servers and browsers over the World Wide Web. **HTTP** makes all Web site addresses begin with "http://". Whenever a URL is entered into a browser, the computer sends an HTTP request to the appropriate Web server. The Web server, which is designed to handle HTTP requests, then sends to the requested HTML page. **HTTPS (HyperText**



**Transport Protocol Secure**) is a type of HTTP that uses a secure socket layer (SSL) for security purposes. Some examples of sites that use HTTPS include banking and investment websites, e-commerce websites and most websites that require a log in.

**HUB:** In a computer network, a point at which circuits are either connected or switched. For example, in a star network, the hub is the central node; in a star/ring network, it is the location of wiring concentrators.

**HUBBLE CLASSIFICATION:** In astronomy, it is a method of classifying galaxies according to their shapes based on four basic types: elliptical, spiral, barred spiral and irregular. It was originally devised in the 1920s by US astronomer Edwin Hubble (1889- 1953).

**HUBBLE CONSTANT (HUBBLE'S CONSTANT):** In astronomy, it is a unit of measurement of the rate at which the universe is expanding, defined as the ratio of the speed of recession of a galaxy to its distance from the observer. Its current value is between 50 and 100 km/sec per megaparsec.

**HUBBLE RADIUS:** The distance from the observer at which the cosmological recession velocity of a galaxy would equal the speed of light, expressed mathematically as  $c/H_0$ , where  $c$  is the velocity of light and  $H_0$  is the Hubble constant. It is the radius of the observable universe and is believed to be between about 11 and 15 billion light-years.

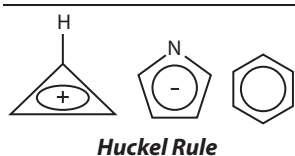
**HUBBLE SPACE TELESCOPE (HST):** Named after the American astronomer, Edwin Hubble, it is orbiting observatory telescope, built and operated jointly by NASA and the European Space Agency, which is equipped with a main mirror 2.4 metres in diameter and used to take pictures from space, avoiding the difficulties arising from the Earth's atmosphere. It was carried into orbit by a space shuttle in 1990. The pictures taken give important information on many aspects of astronomy and cosmology. Some of the scientific results obtained are measurements of the Hubble constant that relates to the age of the universe and the evidence of a black hole at the centre of a galaxy.

**HUBBLE, EDWIN POWELL (1889 - 1953):** US astronomer, who worked at both the Yerkes Observatory and the Mont Wilson Observatory. Most of his studies involved nebulae and galaxies, which he classified in 1926. In 1929 he established the Hubble constant, which he classified in 1926. In studying the galaxies, he made his second remarkable discovery in 1927 that the galaxies were receding from the Milky Way at rates that increased with distance. In 1929, he established the Hubble constant, which enabled him to estimate the size and age of the universe. The Hubble space telescope is named after him.

**HUBBLE'S LAW:** The name for the astronomical observation in physical cosmology first made by American astronomer, Edwin Hubble. It is an empirical law in astronomy stating that the distances between galaxies or clusters of galaxies are continuously increasing and therefore, the universe is expanding. It is a statement of a direct correlation between the distance to a galaxy and its recessional velocity as determined by the red shift. It is usually expressed in the form  $v = Hd$ , where  $v$  is the recessional velocity as determined by the red shift,  $d$  the distance and  $H$  is the Hubble's constant.

**HUCKEL APPROXIMATION:** A set of approximations used to simplify the molecular orbital analysis of conjugated molecules, suggested by Erich Hückel in 1931. It assumes that pi-orbitals ( $\pi$  orbitals) can be treated independently of that of sigma-orbitals ( $\sigma$  orbitals) because the interactions between the pi- and sigma-electrons are negligible in conjugated systems.

**HUCKEL RULE:** The set of rules that estimates whether a planar ring molecule will have aromatic properties. Its formulation was first worked out by physical chemist, Erich Hückel in 1931. It says that an aromatic compound should contain  $4n + 2$  pi-electrons in cyclic planar



arrangement, where  $n$  is any positive integer including zero. To determine whether a compound is aromatic, the number of cyclic planar pi-electrons are counted and equated to  $4n + 2$  to find  $n$ . If  $n$  is an integer, then the compound is aromatic.

**HUE:** One of the main properties of a colour, defined as the degree to which a stimulus can be described as similar to or different from stimuli that are described as red, green, blue and yellow. In painting colour theory, a hue refers to a pure colour without tint or shade. Usually colours with the same hue are distinguished with adjectives referring to their lightness and/or chroma, such as with "light blue", "pastel blue", "vivid blue". Exceptions include brown, which is a dark orange; and pink, a light red with reduced chroma. A hue is an element of the colour wheel. Hues are first processed in the brain in areas in the extended V4 called globs.

**HULL:** 1. The outer coverings of cereal grain, which form bran. 2. The pod of peas or beans. 3. The calyx on a strawberry that stays attached to the fruit when it is picked but is not eaten. 4. The external casing of a rocket, missile or spaceship. 5. The body of a ship, excluding other parts such as the masts and engines.

**HULLER:** A kind of threshing machine which removes seeds from their husks.

**HULSE-TAYLOR PULSAR:** A binary system discovered in 1974 by American physicists Russell Hulse and Joseph Taylor of Princeton University, for which they shared the 1993 Nobel Prize in Physics. It consists of a pulsar orbiting with a companion star and used to demonstrate, by measurement of a period time, that the system was emitting gravitational waves.

**HUM:** This is an undesirable contribution to sound reproduction caused by a nearby electric power circuit or by two vibrating tuning forks of nearby equal frequencies.

**HUMAN (HUMAN BEING):** An individual of the genus Homo, especially a member of the species Homo sapiens, a branch of the tribe Hominini belonging to the family of great apes and therefore belonging to the order Primates, which have limbs capable of performing a variety of functions, five-digit hands adapted for grasping; long pre/postnatal development; etc. Humans are distinguished from other animals by characteristics such as erect posture and bipedal locomotion; manual dexterity; more complex and developed brains capable of abstract reasoning, language and power of articulate speech; ability to build fires and cook food; ability to live in large and organised societies; ability to clothe themselves; ability to create and use simple and very complex tools and machines and other technologies; etc.

**HUMAN CHORIONIC GONADOTROPHIN, HCG:** A peptide hormone secreted by the chorion that prevents the disintegration of the corpus luteum of the ovary and maintains progesterone production, which is critical for a pregnancy in humans.

**HUMAN COMPUTER INTERACTION:** Interactions between a person and a computer through the medium of a user interphase and the underlying processes which produce the interactions. The contributing disciplines include computer science, cognitive science, human factors, software engineering, management science, psychology, sociology and anthropology.

**HUMAN GENOME PROJECT (HGP):** An international scientific research project to map the entire human genome so that the genes could be isolated and sequenced. It involves the production of a DNA library.

**HUMAN IMMUNODEFICIENCY VIRUS (HIV):** Either of two strains of a retrovirus HIV-1 or HVI-2 that destroys the immune system's helper T-cells, the loss of which causes AIDS. Retroviruses produce the enzyme reverse transcriptase, which allows transcription of the viral genome onto the DNA of the H host cell. The retrovirus attacks and gradually destroys the immune system, leaving the host unprotected against infection.

**HUMAN LEUCOCYTE ANTIGEN SYSTEM (HLA SYSTEM):** A series of gene groups, designated HLA-A, HLA-B, HLA-C and HLA-D that code for proteins on the surface of cells. They are responsible

for the regulation of the human immune system by distinguishing the body's own proteins from proteins made by foreign invaders such as viruses and bacteria. The HLA system is used to assess tissue compatibility and people with the same HLA types are said to be histocompatible. Some autoimmune disorders such as psoriasis, type 1 diabetes mellitus, multiple sclerosis and rheumatoid arthritis are linked to specific HLA alleles.

**HUMAN PAPILLOMAVIRUS (HPV):** A family of over 100 viruses some of which infect humans causing warts most commonly on the feet, face, neck and the genital. HPV is also associated with causing tumours of the genital tract including cancer of the cervix. Some types are transmitted through sexual contact.

**HUMAN PLACENTAL LACTOGEN, HPL (PLACENTAL LACTOGEN):** A hormone secreted by the placenta, which disappears from the blood immediately after delivery. It has lactogenic, luteotropic and growth-promoting activity and inhibits maternal insulin activity during pregnancy.

**H HUMAN POPULATION CURVE:** A diagram or graph showing changes in the human population.

**HUMATE:** A salt that is derived from humus. **Humus** is a brown or black complex variable material resulting from partial decomposition of plant or animal matter and forming the organic portion of soil.

**HUME-ROTHERY RULES:** A set of empirical rules put forward in 1926 by William Hume-Rothery to describe how an element could dissolve in a metal, forming a solid solution. There are two sets of rules, one which refers to substitutional solid solutions and another which refers to interstitial solid solutions.

**HUMECTANT:** Any hygroscopic substance employed to sustain the moisture levels of a system within a certain range. Humectants are used in confectionery and food stuff as additives. It often consists of a molecule with several hydrophilic groups, most often hydroxyl groups, but amines and carboxyl groups, sometimes esterified, can be encountered as well; the affinity to form hydrogen bonds with molecules of water is important. Some common humectants are glycerol, mannitol and other polyhydric alcohols usually carbohydrates. Desiccant is the opposite of humectant.

**HUMERUS:** The long bone of the upper arm between the elbow and shoulder in mammals and other vertebrates.

**HUMIC ACID:** Any of a variety of organic acids obtained from humus found in soil, peat, sediment, or aquatic environments. Humic acid is of higher molecular weight than fulvic acid, but just like fulvic acid, it is frequently used as a soil supplement to enhance crop productivity as well as human nutritional supplement. Humic acid, fulvic acid and humic acid are labelled as humic substances.

**HUMIC LAYER:** A layer of highly decomposed organic soil material containing little fibre.

**HUMIDIFIER:** A machine or device that keeps the air moist inside an enclosed space. Examples are point-of-use humidifiers, which are commonly used to humidify a single room and whole-house or furnace humidifiers, which connect to a home's HVAC system to provide humidity to the entire house.

**HUMIDITY:** The amount of moisture water vapour in the air or the condition of having a high amount of moisture in the air. Humidity indicates the likelihood of precipitation, dew or fog. High humidity makes people feel hotter outside in the summer because it reduces the effectiveness of sweating to cool the body by reducing the evaporation of perspiration from the skin. **Absolute humidity** is the quantity of water vapour or moisture in a given volume of the atmosphere or a measurement of the amount of water vapour in the atmosphere. **Relative humidity** is the amount of water vapour that exists in a gaseous mixture of air and water vapour expressed as a percentage.

**HUMIFUSE:** A term used to describe plants that spread along the ground or lie on the surface.

**HUMIFY:** To break down rotting organic waste to form humus hump. **Humification** is the processes by which organic matter decomposes to form humus by the action of decomposers such as bacteria and fungi.

**HUMIN:** The fraction (about 50%) of organic matter in soil that does not dissolve when the soil is treated with dilute alkali.

**HUMMOCK:** A rounded or conical mound within a volcanic landslide or debris avalanche deposit. Hummocks contain a wide range of rock debris, reflecting the variation of deposits that previously formed the flanks of the volcano.

**HUMMOCK AND HOLLOW CYCLE:** A cyclic change in the vegetation that takes place on a developing raised bog. A pool of water on the peat is invaded by *Sphagnum* mosses and a low hummock is built up. Other plants, such as *Calluna vulgaris* (ling) and *Erica tetralix* (crossleaved heath), then establish themselves. When the ling dies the hummock is easily eroded and reduced to a water-filled hollow surrounded by other hummocks in various stages. The cycle can then be repeated. Similar cycles occur in grassland and in the tundra where frost hummocks of peat are formed by frost action and subsequently eroded.

**HUMORAL:** Relating to the blood or other body fluid. For example, **humoral immunity** is immunity conferred by the antibodies present in the blood lymph and tissue fluid.

**HUMORAL IMMUNITY:** The immediate response to an antigen which triggers antibody production. This is distinct from cell-mediated immunity, which involves white blood cells and is a delayed response to an antigen.

**HUMOUR:** 1. Any fluid or moisture of animal bodies, such as the vitreous humour and aqueous humours. 2. Also known as **cardinal humour**, it is any of the four fluids in the human body, namely the blood, phlegm, yellow bile and black bile that ancient physicians believed when balanced and working well kept a person healthy, but if in excess or deficit, caused diseases and disabilities.

**HUMP:** 1. Also called **kyphosis** or **hunchback** it is a deformity of the back in humans caused by an abnormal convex curvature of the upper spine. In severe cases, kyphosis can cause severe pain and discomfort, breathing and digestion difficulties, cardiovascular irregularities, neurological compromise and in the more severe cases, significantly shortened life-spans. It is common in people suffering from osteoporosis and those who have poor posture, which creates an unequal muscle pull on the spine. 2. The rounded flesh on the back or shoulders of an animal such as a camel or certain breeds of cattle.

**HUMPHREYS SERIES:** A series of lines in the infrared spectrum of atomic hydrogen with wave numbers given by  $R_H(1/36) - (1/n^2)$ , where  $R_H$  is the Rydberg constant for hydrogen and  $n$  is any number greater than 6.

**HUMUS:** Dark-brown organic component of the soil consisting of decaying plants, animal remains and excrement and other organic matter which provide nutrients for plants and vitamins for animals as well as helps the soil to hold water. It is usually richer towards the surface and an important source of mineral in soil fertility.

**HUND COUPLING CASES:** Five ways of combining the electron-spin angular momentum, electron-orbital angular momentum and nuclear-rotation angular momentum to form the total angular momentum of a molecule.

**HUND'S RULES:** A set of rules used to determine which is the term symbol that corresponds to the ground state of a multi-electron atom. It was proposed by the German physicist Friedrich Hund, (1896–1997) in 1925. In other words, they are rules in atomic spectra that determine the lowest energy level for a configuration of two equivalent electrons (electrons with the same  $n$  and  $l$  quantum numbers) in a many electron atom. Rule 1 states that the lowest energy state has the maximum multiplicity consistent with the Pauli exclusion principle. Rule 2 states that the lowest energy state has the maximum total electron orbital angular momentum quantum number, consistent with Rule 1. The most stable arrangement of electrons in atomic subshells is the one with the greatest number of parallel spins.

**HUNDREDWEIGHT:** A unit of mass in the f.p.s (foot, pound, second) system, which in British imperial units is 112 lb, which is equal to

50.802345 kg. In U.S. customary units, it is 100 lb, which is equal to 45.359237 kg.

**HUNGARIAN METHOD:** An algorithm to solve the standard assignment problem, anticipating later primal-dual methods. It was developed by Harold Kuhn (1925–2014), an American mathematician in 1955.

**HUNTING:** The oscillation of a gauge needle or engine speed about a mean value.

**HUNTINGTON'S DISEASE, HD (HUNTINGTON'S CHOREA):**

A rare hereditary disease of the nervous system characterised by the progressive degeneration of nerve cells in the brain leading to dementia, involuntary movements (chorea), emotional disturbances and speech disturbances. There is currently no effective cure, but certain medications may be used to control some of the symptoms.

**HURRICANE (TROPICAL STORM):** A revolving storm about a central area of very low pressure, about 650 km across. It is found mainly in tropical regions of the Atlantic or Caribbean, travelling northwest, northeast or north. It has torrential rain, extremely strong wind of above 119 km/h accompanied by towering clouds.

**HURWITZ' THEOREM:** The theorem, developed by German mathematician, Adolf Hurwitz (1859–1919) that if the quadratic form defines a homomorphism into the positive real numbers on the non-zero part of the algebra, then the algebra must be isomorphic to the real numbers, the complex numbers, the quaternions or the octonions.

**HUSBANDRY (HUSBANDRY SYSTEM; ANIMAL HUSBANDRY):**

The activity of looking after breeding and raising livestock, which includes their habitat, diet, medical care, production rates and general welfare.

**HUSK:** The outer membranous covering of some fruits, grains and nuts.

**HUTCH:** A box or cage with a roll of wire netting, especially one used for keeping rabbits.

**HUYGENS, CHRISTIAN (1629–95):** Dutch astronomer and physicist, who worked at the Academy of Science in Paris from 1666 to 1681. Returning home in 1657, he designed the first pendulum clock and made improvements to astronomical telescopes with which he discovered Saturn's satellite, Titan in 1655. His greatest achievement was the principle which states that every point on the front of an advancing wave is itself a source of new waves, from which he developed the wave theory of light. In 1678 he discovered the polarisation of light by double refraction in calcite.

**HUYGENS' CONSTRUCTION (HUYGENS' PRINCIPLE):** A principle that explains what happens when light or other waves encounter a barrier or enter another medium. Every point on a wave front may itself be regarded as a source of secondary waves that radiate from their centres with the same frequency, velocity and wavelength as their parent.

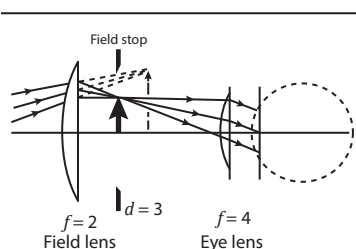
**HUYGENS' EYEPIECE:** It consists of two Plano-convex lenses facing an incident light with focal lengths and separation in the proportion of 2:3:4 as field lens  $f$ : separation  $d$ : and eye lens  $f$ . It is commonly used with telescopes and microscopes.

**HYACINTH (JACINTH):**

1. Red or yellowish-red variety of zircon, used as a gemstone. 2. Any genus of liliaceous Mediterranean plant, especially *Hyacinthus orientalis* having narrow leaves and a terminal of fragrant flowers with shades of blue, purple, red, pink, white and yellow.

**HYALINE:** Transparent or translucent, resembling a glass.

**HYALINE CARTILAGE:** The most common type of cartilage. It is a bluish-white material with a matrix of chondroitin sulfate in which fine collagen fibrils are embedded.



*Huygens' eyepiece*

**HYALOPLASM (CYTOPLASM):** The part of the protoplasm outside the nucleus in which the cisternal elements and membrane-bound organelles lie. The soluble phase of cytoplasm contains the principal components of the cell's biochemical pathways, i.e. ions, dissolved gases and most of the enzymes and substrate molecules for metabolic processes. The cytoplasmic matrix also contains storage products, such as lipid droplets, aleurone grains and starch grains.

**HYALURONIC ACID (HYALURONAN; HYALURONATE):** A glycosaminoglycan that is an integral part of the gel-like substance of animal connective tissue with the chemical formula  $(C_{14}H_{21}NO_{11})_n$ . It binds cells together and serves as a lubricant and shock absorbent in the joints. It may also play a role in the migration of cells at wounds. This activity eases when hyaluronidase breaks down hyaluronic acid. Hyaluronic acid has been isolated from umbilical cord, synovial fluid, skin, certain fowl tumours and other sources.

**HYALURONIDASE (HYALURONATE LYASES):** The enzyme produced by mammals, reptiles, insects and bacteria that breaks down hyaluronic acid, thereby decreasing its viscosity and increasing the permeability of conductive tissue. It is used in medicine to increase the absorption and diffusion of drugs administered by injection or application.

**HYBRID:** A plant or animal produced as a result of a cross between two parents of the same species that are genetically unlike each other or any offspring resulting from the mating of two distinctly homozygous individuals. A **structural hybrid** is the fusion of gametes that have differing structure in at least one chromosome, as a result of structural abnormalities. A **numerical hybrid** results from the fusion of gametes having different haploid numbers of chromosomes; and a **permanent hybrid** is a situation where only the heterozygous genotype occurs, because all homozygous combinations are lethal. Hybrids between different subspecies within a species such as between the Bengal tiger and Siberian tiger are known as **intra-specific hybrids**. **Interspecific hybrids** are between different species within the same genus such as between lions and tigers. Hybrids between different genera such as between sheep and goats are known as **intergeneric hybrids**.

**HYBRID BREAKDOWN:** A type of postzygotic barrier in which there is a reduction in fitness associated not with the progeny of the interspecific cross but instead the progeny of those progeny. In other words, there is a reduction in fitness to the offspring of hybrids rather than to the hybrids themselves. In hybrid breakdown, hybrid offspring are viable and can reproduce, but their own offspring are not viable or are sterile. For example, hybrid zygotes sometimes develop into adults, such as mules (hybrids between female horses and male donkeys), but the adults fail to develop functional gametes and are sterile.

**HYBRID COMPUTERS:** 1. Computers that have both analogue and digital features. The digital features are normally used for logical operations and the analogue features are for solving differential operations. 2. A term used to describe client systems that use a hard drive but also relies on programs that run on the server. Lately, it has been used to describe systems that can serve both as a laptop and a tablet or as a desktop and a laptop. An example is HP Envy Hybrid Computer.

**HYBRID DYSGENESIS:** A syndrome of abnormal traits such as infertility and increased incidence of chromosome mutation that appears in the hybrids between certain strains of the fruit fly, *Drosophila melanogaster* and also observed in the progeny of laboratory-reared organisms, crossed with wild organisms. It is thought to be caused by transposable elements in the wild-type males which are activated in eggs.

**HYBRID INVIABILITY:** A type of postzygotic barrier, in which certain hybrid organisms have lower fitness than the parent organisms. Deleterious genes ensure that the offspring would not be able to survive and usually die before birth.

**HYBRID MATERIAL:** Material composed of an intimate mixture of inorganic components organic components or both types of component.



**HYBRID ORBITAL:** An atomic orbital derived through hybridisation of atomic orbitals with different angular momentum quantum numbers located at a given atom. In other words, it is an atomic orbital formed when two orbitals of different energy levels in the same atom combine before forming a covalent bond. The name is derived from the atomic orbitals that combined. For example,  $sp^3$ -hybrid orbital is formed from one s-orbital and three p-orbitals. They are stronger than the original atomic orbital alone.

**HYBRID PROPELLANT:** A form of chemical rocket propellant in which the fuel and oxidiser are in different physical states. One of the substances is solid, usually the fuel, while the other, usually the oxidiser is liquid. The liquid is injected into the solid, whose fuel reservoir also serves as the combustion chamber.

**HYBRID STERILITY:** A type of postzygotic mechanism, which reduces the ability of some hybrids to produce viable gametes. This is caused by the absence of homologous pairs of chromosomes so that bivalents cannot form during meiosis. The resulting gametes are thus aneuploid. A hybrid zygote sometimes develops into adults, for example, mules (hybrids between female horses and male donkeys), but the adults fail to develop functional gametes and are sterile. Hybrid sterility may be overcome by polyploidy.

**HYBRID SWARM:** A continuous and variable series of hybrids between two original parental forms. The genetic and phenotypic characteristics of individuals range between the two parent types. It occurs where a barrier to reproduction between two distinct populations has broken down.

**HYBRID VIGOUR (HETEROSIS):** Offspring of greater strength and potential which results, when two different breeds or varieties are crossed with certain strains of laboratory females, especially if the parents have previously been inbred. For example, a mule, which is the offspring of a mare and a donkey has greater strength and resistance to disease and longer lifespan than either parent. It contributes to scientific agriculture in providing hybrid corn and vigorous, high-yielding hybrids in other plants and in livestock.

**HYBRID ZONE:** A zone that exists between the populations of two related species, whose geographical ranges overlap and which contains hybrid resulting from mating between the two species. The hybrid which has a reduced fertility will eventually act as a barrier between further interspecific mating and hence will resist gene flow between the two species. For a hybrid zone to be stable, the offspring produced by the hybrids have to be less fit than members of the parent species.

**HYBRIDISATION (HYBRIDIZATION):** 1. Production of one or more hybrid organism by mating of genetically different parents. In horticulture, the hybridisation of normally self-pollinating species involves the removal of the anthers (emasculation) and the artificial transfer of pollen from another plant. A paintbrush is often used to place the foreign pollen on the stigma. 2. A process by which an atom forms two or more sets of similar orbitals that have similar energies from different types of atomic orbitals. In other words, it is the process of mixing up nonequivalent orbitals of the same atom to give a new set of orbitals of equivalent energy. Hybrid orbitals are often used in organic chemistry to describe the bonding molecules containing tetrahedral ( $sp^3$ ), trigonal ( $sp^3$ ) and digonal ( $sp$ ) atoms.

**HYBRIDOMA:** A type of hybrid cell invented by Cesar Milstein, Georges J. F. Köhler and Niels Kaj Jerne in 1975 that is produced by fusing a specific antibody-producing B cell with a myeloma (B cell cancer) cell that is selected for its ability to grow in tissue culture and for an absence of antibody chain synthesis. The resulting hybrid cell line is able to produce large amounts of normal antibody which is described as monoclonal as it results from a single cloned cell line.

**HYDATHODE:** Specialised secretory tissue found in seedlings and herbaceous plants growing in moist places which exude water in liquid form under conditions of high humidity, where transpiration is low. They occur at the apices of veins, at the tips of leaves or in their margins.

**HYDRA:** An aquatic animal with a double-layered tube-like body and with six to ten tentacles around the mouth used for capturing food. It is a hollow animal, closed at one end, which is known as the foot and opening at the other into the mouth. It moves either by gliding on the foot, after the fashion of a snail or by somersaulting.

**HYDRATE:** A chemical compound that has discrete water molecules combined with it or a compound which has water chemically combined within it. For example, ethanol,  $CH_3-CH_2-OH$ , can be considered as a hydrate of ethene,  $CH_2=CH_2$ , formed by the addition of H to one C and OH to the other C. **Hydrated** is any compound, which is chemically combined with water, especially existing in the form of a hydrate.

**HYDRATION:** A chemical reaction in which a compound becomes surrounded with water molecules. Water is added to a compound to form a new substance or to form different crystals, held together by relatively weak forces or existing as a definite compound. An example is the hydration of ethene:

$H_2C=CH_2 + H_2O \rightarrow CH_3-CH_2-OH$ . It can also be defined as the process an ion or a molecule is stabilised by water molecules, arranged in a specific manner, an example is:  $Al^{3+} + 6H_2O \rightarrow [Al(H_2O)_6]^{3+}$ . Under certain conditions of temperature and pressure some salts form solid hydrates, when exposed to water vapour. Water is lost from these compounds when they are heated or when the water vapour pressure falls below a minimum value. Solids forming hydrates at low pressures are used as drying agents.

**HYDRATION ENERGY:** The change of energy that accompanies the hydration of a mole of gas and ions.

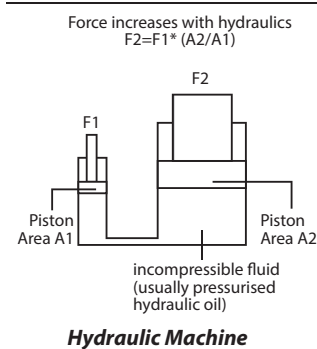
**HYDRAULIC ACTION:** An erosive force exerted by water, which is distinct from the forces exerted by rocky particles carried by water. It is caused by the force of moving water currents rushing on to the rocks below. Hydraulic action is one component of fluvial erosion.

**HYDRAULIC CONDUCTIVITY:** The rate at which water can move through pore spaces or soil fractures, rock or vascular plants depending on the intrinsic permeability of the material and on the degree of saturation. It may be used as a factor in soil management. Its symbol is  $K$ . It is a constant for a given porous medium. A very porous medium with little resistance to flow will have a high value for  $K$ , while a tightly packed medium with high resistance to flow will have a low value for  $K$ .

**HYDRAULIC GRADIENT:** A line that joins the points of highest elevation of water in a series of vertical open pipes rising from a pipeline in which water flows under pressure. It is a vector gradient between two or more hydraulic head measurements over the length of the flow path.

**HYDRAULIC HEAD:** A measure of the height of a static water column above an arbitrary point, such as sea level of the column of water that can be supported by the hydraulic pressure at a given point in a groundwater system. It is usually expressed in metres. The higher the water level or hydraulic head, the more energy that the water has at a specific location. Hydraulic heads provide an indication of the direction of groundwater flow through aquifer systems; and is also used to determine hydraulic gradients.

**HYDRAULIC MACHINE:** A machine used to transfer energy from input to output and powered by a motor activated by the confined flow of a stream of liquid, such as oil or water under pressure. A small force (effort) acting on a small area in a hydraulic press or jack can exert a large force over a large area. The fluid is controlled directly or automatically by control valves and distributed through hoses and tubes. Referring to the diagram,  $F_1$  is the force (the effort) applied to overcome the resistance or load  $F_2$ . When force



F1 is applied to the piston A1, pressure is transmitted equally to push piston A2 upwards (Pascal's Principle).

**HYDRAULIC PRESS:** A device in which a force applied to a small piston creates a pressure which is transmitted through an incompressible fluid to a larger piston where it gives rise to a larger force. It was invented by the British engineer, Joseph Bramah, for which he was granted a patent in 1795. A hydraulic press is used for almost all industrial purposes. It is used for transforming metallic objects into sheets of metal; to shape, extrude or stamp metals and to test materials under high pressures. It is also used for the thinning of glass, making powders in the cosmetic industry and for forming the tablets for medical use. As illustrated in the previous diagram, the system comprises of two cylinders, the compressible fluid (usually oil) in the cylinder having a small diameter, known as the **slave cylinder**. The piston in this cylinder is pushed so that it compresses the fluid in it that flows through a pipe into the larger cylinder, known as the **master cylinder**. The pressure is exerted on the larger cylinder and the piston in the master cylinder pushes the fluid back to the original cylinder. The force F1 applied on the fluids by the smaller cylinder results in a larger force F2 when pushed in the master cylinder, expressed as  $F_2 = F_1(A_2/A_1)$ , where A1 and A2 are the cylinder areas of the slave and master cylinders respectively.

**HYDRAULIC RADIUS (HYDRAULIC MEAN):** The ratio of the cross-sectional area of a river or stream to the length of the wetted perimeter, which is a measure of a river's ability to convey water. It is used by water engineers to assess the likelihood of flooding.

**HYDRAULICS:** 1. The study of water or other fluids at rest or in motion, particularly with respect to their engineering use. In other words, it is the practical application of hydrostatics and hydrodynamics to engineering. It targets the practical problems of collecting, storing, measuring, transporting, controlling and using water and other fluids and their applications in engineering. It is practical or applied hydrodynamics. 2. A mechanical device or system using hydraulic components.

**HYDRAZIDINES:** Compounds,  $RC(NHNH_2)_2=NNH_2$ , derived from carboxylic acids by replacing  $-OH$  by  $-NHNH_2$  (or *N*-substituted analogues) and  $=O$  by  $=NNH_2$  (or substituted analogues). A specific hydrazidine is named as a hydrazide hydrazone, for example, hexanohydrazide hydrazone.

**HYDRAZINE (DIAZANE):** A colourless liquid or white crystalline solid, soluble in water and ethanol with the molecular formula  $N_2H_4$ , density  $1.021 \text{ g cm}^{-3}$ , melting point  $2^\circ\text{C}$  (275 K) and boiling point  $114^\circ\text{C}$  (387 K). Physically, it is similar to water, but chemically it is reducing, decomposable, basic and bifunctional. Hydrazine is prepared by the Raschig synthesis in which ammonia reacts with sodium(I) chlorate (sodium hypochlorite) to give  $NH_2Cl$ , which then undergoes further reaction with ammonia to give  $N_2H_4$ . It is a powerful reducing agent and reacts violently with many oxidising agents. It is used as a rocket propellant, to inhibit corrosion, for rubber curing and foam-rubber production.

**HYDRAZO COMPOUNDS:** Compounds containing the divalent hydrazo group,  $-NH-NH-$ , such as hydrazoarenes (1,2-diarylhydrazines or 1,2-diaryldiazanes, usually with both aryl groups the same) and their *N*-substituted derivatives  $Ar-NR-NR-Ar$ .

**HYDRAZONE:** A member of a group of organic compounds that contain the group  $-C=NNH_2$  and related to ketones and aldehydes by the replacement of the oxygen with the  $NNH_2$  functional group. Hydrazones are formed by the action of hydrazine on an aldehyde or ketone and are used in identifying them.

**HYDRIC LAYER:** A layer of water in the control section of organic soils, extending from a depth of not less than 40 cm to a depth of more than 160 cm.

**HYDRIC SOIL:** A soil that formed under conditions of saturation, flooding, or ponding long enough during the growing season to

develop anaerobic conditions in the upper part. Hydric soils along with hydrophytic vegetation and wetland hydrology are used to define wetland boundaries.

**HYDRIDE:** An anion of hydrogen ( $H^-$ ), which contains hydrogen and another element only and in which the hydrogen is the more electronegative. There are three basic types of hydrides: saline or ionic, metallic and covalent. Saline hydrides, such as sodium hydride ( $NaH$ ) and calcium hydride ( $CaH_2$ ), are often used as portable sources of hydrogen gas ( $H_2$ ). Metallic hydrides, such as titanium hydride ( $TiH_2$ ), are alloy-like materials with some properties of metals, such as lustre and electrical conductivity. Covalent hydrides are mostly compounds of hydrogen and nonmetallic elements; they include water, ammonia, hydrogen sulfide ( $H_2S$ ) and methane.

**HYDRO:** 1. Of water or liquid, as for example, hydroelectricity. 2. Combined with hydrogen, as for example, hydrochloric.

**HYDROBROMIC ACID:** A colourless, acidic, aqueous solution of hydrogen bromide formed by dissolving the diatomic molecule hydrogen bromide in water. Its chemical formula is  $HBr$ , density  $1.49 \text{ g/cm}^3$ , melting point,  $-11^\circ\text{C}$  (262 K) and boiling point  $122^\circ\text{C}$  (395 K) at 700 mmHg. It is a strong acid and dissociates completely in aqueous solutions:  $HBr_{(aq)} \rightarrow H^+_{(aq)} + Br^-_{(aq)}$ . Hydrobromic acid can be prepared in the laboratory via the reaction of  $Br_2$ ,  $SO_2$  and water:  $Br_2 + SO_2 + 2H_2O \rightarrow H_2SO_4 + 2HBr$ . Also, laboratory preparations involve the production of anhydrous  $HBr$ , which is then dissolved in water. Industrially, it is prepared by reacting bromine with either sulfur or phosphorus and water. It can also be prepared by treating bromides with non-oxidising acids like phosphoric or acetic acids. It is used for the production of inorganic bromides, especially the bromides of zinc, calcium and sodium.

**HYDROCARBON:** A compound including alkanes, alkenes and alkynes made up of hydrogen and carbon atoms arranged in rows, rings or both and connected by single, double or triple bonds. In other words, it is an organic compound made up of only carbon and hydrogen. Hydrocarbons are classified into two major groups: open chain or aliphatic hydrocarbons and cyclic hydrocarbons. Open chain or aliphatic hydrocarbons have carbon atoms arranged in chains that have open ends and may be connected in single bonds (alkanes), double bonds (alkenes) and triple bonds (alkynes). Cyclic hydrocarbons are classified into alicyclic or aromatic hydrocarbons. Hydrocarbons are obtained industrially mainly from refining crude oil. Natural hydrocarbons derived from decomposed organic matter provides an abundance of carbon and hydrogen which, when bonded, can catenate to form limitless chains.

**HYDROCARBYL GROUPS:** Univalent groups formed by removing a hydrogen atom from a hydrocarbon, for example, ethyl and phenyl.

**HYDROCARBYLENE GROUPS:** Divalent groups formed by removing two hydrogen atoms from a hydrocarbon, the free valencies of which are not engaged in a double bond, for example, 1,3-phenylene,  $-CH_2CH_2CH_2-$  propane-1,3-diyl,  $-CH_2-$  methylene.

**HYDROCELE:** Accumulation of clear fluid in a sac that surrounds a testicle that causes swelling in the scrotum. It may be caused by inflammation or injury within the scrotum. It is treated by draining or by excision of the sac.

**HYDROCEPHALUS:** An abnormal buildup of cerebrospinal fluid (CSF) in the ventricles of the brain. It exerts dangerous pressure on brain tissue. It can be due either to obstruction, preventing the CSF from draining away or to a failure of natural reabsorption. In babies, it is congenital, often associated with spina bifida and causes a characteristic enlargement of the head. In adults, it may arise from brain injury or disease. Hydrocephalus is treated by insertion of a shunting system to drain the CSF harmlessly into the abdominal cavity.

**HYDROCHLORIC ACID:** A clear, colourless, fuming, poisonous, highly acidic aqueous solution of hydrogen in water with chemical

formula HCl. The concentrated acid is about 35% hydrogen chloride and is corrosive. It is a strong acid and dissociates completely in aqueous solution as:  $\text{HCl}_{(\text{aq})} \rightarrow \text{H}^+_{(\text{aq})} + \text{Cl}^-_{(\text{aq})}$ . It is found in the stomach in dilute form. It is used as a chemical intermediate and in petroleum production ore reduction, food processing, pickling and metal cleaning.

**HYDROCHLORIDE (RANITIDINE; ZANTAC; ZINETAC):** The name for an anti-ulcer drug with the chemical formula  $\text{C}_{13}\text{H}_{22}\text{N}_4\text{O}_3\text{S}$ . It is a histamine  $\text{H}_2$ -receptor antagonist that inhibits stomach acid production. It is used to treat peptic ulcers and relieve heart burns.

**HYDROCOOLING:** Removal of heat by immersing in cold water.

**HYDROCRACKING UNIT:** The thermal decomposition of heavy (high molecular weight) hydrocarbons to smaller (low molecular weight) hydrocarbons; employing high pressures of hydrogen and a special catalyst. Sulfur compounds in the fuel are reduced to  $\text{H}_2\text{S}$  and the final hydrocarbon product can be obtained relatively sulfur-free.

**HYDROCYANIC ACID (PRUSSIC ACID):** A solution of hydrogen cyanide gas in water, which is a linear molecule, with a triple bond between carbon and nitrogen with the chemical formula HCN, relative density 0.699 (liquid at 22 °C, 295 K), melting point -14 °C (259 K) and a boiling point of about 26 °C (299 K). It is a colourless, highly poisonous volatile liquid smelling of bitter almonds. It is used as inhalation rodenticide and also used as a precursor in polymers, pharmaceuticals and many other chemical compounds.

**HYDRODYNAMICALLY EQUIVALENT SPHERE:** A hypothetical sphere, impenetrable to the surrounding medium, displaying in a hydrodynamic field the same frictional effect as an actual polymer molecule. Its size may be different for different types of motion of the macromolecule, for example, for diffusion and for viscous flow.

**HYDRODYNAMICS:** The scientific field that studies non viscous liquids such as water, alcohol and ether in motion or low-velocity gas flows where the gas can be considered as being essentially incompressible.

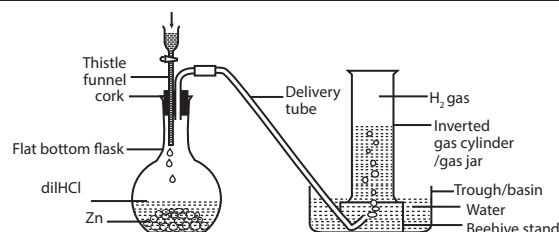
**HYDROELECTRIC POWER (HYDROELECTRICITY):** Electricity generated by the use of the energy of falling water, often behind a dam or the electrical energy produced by generators powered by turbines that are driven by moving water. Water is stored in a reservoir and piped to turbines that convert the energy in falling or fast-flowing water to mechanical energy which drives generators to produce electricity. The advantages of hydroelectric power over other sources such as fossil fuels and nuclear fission are that it is continually renewable and produces no pollution. Examples of facilities generating hydroelectric power are Akosombo (maximum 1,020 MW) in Ghana, the Three Gorges Dam in Hubei, China, the world's largest hydroelectric station with a generating capacity of 22,500 MW, Otaipu Dam in Brazil (14,000 MW) and the Aswan Dam in Egypt (2,100 MW).

**HYDROFLUORIC ACID (FLUORIC ACID; FLUORHYDRIC ACID):** A colourless, corrosive aqueous solution of hydrogen fluoride in water. Its salts are fluorides with the chemical formula, HF. It is a weak acid and ionises partially in aqueous solution as:  $\text{HF}_{(\text{aq})} \rightleftharpoons \text{H}^+_{(\text{aq})} + \text{F}^-_{(\text{aq})}$ . It dissolves glass by reacting with silicon dioxide ( $\text{SiO}_2$ ), the major component of most glass, to form silicon tetrafluoride gas and hexafluorosilicic acid. It is used for etching glass.

**HYDROFOIL:** A flat or curved wing-shaped blades, attached by struts to the hull of a boat that lifts it above the water's surface. The foil develops lift just like an aircraft wing as it passes through the water; the hull is raised above the surface and the drag (resistance experienced by a body moving through a fluid) caused by the vessel's contact with the water is thereby reduced, resulting in higher speeds. Hydrofoil vessels are capable of attaining speeds of more than 113 km/hr and the vessels do not pitch and roll. They are used for ferries.

**HYDROGEL:** Gel in which the swelling agent is water. The network component of a hydrogel is usually a polymer network. A hydrogel in which the network component is a colloidal network may be referred to as an **aquagel**.

**HYDROGEN:** A colourless, odourless, gaseous non metallic element with the symbol H and atomic number 1, atomic weight 1.00794, density at 0 °C is 0.08987 gram per litre, melting point -259.14 °C (14 K) and boiling point -252.8 °C (20.35 K). It is the lightest and the most abundant of all elements but naturally occurring elemental hydrogen is relatively rare on Earth. It constitutes about 75% of the universe's elemental mass. Stars are mainly composed of hydrogen in its plasma state. It is used in the production of ammonia, reduction of metal ores to metal and many industrial processes. Combustion of hydrogen with oxygen gives water as the sole product. It gives a squeaky pop with lighted splint.

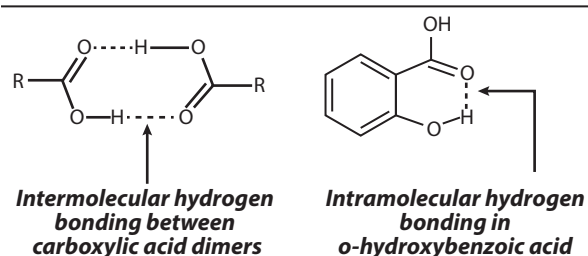


**Preparation of hydrogen from zinc with dilute acids**

**HYDROGEN AZIDE (HYDRAZOIC ACID; AZOIMIDE):** A colourless, volatile and extremely explosive liquid, a compound of nitrogen and hydrogen with the chemical formula  $\text{HN}_3$ , density 1.09 g/cm<sup>3</sup>, melting point -80 °C (193 K) and boiling point 37 °C (310 K). It is normally referred to as ammonia, which is highly toxic and a powerful reductant. It explodes in the presence of oxygen and other oxidising agents. It is prepared by the reaction of sodium amide and sodium nitrate at 175 °C (448 K) followed by distillation of a mixture of the resulting sodium azide and a dilute acid.

**HYDROGEN BOMB:** A bomb that works on the principle of nuclear fusion. Large-scale explosion results from the thermonuclear release of energy when hydrogen nuclei are fused to form helium nuclei. For the hydrogen bomb reaction to become self-sustaining, a critical temperature of about  $3.5 \times 10^7$  K must be attained with the aid of the enormous temperature created by a fission explosive.

**HYDROGEN BOND:** A type of attractive (dipole-dipole) interaction between a hydrogen atom and an electronegative atom (fluorine, nitrogen or oxygen) in one molecule and a hydrogen atom bonded to another electronegative atom in another molecule. Hydrogen bonds can occur between molecules or within parts of a single molecule. A hydrogen bond tends to be stronger than van der Waals forces, but weaker than covalent bonds or ionic bonds.



**HYDROGEN BROMIDE:** A pale yellow or colourless gas with the chemical formula HBr, which dissolves in water to form hydrobromic acid. Its melting point is -88.5 °C (184.5 K) and boiling point -67 °C (206 K). It can be made by combination of hydrogen and bromine in the presence of a catalyst such as platinum. It forms hydrobromic acid in water solution. It has many uses in chemical synthesis. For example,



it is used for the production of alkyl bromides from alcohols:  
 $\text{ROH} + \text{HBr} \rightarrow \text{RBr} + \text{H}_2\text{O}$ .

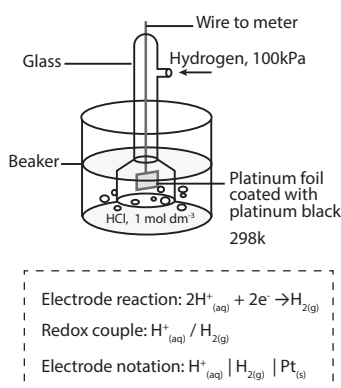
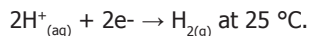
**HYDROGEN CARBONATE [BICARBONATE; HYDROXIDODIOXIDOCARBONATE(1-)]:** The acid salts of trioxocarbonate (IV) acid ( $\text{H}_2\text{CO}_3$ ) in which one hydrogen atom has been replaced usually by a metal. Its chemical formula is  $\text{HCO}_3^-$ . It forms acid salt.

**HYDROGEN CARRIER (HYDROGEN ACCEPTOR):** A molecule that accepts hydrogen ion or atom, becoming reduced in the process. The electron transport system, whose function is to generate energy in the form of adenosine triphosphate (ATP) during respiration, involves a series of hydrogen carriers.

**HYDROGEN CHLORIDE:** A colourless, fuming, non-flammable, corrosive gas which is very soluble in water with the chemical formula  $\text{HCl}$ , density 1.268, melting point  $-114.22^\circ\text{C}$  (159 K) and boiling point  $-85.05^\circ\text{C}$  (188 K). Hydrogen chloride is prepared commercially by the reaction of sulfuric acid with sodium chloride (common salt). It is also produced as a byproduct of the manufacture of chlorinated organic chemicals. It is a strong acid and dissociates fully in solution (hydrochloric acid). It is used mainly in the production of hydrochloric acid. It is also used in the hydrochlorination of rubber, in the production of vinyl and alkyl chlorides, etc.

**HYDROGEN CYANIDE (HCN):** A colourless, volatile, extremely poisonous flammable liquid or gas at normal temperatures, with the chemical formula  $\text{HCN}$ . It is completely miscible with water, burns in air, smells of bitter almonds and polymerises on standing. Hydrogen cyanide forms salts known as cyanides ( $\text{RCN}$ ) and isocyanides ( $\text{RNC}$ ). It is used in organic and synthetic fibre synthesis and as a fumigant, and is also important in the prebiotic synthesis of amino acids and purines.

**HYDROGEN ELECTRODE (STANDARD HYDROGEN ELECTRODE, SHE):** A redox electrode which forms the basis of the thermodynamic scale of oxidation-reduction potentials. It is a half-cell consisting of platinum electrode, in a solution of hydrogen ion saturated with hydrogen gas. It is immersed in acid solution and used as a standard electrode with a potential of zero to measure hydrogen ion activity. The standard is determined by the potential of a platinum electrode in the redox half reaction



*Hydrogen electrode*

**HYDROGEN FLUORIDE:** A colourless, corrosive fuming liquid, with the chemical formula  $\text{HF}$ , density 1.15 g/L, gas ( $25^\circ\text{C}$ , 298 K) 0.99 g/ml, liquid ( $19.5^\circ\text{C}$ , 292.65 K), melting point  $-83.6^\circ\text{C}$  (90 K) and boiling point  $19.5^\circ\text{C}$  (293 K). It dissolves in water to form hydrofluoric acid. It is obtained by an action of sulfuric acid on a metal fluoride. It is used in the manufacture of hydrofluoric acid, as a reagent, catalyst and fluorinating agent and in the refining of uranium and the preparation of many fluorine compounds.

**HYDROGEN HALIDES (HYDROHALIC ACIDS):** Gaseous compounds formed between hydrogen and the halogens (fluorine, chlorine, bromine, iodine, astatine and ununseptium).

**HYDROGEN IODIDE:** A colourless, poisonous gas with the chemical formula  $\text{HI}$ , density  $2.85 \text{ g cm}^{-3}$  (at  $-47^\circ\text{C}$ , 226), melting point  $-50.8^\circ\text{C}$  (222 K) and boiling point  $-35.1^\circ\text{C}$  (238 K). It can be made by the reaction of hydrogen and iodine in the presence of a platinum catalyst. It is used in organic and inorganic synthesis as one of the primary sources of iodine and as a reducing agent.

**HYDROGEN ION:** A general term for all ions of hydrogen and its

isotopes. Depending on the charge of the ion, it can be positively charged ions ( $\text{H}^+$ ) or negatively charged ions ( $\text{H}^-$ ). A characteristic of an acid is the production of hydrogen ions, hydrated to hydroxonium when in aqueous solution. The amount of hydrogen ion present in a water solution is used as a measure of the acidity of a substance; the higher the concentration of hydrogen ion the more acidic the solution and the lower the pH.

**HYDROGEN MOLECULE ION:** The simplest type of molecule ion ( $\text{H}_2^+$ ), consisting of two hydrogen nuclei and one electron of two positively-charged protons and one negatively-charged electron, that can be formed from ionisation of a neutral hydrogen molecule. In the Born–Oppenheimer approximation, in which the nuclei are regarded as being fixed, the Schrödinger equation for the hydrogen molecule ion can be solved exactly. This enables ideas and approximation techniques concerned with chemical bonding to be tested quantitatively.

**HYDROGEN OXIDISING/OXIDIZING BACTERIA:** Bacteria that oxidise hydrogen as a source of energy and synthesise carbohydrates, using carbon dioxide as their source of carbon in the absence of other organic compounds.

**HYDROGEN PEROXIDE (DIHYDROGEN DIOXIDE; DIOXIDANE):** A colourless liquid with the chemical formula  $\text{H}_2\text{O}_2$ , density  $1.450 \text{ g/cm}^3$  at  $20^\circ\text{C}$ , 293 K (pure), melting point  $-0.43^\circ\text{C}$  (273 K) and boils at  $150^\circ\text{C}$  (423 K) with sometimes violent, explosive decomposition. Hydrogen peroxide is prepared commercially by oxidation of alkylhydroanthraquinones and by electrolysis of ammonium bisulfate. It can also be prepared by the reaction of barium peroxide with sulfuric acid and is prepared (with acetone) by oxidation of isopropanol. It is used in dilute aqueous solutions as an oxidising agent, as a mild antiseptic, a bleaching agent and a laboratory reagent. It readily decomposes to give oxygen.

**HYDROGEN SPECTRUM:** The emission spectrum of atomic hydrogen, characterised by lines corresponding to radiation quanta of sharply defined energy. Its graph is obtained by plotting the frequencies at which the lines occur against the ordinal number that characterises their position in the series of lines to give a smooth curve.

**HYDROGEN SULFATE:** A compound containing the hydrogen tetraoxosulfate(VI) ion with the chemical formula  $\text{HSO}_4^-$ , density  $1.84 \text{ g/cm}^3$  (liquid), melting point  $10^\circ\text{C}$  (283K) and boiling point  $337^\circ\text{C}$  (610 K). It is an acid salt. It is formed when one hydrogen atom is removed from sulfuric acid ( $\text{H}_2\text{SO}_4$ ) by reaction with a metal, metal salt or organic group. It is used in lead-acid batteries for cars and other vehicles, mineral processing, fertiliser manufacturing, oil refining, wastewater processing and chemical synthesis.

**HYDROGEN SULFIDE:** A colourless, poisonous, flammable gas with a sweetish sickly odour similar to rotten eggs. Its chemical formula is  $\text{H}_2\text{S}$ , relative density 1.54 (liquid), melting point  $-85.5^\circ\text{C}$  (187.65 K) and boiling point  $-60.7^\circ\text{C}$  (212.45 K). It is slightly soluble in water and soluble in carbon disulfide. It is produced when organic matter containing sulfur decomposes anaerobically. Hydrogen sulfide is found naturally in volcanic gases and in some mineral waters. It is a part of many unrefined carbonaceous fuels such as natural gas, crude oil and coal. It is obtained as a by-product of refining such fuels. It may be made by reacting hydrogen gas with molten sulfur or with sulfur vapours or by treating a metal sulfide, for example, ferrous sulfide ( $\text{FeS}$ ) with an acid. It is used as an antiseptic, bleach and a reagent. It turns moist litmus paper red and lead ethanoate paper black.

**HYDROGEN SULFITE (BISULFITE; SODIUM HYDROGEN SULFITE):** An acidic salt containing the ion  $\text{HSO}_3^-$  with the chemical formula  $\text{NaHSO}_3$ , density  $1.48 \text{ g/cm}^3$  and melting point  $150^\circ\text{C}$  ( $423 \text{ K}$ ). It is used in winemaking to kill yeasts, fungi and bacteria in the grape juice before fermentation to prevent oxidation and to preserve flavour in bottled wine. It is also used as a reducing agent in the chemical industries; in the analysis of methylation status of cytosines in DNA; and also in wastewater treatment to neutralise the residual chlorine before discharging the treated effluent.

**HYDROGEN TRIOXIDE (DIHYDROGEN TRIOXIDE; DIHYDROXY ETHER; TRIOXIDANE;  $\mu$ -TRIOXIDANEDIIDODIHYDROGEN):** A relatively stable inorganic compound of hydrogen and oxygen with the molecular formula  $\text{H}_2\text{O}_3$  present in the atmosphere and possibly also in living tissue. It was first synthesised in 1994 and is produced in a reaction similar to that used for the commercial production of hydrogen peroxide ( $\text{H}_2\text{O}_2$ ) but ozone ( $\text{O}_3$ ) is used instead of oxygen. Hydrogen trioxide is stable at low temperatures but begins to decompose slowly at  $-40^\circ\text{C}$  forming the high-energy form of oxygen, singlet oxygen.

**HYDROGEN TRIOXOCARBONATE(IV) ION:** The International Union of Pure and Applied Chemistry's (IUPAC) name for **hydrogen carbonate ion** with the chemical formula  $\text{HCO}_3^-$ . It is the acid salts of trioxocarbonate (IV) acid ( $\text{H}_2\text{CO}_3$ ) in which one hydrogen atom has been replaced usually by a metal.

**HYDROGEN-OXYGEN FUEL CELL:** An electrochemical cell that converts a source fuel into an electrical current, in which hydrogen fuel is the reducing agent and oxygen is the oxidising agent. Fuel cells can be made from different fuels and oxidants such as hydrocarbons and alcohols; oxidants include chlorine and chlorine dioxide. The difference between fuel cells and batteries is that fuel cells consume reactant from an external source, which must be replenished representing a thermodynamically open system; batteries store electrical energy chemically and represent a thermodynamically closed system.

**HYDROGENATION:** A chemical reaction in which unsaturated organic compounds are converted into saturated compounds by the addition of hydrogen. It is usually done in the presence of a catalyst nickel, palladium or platinum. Hydrogenation is used in vegetable oils to produce many edible fats, such as margarine and also in the production of synthetic petrol from coal.

**HYDROGENATION OF COAL:** Industrial synthesis of mineral oil from coal by hydrogenation.

**HYDROGENATION OF OIL:** The manufacture of edible fats, such as margarine by the hydrogenation of liquid vegetable oils to produce edible fats.

**HYDROGENIC:** 1. Formed by the agency of water; or developed under the dominant influence of water. 2. Describing species which resemble the hydrogen nuclide in respect of the number of electrons they contain. Hydrogenic species contain only a single electron just like hydrogen, for example,  $\text{H}$ ,  $\text{He}^+$ ,  $\text{Li}^{2+}$ ,  $\text{C}^{5+}$ .

**HYDROGRAPH:** A graph showing how the discharge of a river varies with a period of time; or a graph showing the volume of water reaching a particular outfall or location in a sewerage network, related to time.

**HYDROID:** 1. A colony formed by hydrozoan coelenterates having a polyp rather than a medusoid form as the dominant stage of the life cycle. 2. A specialised elongated water-conducting cell found in the cortex of certain mosses. They are aligned end to end. At maturity they lose their cytoplasm and their end walls partially break down, becoming permeable to water and forming a rudimentary vessel.

**HYDROLASE:** Any of a class of enzymes that catalyses the hydrolysis of a chemical bond or the addition or removal of water from a

molecule. It plays an important role in the construction and breakdown of storage materials such as starch.

**HYDROLOGICAL CYCLE (WATER CYCLE):** The movement of water between the Earth's surface and its atmosphere and its transformation between the gaseous (vapour), liquid and solid forms. The processes involved are condensation by precipitation, evaporation, movement and storage of water overland or underground and the horizontal transport of moisture. The length of time any water stays in the atmosphere is about 11 days.

**HYDROLOGY:** The study of the movement, location, chemical and physical properties of inland water, both frozen and liquid above and below ground. It also includes its reaction with the environment and how it relates to living things.

**HYDROLYSIS:** A chemical reaction in which water ( $\text{H}_2\text{O}$  or  $\text{HOH}$ ) and another reactant exchange functional groups to form two products, one containing the H and the other the OH; or a chemical reaction in which a compound reacts with water, causing decomposition and the production of two or more other compounds. It may also be defined as a chemical reaction between an ion and water molecule resulting in proton transfer causing acidity or alkalinity of the solution. An example is the conversion of starch to glucose; an ester to a carboxylic acid and an alcohol. It is one of the mechanisms for the breakdown of food by the body or the basic means of virtually all processes of digestion of proteins, fats and many other compounds. In hydrolyses of compounds with ionic bonds, the non-water reactants are salts, acids or bases, participating in dissociation reactions.

**HYDROMETEOR:** A precipitation product, such as rain, snow, fog or clouds, formed from condensation of the water vapour in the atmosphere, which can occur either in the atmosphere or on the surface of the Earth.

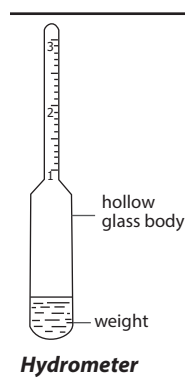
**HYDROMETER:** An instrument used to measure the specific gravity or density (weight per unit volume) of liquids compared with that of water. It usually consist of a sealed, graduated tube, weighted at one end, that sinks in a fluid to a depth used as a measure of the fluid's specific gravity. Its uses include checking the state of milk, car batteries, etc. Its unit is grams per cubic centimetre ( $\text{g/cm}^3$ ). The hydrometer should give a numeric reading directly from the instrument. A good hydrometer is accurate to  $\pm 0.005$  points so 1.265 could read from 1.260 - 1.270. The instrument accuracy should be known. In using a hydrometer, one should have an idea of what reading one is expecting. For example, a lead acid battery cell is fully charged with a specific gravity of 1.265 - 1.275. Reading is taken at where the stem floats in liquid. The specific gravity will vary depending on the temperature inside the batteries and the instruction manual for the battery will tell you what correction to apply.

**HYDROMORPHIC SOIL:** A waterlogged soil found in bogs and marshes with reduced amounts of iron and other elements. It has a grey and mottled colour.

**HYDRON:** The hydrogen cation  $\text{H}^+$ , a term used when it is not relevant to specify the isotope, as is usually the case of compounds in which the hydrogen is in natural abundance. When the isotope is relevant, then proton ( $^1\text{H}^+$ ) or Triton ( $^3\text{H}^+$ ) should be used.

**HYDROPHILIC:** Of, relating to or having a strong affinity for water.

**HYDROPHILY:** A pollination process in which the pollen is carried by water in rivers and streams either by distributing their pollen to the surface of water or beneath the surface.



**HYDROPHOBIA:** 1. An abnormal or unnatural fear of water. 2. Also known as **rabies**, it is an acute viral disease of the nervous system of warm-blooded animals that can be fatal if it affects the brain. It is usually carried by wild animals such as bats and raccoons and transmitted through the saliva of an animal such as a dog that has been infected. Symptoms include fever, aching muscles and headache, potentially progressing to inflammation of the brain, confusion, seizures, paralysis, coma and death. It is called hydrophobia because one of its symptoms is that the affected animal or person has difficulty swallowing and so fears water.

**HYDROPHOBIC:** Lacking an affinity for water; insoluble in water; repelling water. Examples include the alkanes, oils, fats and greasy substances in general. In water, they often cluster together forming micelles. Hydrophobic materials are used for oil removal from water, the management of oil spills and chemical separation processes to remove non-polar from polar compounds.

**HYDROPHOBICITY:** The association of non-polar groups or molecules in an aqueous environment which arises from the tendency of water to exclude non-polar molecules.

**HYDROPHONE:** A microphone and ancillary equipment designed to be used underwater to pick up waterborne sounds. It is often used to detect enemy submarines and listen to the sounds made by whales. Its power is often supplied by a piezoelectric transducer that generates electricity when subjected to a pressure change, converting sound signal into an electrical signal since sound is a pressure wave.

**HYDROPHYTE (AQUATIC PLANTS; HYDROPHYTIC PLANTS; HYDROPHYTES):** Any plant adapted to living in or on aquatic environments. Structural modification of hydrophytes include the reduction of mechanical and supporting tissues and vascular tissue, the absence or reduction of a root system and the specialised leaves that may either be floating or finely divided with little or no cuticle. Examples of hydrophytes include ceratophyllum, utricularia, water-lilies and certain pondweed.

**HYDROPOLYSULFIDES:** Compounds having the structures  $RS_2H$ ,  $RS_3H$  ...  $RS_nH$ , in which  $S_n$  is a chain of sulfur atoms and R is hydrocarbyl. Some people exclude hydrodisulfides from the class hypopolysulfides.

**HYDROPONICS:** The cultivation of plants in a nutrient liquid with or without the mechanical support of an inert medium such as sand or gravel.

**HYDROPS UTERI:** A disease of livestock caused by excessive accumulation of serous fluid in the tissues or in a body cavity.

**HYDROSERE:** The sequence of plant communities such as water-lilies and pondweeds that colonise the site when shallow open water becomes gradually silted up and transformed into forest or bog.

**HYDROSOL:** Aqueous solution of a colloid.

**HYDROSPHERE:** All the water on the Earth's surface together with the water vapour in the atmosphere. It includes all liquid and frozen surface waters, groundwater held in soil and rock and atmospheric water vapour.

**HYDROSTATIC:** 1. In mathematics, of a Cartesian tensor, equal to the product of a scalar field and the identity tensor, such as a tensor representing the pressure at a point in a fluid at rest. 2. Of a stress, such that the tangential component of the stress vector is zero and the normal component is independent of the unit outward normal for all possible surfaces. **Hydrostatics** is the branch of hydrodynamics that deals with the statics of fluids, usually confined to the equilibrium and pressure of liquids.

**HYDROSTATIC SKELETON (HYDROSKELETON):** The system of support found in cold-blooded organisms and soft-bodied invertebrates consisting of a fluid-filled cavity, the coelom, surrounded by muscles, which relies on the incompressibility of fluid contained

within the body cavity. In the earthworm, the coelomic fluid is under pressure within the coelom and therefore provides support for the internal organ. It also aids in the locomotion of echinoderms (starfish, sea urchins), cnidarians (jellyfish), annelids (earthworms), nematodes and other invertebrates.

**HYDROSTATICS:** The branch of physics that study fluids at rest and under pressure (such as in storage mediums, for example, tanks, dams and bulkheads) and hydraulic machinery. One basic characteristic of any fluid at rest is that the force exerted on any particle within the fluid is the same in all directions. Its practical application includes shipbuilding and dam design. It was founded by Archimedes.

**HYDROTHERMAL ACTIVITY:** Any process occurring around active or dormant volcanoes involving the action of high-temperature waters that gives rise to the circulation of heated fluids that migrate outwards and move through the surrounding rock, especially one in which the hot water or magma deposits any minerals they carry after cooling, as well as the formation of hot springs and geysers.

**HYDROTHERMAL:** Relating to or produced by hot water. For example, hydrothermal solutions, from which so many minerals are deposited, are solutions of hot water escaping from subterranean sources and may have the temperatures of superheated steam or be as cool as bath water.

**HYDROTHERMAL VEIN:** A deposit of crystallised mixture of minerals such as metallic ores and gemstone in a crack in a rock precipitated through the action of hydrothermal activity. Igneous activity often gives rise to the circulation of heated fluids that migrate outwards and move through the surrounding rock. After cooling, the hot water or magma deposits any minerals it carries.

**HYDROTHERMAL VENT (BLACK SMOKER):** A spring of hot water on the ocean floor, found mostly along mid-oceanic ridges. Heated water of temperatures up to 350 °C (623 K) composed of iron, sulfur and other materials exit from vents in the Earth's crust to form dark clouds. Some vents have been found in the Pacific and Atlantic Oceans.

**HYDROTROPISM:** The process of growth of plant parts starting by the root cap sensing water and sending a signal to the elongating part of the root. Thus most roots show positive hydrotropism, growing towards moister regions of the soil, while hypocotyls and the reproductive organs of certain fungi exhibit negative hydrotropism. If water is in short supply hydrotropism exerts a stronger influence on the direction of root growth than geotropism.

**HYDROUS MICA (HYDROMICA):** The groups of clay minerals that have higher lattice water content and lower potassium content than ideal mica found mostly in shales and mudstones. It is also present among the clay minerals in soils. They have a lattice structure which does not expand on absorption of water. They include illite, brammillite (a sodium illite), and glauconite.

**HYDROXIDE ION (HYDROXYL ION; HYDROXYL):** A diatomic anion  $OH^-$ , found in all alkalis and in alkaline solution consisting of bonded oxygen and hydrogen atoms, usually derived from the dissociation of a base; or any inorganic chemical compound containing one or more hydroxyl (OH) group and generally combined with a metal. Examples of hydroxides are sodium hydroxide (NaOH), potassium hydroxide (KOH) and calcium hydroxide ( $Ca(OH)_2$ ).

**HYDROXOACID:** A type of acid in which the acidic hydrogen is on a hydroxyl group attached to an atom that is not attached to an oxo ( $=O$ ) group. An example is  $Si(OH)_4 + H_2O \rightarrow Si(OH)_3(O)^- + H_3O^+$ .

**HYDROXONIUM:** The ion, with the chemical formula  $H_3O^+$  that forms when an acid dissolves in water. It is found in all aqueous solutions of acids. A proton from the acid joins with a molecule of water.

**HYDROXYL GROUP:** A functional group that consists of an atom of hydrogen and an atom of oxygen bonded together and covalently bonded to an organic molecule. Examples of compounds containing hydroxyl groups include alcohols and phenols.



**HYDROXYLAMINE (AMINOL; AZANOL; HYDROXYAMINE; HYDROXYAZANE; HYDROXYLAZANE; NITRINOUS ACID):** A colourless solid, with the chemical formula  $\text{NH}_2\text{OH}$ , density  $1.21 \text{ g cm}^{-3}$  (at  $20^\circ\text{C}$ ), melting point  $33^\circ\text{C}$  ( $306 \text{ K}$ ) and decomposes at  $306 \text{ K}$ . It is also a functional group where one or more of the hydrogens are replaced by substituents. It explodes on heating and may be employed as an oxidising agent or reducing agent. Hydroxylamine is made by the reduction of nitrates or nitrites and is used in the manufacture of nylon. With aldehydes and ketones it forms oximes.

**HYDROXYLATION:** The process by which a hydroxyl group ( $-\text{OH}$ ) is introduced into an organic compound, for example, alkenes can be hydroxylated using potassium permanganate or lead ethanoate to give alcohols. Hydroxylation is the first step in the oxidative degradation of organic compounds in air and very important in detoxification since hydroxylation converts lipophilic compounds into water-soluble (hydrophilic) products that are more readily excreted.

**HYDROXYPROLINE:** A crystalline amino acid obtained from gelatin or collagen, with the chemical formula  $\text{C}_5\text{H}_9\text{NO}_3$ . An increase in urinary hydroxyproline is an indication of damage and breakdown of connective tissue.

**HYGIENE:** The study and practice of how to keep good health and prevent the spreading of diseases by paying attention to cleanliness.

**HYGROMETER:** An instrument used to measure relative humidity or water vapour content of a gas, usually air. The temperature at which dew or frost forms is a measure of the absolute humidity (the weight of water vapour per unit volume of air or other gas at the temperature before cooling). The most common type of hygrometer is the dry- and wet-bulb psychrometer, one of many instruments used for measuring the water vapour content or relative humidity of the atmosphere. It consists of two identical mercury or electrical thermometers, with one of them having a wet bulb and the other a dry bulb. The wet bulb thermometer has a wet cotton or linen wick wrapping around its bulb. Evaporating water from the wick absorbs heat from the thermometer bulb, causing the temperature reading to be lower than that of the dry bulb. The difference between the dry bulb and wet bulb temperatures gives the Wet Bulb Depression and the relative humidity can now be read from the relative humidity table. Another type of hygrometer for measuring relative humidity is the **Hair Hygrometer**, which uses human or horse hair with the oils removed attached to levers that magnify a small change in hair length. As the relative humidity increases, the hair becomes longer and as the humidity drops it becomes shorter. This is recorded by a pen and rotating cylinder, known as a hygrograph. The temperature at which dew forms, called the dew point, is determined by a dew-point or condensation-type hygrometer, which is basically a mirror, usually of polished metal, cooled until dew or frost forms on it. The most accurate way to measure humidity is with an electric hygrometer, in which a known volume of gas passes over a hygroscopic or moisture-absorbing material such as phosphorus pentoxide. It is weighed before and after to determine how much water it took out of the gas.

**HYGROMETRY (MOISTURE ANALYSIS):** The measurement or indication of the water content of the ambient air or of a sample of gas.

**HYGROSCOPIC:** A term describing the ability of a substance to attract water molecules from the surrounding environment through either absorption or adsorption. A substance can absorb up to 70% of its own mass of water without dissolving or getting wet. Examples are sodium chloride, sugar, honey, glycerol, ethanol, methanol, diesel fuel, sulfuric acid, etc.

**HYGROSCOPIC COEFFICIENT (AIR DRY CONDITION):** One of the energy statuses of soil water in which evaporative forces have removed water down to a layer held with 31 bars of tension. It is the

maximum amount of hygroscopic water absorbed by 100 g of dry soil under standard conditions of humidity (50 % relative humidity) and temperature ( $15^\circ\text{C}$ ).

**HYGROSCOPIC WATER:** Water that is absorbed by a dry soil from the atmosphere and held too tightly by soil particles by forces of adhesion and not available in amounts sufficient for plants to survive.

**HYMATOMELANIC ACID:** The fraction of humus that is soluble in alcohol, after having been extracted with alkali and precipitated with acid, and which, when the alcohol is distilled, forms a brittle mass that is insoluble in alcohol.

**HYMEN (VAGINAL CORONA):** A fold of mucous membrane that covers the external opening of the vagina at birth. It normally perforates at puberty to allow the flow of menstrual blood but if the opening is small it may be ruptured at first intercourse.

**HYMENIUM:** The spore-bearing layer of the fruiting body of certain fungi, containing asci or basidia. In the ascomycetes, the hymenium lines the ascocarp while in basidiomycetes, it lines the basidiocarp.

**HYMENOPTERA:** An order of insects of over 130,000 species that comprises the ants, bees, wasps, etc. They generally have a narrow waist between the thorax and abdomen. Females typically have a special ovipositor for inserting eggs into hosts or otherwise inaccessible places. The young develop through complete metamorphosis.

**HYOID APPARATUS:** A group of bones which suspend the tongue and larynx comprising the upper throat bones of the tongue and connective tissues located just below the tongue, above the thyroid cartilage.

**HYOID ARCH:** The second of seven bony V-shaped arches that supports the gills of fish. The dorsal part is specialised as the hyomandibular, which is involved in jaw suspension. In tetrapods, the dorsal part of the hyoid arch has evolved to form the stapes. The ventral part forms the hyoid bone, which supports the tongue.

**HYOSTYLIC JAW SUSPENSION:** A type of jaw suspension seen in most fishes in which the upper jaw is not directly connected to the cranium.

**HYPANTHIUM:** A cup-like floral tube formed from the extension from the fusion of the basal parts of the calyx, corolla and stamens that surround the carpels. This is seen in flowers of cherries or roses.

**HYPER-:** A prefix, denoting over, above, high, etc. An example is hypersonic. In mathematics, it denotes an entity of more than three dimensions, for example, hypercube.

**HYPERACTIVITY:** A condition of excessive activity in young children, which may continue into the adulthood. It is usually characterised by restlessness, inability to remain focused on tasks or activities and difficulty in learning.

**HYPERALGESIA:** The excessive sensitivity to pain or painful stimulus that precedes pain. It may be caused by damage to nociceptors or peripheral nerves. A related term, **allodynia** is pain due to a stimulus that does not normally provoke pain.

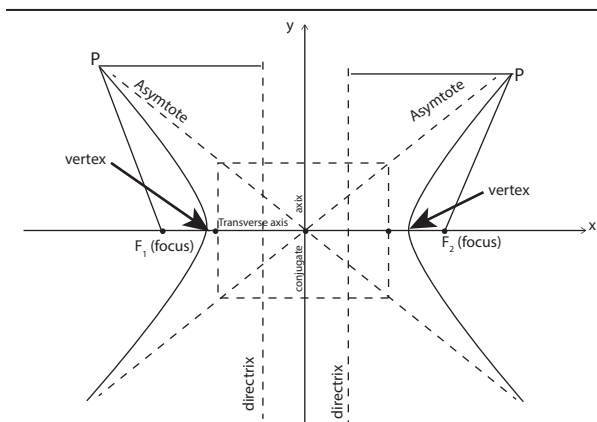
**HYPERBARIC CHAMBER:** A specially equipped pressure vessel used in medicine and physiological research to administer oxygen at elevated pressures. The high oxygen pressures achieved may destroy the anaerobic bacteria (*Clostridia*) responsible for gas gangrene. It is also used for aeroembolism in decompression.

**HYPERBOLA:** The set of all points in a plane such that for equal point of the set, the absolute value of the difference of the distances to two fixed points, called the foci in the plane is constant. For example, if  $P$  is a point on the curve, then  $|PF_1 - PF_2| = k$ , where  $k$  is a constant and  $F_1$  and  $F_2$  are the foci. It is an open curve formed by cutting a circular cone with a plane so that the angle between the plane and the base is greater than the angle between the base and the side of the cone. In other words, it is an open curve that has no ends and has two separate curves in mirror image. Its graph is symmetrical with respect to both coordinate axes and the origin and has no  $y$ -intercepts. A hyperbola with foci  $F_1(-c, 0)$  and  $F_2(c, 0)$  and difference of focal

radii  $2a$  has equations of the form,  $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ , where  $b^2 = c^2 - a^2$ .

The equations of its asymptotes are  $y = \pm \frac{b}{a}x$ . If the foci are on the  $y$ -axis at  $F_1(0, -c)$  and  $F_2(0, c)$ , the equations of the hyperbola is of the form  $\frac{y^2}{a^2} - \frac{x^2}{b^2} = 1$ , where  $b^2 = c^2 - a^2$  and the asymptotes are  $y = \pm \frac{a}{b}x$ . There are hyperbolas that are not symmetric to the  $x$

and  $y$  axes, for example, the equation of the form  $xy = k$ ;  $k \neq 0$  is a hyperbola symmetric to the lines  $y = -x$  and also  $y = x$ . In this case,  $xy = c^2$  and the asymptotes are perpendicular and the coordinate axes are the asymptotes. The **directrices** (plural of directrix) of a hyperbola are two parallel lines which are perpendicular to the major axis of a hyperbola. The directrices are between the two parts of a hyperbola and can be used to define it as the locus of points such that the ratio of the distance to the nearer focus to the distance to the nearer directrix equals a constant, called **eccentricity** that is greater than one. The **asymptote** of a curve is a line such that the distance between the curve and the line approaches zero as they tend to infinity. The asymptotes are the boundaries of a hyperbola and are used as guidelines in sketching the graph of a hyperbola. The **conjugate axis** (the minor axis, indicated by  $y$  in the diagram) of a hyperbola is the axis perpendicular to the transverse axis at a point equidistant from the foci; it passes through the centre of the hyperbola and is perpendicular to the transverse axis. The segment of this axis is equal to one side of a rectangle that has its other side equal to the transverse axis and diagonals that are along the asymptotes of the hyperbola. The transverse axis (the major axis, indicated by  $x$  in the diagram) of a hyperbola is the axis along the direction the hyperbola opens or the axis that passes through the two foci. The **vertices** are points of intersection of the hyperbola and the transverse axis.



A diagram showing a Hyperbola with its various parts on the cartesian plane

**HYPERBOLIC FUNCTIONS:** A set of functions,  $\sinh$ ,  $\cosh$ , and  $\tanh$ , that have similar properties to trigonometric functions but are related to the hyperbola in the manner that trigonometric functions are also related to the circle. For an angle  $x$ , the hyperbolic functions are defined as:  $\sinh x = \frac{1}{2}(e^x - e^{-x})$ ;  $\cosh x = \frac{1}{2}(e^x + e^{-x})$ ;  $\tanh x = \frac{(e^x - e^{-x})}{(e^x + e^{-x})}$ .

**HYPERBOLIC GEOMETRY (LOBACHESKIAN GEOMETRY):** The branch of non-Euclidean geometry that replaces the parallel postulate of Euclidean geometry with the postulate that two distinct lines may be drawn parallel to a given line through a point not on the given line. In hyperbolic geometry, contrary to the parallel postulate, there exists a line  $m$  and a point  $p$  not on  $m$  such that at least two distinct lines parallel to  $m$  pass through  $p$ . As a result, the sum of the angles

of a triangle is less than  $180^\circ$  and, for a right triangle, the square of the hypotenuse is greater than the sum of the squares of the other two sides.

**HYPERBOLIC ORBIT:** An open orbit in the shape of a hyperbola, the eccentricity of which is greater than 1. A hyperbolic orbit is followed by an object which escapes from the gravitational field of a larger body.

**HYPERBOLIC SPIRAL:** A transcendental plane curve, whose radius vector is inversely proportional to the polar angle. Its polar coordinate is given by  $r\theta = a$ .

**HYPERBOLIC VELOCITY:** A velocity that exceeds the escape velocity of a celestial body, such as a star. It allows an object, such as a spacecraft or rogue planet, to break free of the gravitational attraction of the larger body by following an open, hyperbolic orbit.

**HYPERBOLOID:** A quadratic surface having two basic forms: (i) a **hyperboloid of one sheet**, generated by spinning a hyperbola around its conjugate axis. It is a ruled surface with the cross-sections parallel to the  $x$ - $y$  plane elliptical and those parallel to the other

two coordinate planes hyperbolic. Its equation is  $\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1$ ,

where  $a$ ,  $b$  and  $c$  are constants and the axes of symmetry are taken as the coordinate axes. (ii) The hyperboloid of two sheets consists of two parts separated by a finite distance; produced by rotating a

hyperbola about its transverse axis. Its equation is  $\frac{x^2}{a^2} - \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1$ ,

where  $a$ ,  $b$  and  $c$  are constants and the axes of symmetry are taken as the coordinate axes.

**HYPERCALCAEMIA:** The presence in the blood of an abnormally high level of calcium. There are many causes including excessive intake of vitamin D, elevated parathyroid gland activity (hyperparathyroidism), malignant disease or other conditions. It can cause a number of symptoms, including loss of appetite, nausea, thirst, fatigue, muscle weakness, restlessness and confusion.

**HYPERCHARGE:** A quantised property of baryons analogous to electric charge, which gives a formal method of accounting for the non-occurrence of certain expected decays by means of strong interaction. It is not conserved in weak interactions but analogous to electric charge in some respects. Pion has hypercharge value of 0, quarks have fractional hypercharges and nucleons have a hypercharge of  $\pm 1$ .

**HYPERCHROMIC EFFECT:** The increase in the intensity of a spectral band due to substituents or interactions with the molecular environment.

**HYPERCOMPLEX NUMBERS:** Any number such as exterior algebras, group algebras, matrices, octonions and quaternions without the properties of the real and complex numbers.

**HYPERCONJUGATION:** The mechanism through which electron deficient atoms are stabilised by interaction of their empty sigma orbitals with adjacent sigma or pi-electrons in a molecule, for example, the interaction of a methyl group with the benzene ring in toluene.

**HYPERCOORDINATION:** A property of main-group atoms in molecular entities to acquire coordination numbers greater than four (which would comply with the Lewis octet rule). It may be associated with hypervalency, but usually is referred to peculiar atomic centres in the electron-deficient species with multicentre  $\sigma$ -bonding, in which the bonding power of a pair of electrons is spread over more than two atoms. An example of a hypercoordinated atom is the five-coordinate carbon atom in the methanium cation, where three C-H bonds may be regarded as normal two centre - two electron bonds and the bonding in the remaining  $\text{CH}_2$  fragment is governed by the three-centre, two-electron bond.

**HYPERCUBE:** A higher dimensional analogue of a cube. In other words, it is a figure of four or more dimensions usually in the Euclidean space with all of its sides equal and all angles rightangles. The hypersurface of a tesseract consists of eight cubical cells. A four-dimensional hypercube is known as a **tesseract**.

**HYPERELASTIC (GREEN ELASTIC):** Of a body, such that there is a scalar function of the deformation gradient whose rate of change is the stress-power per unit volume. Although all elastic bodies in nature have been found to be Green-elastic, in principle a Green-elastic body is elastic but not elastic.

**HYPERFACTORIAL:** A number such as 108, which is equal to  $3^3 \times 2^2 \times 1^1$ . In general, the  $n$ -th hyperfactorial  $H(n)$  is given by  $H(n) = n^n (n-1)^{n-1} \dots 3^3 2^2 1^1$ . The first eight hyperfactorials are 1, 4, 108, 27648, 86400000, 4031078400000, 3319766398771200000 and 55696437941726556979200000.

**HYPERFINE COUPLING:** The interaction between the spin magnetic moment of an unpaired electron and the nuclear spin magnetic moments resulting in the splitting of the  $\alpha$  (spin up) and  $\beta$  (spin down) energy levels in an external magnetic field and, thus, in the multiplet pattern of the ESR spectra of radical-like species and transition metal compounds. Two main contributions to the hyperfine coupling, Fermi contact and dipolar interactions are usually considered.

**HYPERFINE STRUCTURE:** A closely spaced structure of the spectrum lines forming a multiplet component in the spectrum of an atom or molecule or of a liquid or solid. The name is a reference to the *fine structure* which results from the interaction between the magnetic moments associated with electron spin and the electrons' orbital angular momentum. Hyperfine structure, with energy shifts typically orders of magnitude smaller than the fine structure, results from the interactions of the nucleus with internally generated electric and magnetic fields.

**HYPERGEOMETRIC DIFFERENTIAL EQUATION (GAUSS' HYPERGEOMETRIC EQUATION):** The differential equation, arising in many physical contexts, defined by  $x(x-1)y'' + [(a+b+1)x-c]y' + aby = 0$ , the solution  $c$  not a positive integer and  $|x| < 1$  can be expressed as  $y = k_1 F(a, b; c; x) + k_2 x^{1-c} F(a-c+1, b-c+1; 2-c; x)$ , where  $F(a, b; c; x)$  is the hypergeometric function.

**HYPERGEOMETRIC DISTRIBUTION:** In statistics, it is the distribution of the number of elements with a given property in a sample size  $n$ , chosen from a population of size  $N$ , of which  $D$  has the property and the distribution has the parameters  $D$ ,  $N$ ,  $n$  related as

$$P(X=d) = \frac{\binom{D}{d} \binom{N-D}{n-d}}{\binom{N}{n}}. \text{ The distribution is discrete, existing only}$$

for nonnegative integers less than the number of samples or the number of possible successes, whichever is greater.

**HYPERGEOMETRIC FUNCTION:** One of a class of functions that can be represented as power series of which the coefficients are appropriately defined products and quotients of factorials and gamma function values. The sum of the hypergeometric series, which is defined for  $|x| < 1$  by the power series

$${}_2F_1(a, b, x) = 1 + \frac{ab}{1!c}x + \frac{a(a+1)b(b+1)}{2!c(c+1)}x^2 + \dots = \sum_{n=0}^{\infty} \frac{(a)_n(b)_n}{(c)_n n!} x^n.$$

Many common functions can be written as hypergeometric functions.

**HYPERGEOMETRIC SERIES:** A power series of which the coefficients are products and quotients of factorials and gamma functions. If  $F(a,$

$b; c; x)$  is the hypergeometric function,  $F(a, b; c; x) = \sum_{n=0}^{\infty} \frac{(a)_n(b)_n}{(c)_n n!} x^n$

is valid at least for  $z$  in the unit disk, where  $(a)_n$  is the Pochhammer symbol or rising factorial defined as a ratio of Gamma functions.

**HYPERGLYCAEMIA (HYPERGLYCEMIA):** An excessive concentration of glucose in the blood stream, which occurs in diabetes mellitus due to irregular self-administration of insulin, by

insulin resistance or defective insulin response in the body, by stress or infection and by the activation of certain autoimmune processes characteristic of type 1 diabetes.

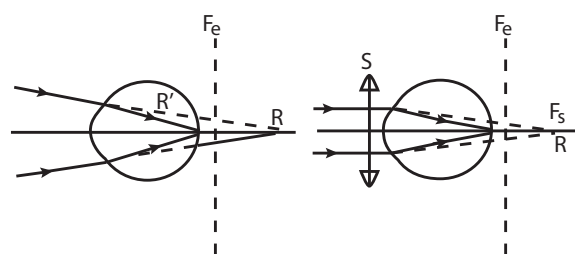
**HYPERGOLIC PROPELLANT:** A form of liquid propellant in which the fuel ignites spontaneously upon contact with an oxidiser, typically nitrogen tetroxide ( $N_2O_4$ ) or nitric acid ( $HNO_3$ ) thereby eliminating the need for an ignition system. Hypergolics remain liquid at normal temperatures, they do not pose the storage problems of cryogenic propellants. Its disadvantage is that it is highly toxic and must be handled with extreme care. Examples of hypergolic fuels include hydrazine, monomethyl hydrazine (MMH) and unsymmetrical dimethyl hydrazine (UDMH).

**HYPERGRAVITY:** The condition experienced in an intense gravitational field, where the force of gravity exceeds that on the surface of the Earth. This is expressed as being greater than 1g. Hypergravity conditions are created on Earth for research on human physiology in aerial combat and space flight, as well as testing of materials and equipment for space missions.

**HYPERLINK:** A word, phrase or image that links you to a new document or a new section within the current document, which is typically activated by clicking on a highlighted word or image on the screen. Hyperlinks are found in nearly all Web pages, allowing users to click their way from page to page. Text hyperlinks are often blue and underlined, but do not have to be. When you move the cursor over a hyperlink, whether it is text or an image, the arrow should change to a small hand pointing at the link. When you click it, a new page or place in the current page opens.

**HYPERMEDIA:** An extension of hypertext that allows images, movies and Flash animations to be linked to other content. For example, a user reading an encyclopedia entry on jazz could also hear excerpts from recordings, read biographies of jazz artists and view photos of them.

**HYPERMETROPIA (HYPEROPIA; FARSIGHTEDNESS; LONGSIGHTEDNESS):** A vision defect in which the lens of the eye is unable to focus on near objects. As an object moves toward the eye, the eye must increase its optical power to keep the image in focus on the retina. If the power of the cornea and lens is insufficient, the image will appear blurred. It is caused usually when the eyeball is too short or the lens cannot become round enough. It can be corrected by using converging spectacle lenses or contact lenses to move the image back to the retina or through the surgical procedure known as LASIK (laser in situ keratomileusis).



**Hyperopia (Long sightedness)**

**HYPERMORPHOSIS:** A type of phylogenetic development leading to disruption of the relationship of an organism to its environment as a result of hypertrophy of certain organs. Examples of hypermorphosis are the upper canine teeth of the fossil saber-toothed tiger (*Machairodus*), the horns of the giant reindeer and the tusks of the contemporary boar (the babirusa). A frequent case of hypermorphosis is the general enlargement of body dimensions, resulting in disruption of the coordination of certain organs. Hypermorphosis indicates a lag in the evolution of the organism to adapt to changes in the conditions of its existence. When there is considerable manifestation of hypermorphosis, the consequence is extinction.



**HYPERNOVA:** An explosive event in which an exceptionally large star collapses at the end of its lifespan with the formation of a black hole. The radiation output of a nearby hypernova could cause serious harm to Earth, however no known hypergiant is located close enough to Earth to pose a threat.

**HYPERON:** A member of a group of short-lived elementary particles which are greater in mass than the neutron.

**HYPERPARASITE:** An organism that is parasitic on other parasites. Examples are insects that lay their eggs inside and near parasitoid larvae, which are themselves parasitizing the tissues of a host, usually an insect larva.

**HYPERPLANE:** A translate of the null space of any linear functional; a three dimensional space in four dimensions or more generally an ( $n-1$ )-space in  $n$  dimensions. In other words, if a space is 3-dimensional then its hyperplanes are the 2-dimensional planes, while if the space is 2-dimensional, its hyperplanes are the 1-dimensional lines.

**HYPERPLASIA:** Excessive formation of cells within an organ or tissue due to an increase in the number of its component cells.

**HYPERPOLARISATION:** Any change in the amount of electrical charge separated by the cell membrane and hence in the strength of the transmembrane potential and that makes it more negative. For example, a neuron membrane is hyperpolarised if a stimulus increases its voltage from the resting potential of  $-70$  mV, reducing the chance that the neuron will transmit a nerve impulse.

**HYPERREAL NUMBER (NONSTANDARD REALS):** Any of the set of numbers formed by the addition of infinite numbers and infinitesimal numbers to the set of real numbers. Because hyperreals represent an extension of the real numbers,  $R$ , they are usually denoted by  ${}^*R$ .

**HYPERSENSITIVITY:** 1. Increase or abnormal sensitivity to compounds which can elicit a specific immune response accompanied by tissue damage. Hypersensitivity reaction includes allergies and anaphylaxis and may take a form such as asthma, hives, hay fever, anaphylactic reactions to certain foods or insect venoms, some forms of eczema and kidney diseases and skin reactions to poison ivy antigens and many other substances. 2. A reaction by the tissue of a plant in response to invasion by pathogenetic viruses, bacteria, fungi or other organisms. The reaction varies according to the type of infection agent, but typically involves the syntheses of enzymes that disarm or degrade pathogens and strengthening of cell walls in the vicinity of the attack by deposition of a lignin and other materials.

**HYPERSENSITISATION (HYPERSENSITIZATION):** A way of treating photographic emulsions to prevent their sensitivity falling off rapidly when used for long exposures. The commonest method is to soak the film in a chemically reducing gas: 8% hydrogen and 92% nitrogen mixture is popular for amateur use.

**HYPERSONIC:** A term denoting speeds of Mach 5 or moving at a speed of at least five times the speed of sound. **Mach number** is the ratio of the relative speed of a fluid and a rigid body to the speed of sound in that fluid.

**HYPERSPACE:** 1. A space that has more than three dimensions. 2. A fictional space in which the usual laws of physics may be circumvented allowing faster-than-light (FTL) travel or time travel.

**HYPERSPHERE (4-SPHERE):** A higher-dimensional analogue of a sphere. A hypersphere of radius  $R$  in  $n$ -dimensional Euclidean space consists of all points at distance  $R$  from a given fixed point (the centre of the hypersphere). It is analogous to the sphere in that all points on the surface are equidistant from the central point.

**HYPERSTHENE:** An important rock-forming silicate of magnesium and iron,  $(Mg, Fe)SiO_3$ , that crystallises in the orthorhombic system.

**HYPERTENSION (HIGH BLOOD PRESSURE):** A chronic medical condition with abnormally high blood pressure in the arteries. Persistent hypertension is one of the risk factors for strokes, heart attacks, heart failure and arterial aneurysm and is a leading cause of chronic kidney failure. Although a certain amount of pressure is needed to carry blood through the body, if that pressure is too high,

hypertension or high blood pressure develops. Blood pressure is measured by an instrument called sphygmomanometer and recorded as the systolic number over the diastolic number, indicating pressure as measured in millimetres of mercury on the gauge (mm Hg). In general, someone has high blood pressure if several measurements of blood pressure show readings of 140 systolic or 90 diastolic (140/90) or higher. Some of the factors that can contribute to hypertension include age (blood pressure usually increases with age), diet, excessive alcohol consumption, lack of exercise, obesity, sleep apnoea and stress.

**HYPertext:** A computer-based text retrieval system that enables a user to access particular locations in webpages or other electronic documents by clicking on links within specific webpages or documents. For example, when you select an object, you can see all the other objects that are linked to it. Hypertext systems are particularly useful for organising and browsing through large databases that consist of disparate types of information.

**HYPertext MARKUP LANGUAGE (HTML):** The language that is used to create documents on the Web. Web pages must conform to the rules of HTML in order to be displayed correctly in a Web browser. The HTML syntax is based on a list of tags that describe the page's format and what is displayed on the Web page. HTML supports inclusion of audio, video and animation into an electronic document through helper software such as ActiveX, Java applets, Quick-Time.

**HYPertext TRANSFER PROTOCOL (HTTP):** The protocol used to transfer hypertext requests and information between servers and browsers over the World Wide Web. HTTP makes all Web site addresses begin with "http://". Whenever a URL is entered into a browser, the computer sends an HTTP request to the appropriate Web server. The Web server, which is designed to handle HTTP requests, then sends to the requested HTML page. HTTPS (HyperText Transport Protocol Secure) is a type of HTTP that uses a secure socket layer (SSL) for security purposes. Some examples of sites that use HTTPS include banking and investment websites, e-commerce websites and most websites that require a log in.

**HYPERTHERMOPHILES:** Microbes that live and reproduce at very high temperatures such as hot springs, in the temperature range of 60 to 113 °C. The majority are archaea, although some cyanobacteria and anaerobic photosynthetic bacteria grow well at temperature range of 70 to 75 °C.

**HYPERTHYROIDISM:** Over activity of thyroid gland causing an overproduction of thyroid hormones. This condition causes increase in basal metabolic rate, with such symptoms as rapid pulse and weight loss. It is sometimes associated with goitre.

**HYPERTONIC SOLUTION:** A solution that has higher osmotic pressure or has more solutes than another solution to which it is compared. Such a solution will gain water by osmosis. In other words, it is a solution with a greater solute concentration than normal body cells so that the water is drawn out of the cells by osmosis. When a cell's cytoplasm is bathed in a hypertonic solution the water will be drawn out of the cell and into the solution by osmosis. If water molecules continue to diffuse out of the cell, it will cause the cell to shrink. Examples of hypertonic solutions are saline solution; a solution of 5% dextrose (sugar) and 0.45% sodium chloride; corn syrup is a hypertonic solution in relation to the cells within a living organism, because the corn syrup solution contains more solutes inside it than the body cells; and D10W (dextrose 10% in water), because its glucose concentration is higher than 5%.

**HYPERTROPHY:** An increase in volume of an organ or tissue due to an increase in the size of its component cell. It often occurs in response to an increase workload in an organ which may result in malfunction or disease. An example is the enlargement of the cells of the uterus during pregnancy.

**HYPERTVALENCY:** The ability of an atom in a molecular entity to expand its valence shell beyond the limits of the Lewis octet rule. Hypervalent compounds are common for the second and subsequent

row elements in groups 15-18 of the periodic table. A description of the hypervalent bonding implies a transfer of the electrons from the central (hypervalent) atom to the nonbonding molecular orbitals which it forms with (usually more electronegative) ligands. A typical example of the hypervalent bond is a linear three-centre, four-electron bond, e.g. that of  $F_{ap}-P-F_{ap}$  fragment of  $PF_5$ .

**HYPERVELOCITY GUN:** Test equipment for accelerating particles to the velocity range of meteoroids in space. The equipment is used in investigations of meteoroid damage to spacecraft and methods of meteoroid protection.

**HYPERVELOCITY IMPACT:** A collision of an object into a surface at a speed of at least 1 to 2 km/s, which results in the impact rock behaving like a fluid. Hypervelocity impacts result in craters that are generally circular, as the crater is typically much larger than the impacting body.

**HYPERVENTILATION:** An increase in the amount of air taken into the lung, caused by an increase in the depth or rate of breathing resulting in the loss of carbon dioxide from the blood. It is characterised by low blood pressure, dizziness, lightheadedness, weakness, shortness of breath, a sense of unsteadiness, muscle spasms in the hands and feet, a tingling sensation around the mouth and fingertips and sometimes fainting.

**HYPervisor (VIRTUAL MACHINE MANAGER):** A software program that allows multiple operating systems, called guests to share a single computer system, called host. The hypervisor manages the system's processor, memory, and other resources to allocate what each guest or operating system requires.

**HYPHA:** A delicate usually long, branching filament, the main mode of vegetative growth, many of which collectively form the mycelium and fruiting bodies of a fungus.

**HYPHO-:** An affix used in names to designate an open structure, especially a boron skeleton, more closed than a klado structure but more open than an arachno structure.

**HYPNOSIS:** An artificially induced mental state that resembles sleep but is induced. It is characterised by an individual's loss of critical powers and consequent openness to suggestion. It may be induced by an external agency or by the individual herself (autohypnosis). Hypnosis has been widely used in medicine (usually to induce analgesia) and especially in psychiatry and psychotherapy.

**HYPHO-:** A prefix denoting under, low, below, for example, hypoglycaemia or low blood glucose.

**HYPHO-PHASE:** The denser phase in an extraction system. This term is often used when two non-aqueous phases are present or when the solvent is an aqueous phase.

**HYPOBROMOUS ACID (BROMANOL; BROMIC(I) ACID; HYDROXIDOBROMINE; OXOBROMATE(I) ACID;):** A yellow liquid, with the chemical formula  $HBrO$ . A weak unstable acid and a strong oxidising agent, it occurs only in solution.

**HYPOCAMPUS:** A part of the vertebrate brain associated with memory consisting of two ridges in each cerebral hemisphere of the brain. It is highly developed in advanced mammals and its functions appear to be related to the expression or response that generates emotions such as anger and fear and short-term memory and forming part of the limbic system.

**HYPOCENTRE (FOCUS):** A point beneath the Earth's surface, directly below the epicentre where the vibrations of an earthquake originate; the rocks start to fracture at this spot, then vibrating waves radiate outward in all directions.

**HYPOCHLORITE:** Salt of hypochlorous acid. Examples are sodium hypochlorite and calcium hypochlorite used for bleaches and disinfectants.

**HYPOCHLOROUS ACID (CHLORIC(I) ACID; CHLORANOL, HYDROXIDCHLORINE; HYDROGEN HYPOCHLORITE; CHLORINE HYDROXIDE):** Weak liquid acid with the chemical formula  $HClO$ , stable only in solution or a weak unstable greenish yellow acid that occurs only in solution or in its salts. Its salts are hypochlorites. It is obtained by dissolving chlorine in water. It is used as a bleach, an oxidiser, a deodorant and a disinfectant.

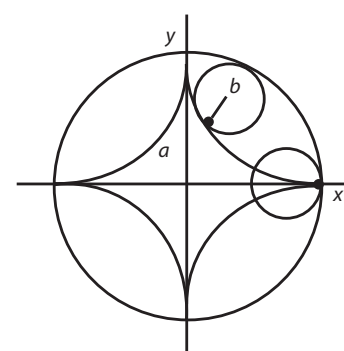
**HYPOCHONDRIA (HYPOCHONDRIASIS):** The persistent conviction that one is or is likely to become ill, often involving symptoms when illness is neither present nor likely and persisting despite reassurance and medical evidence to the contrary.

**HYPOCOTYL:** The region of a germinating seedling of a seed plant, which is the stem beneath the stalk of the seed leaves and directly above the young root of an embryo plant. It sends out a shoot called a **radicle** that becomes the primary root and penetrates down into the soil. It grows rapidly in seedling showing epigeal germination.

**HYPOCRATERIFORM:** 1. Shaped like a salvar. 2. In botany, it is an epithet applied to a corolla consisting of a straight tube surmounted by a flat and spreading limb.

**HYPOCYCLOID:** The path described by a point on the circumference of a circle as it rolls around the inside of another coplanar circle, of which the form depends on the radii of the circles. A common hypocycloid has the

equation  $x^{\frac{2}{3}} + y^{\frac{2}{3}} = 3^{\frac{2}{3}}$ .



*Hypocycloid*

**HYPODERMIC SYRINGE:** A type of piston pump consisting of glass or plastic tube into which a plunger has been inserted that fits tightly. A thin hollow needle is attached to the plunger and used to withdraw fluids especially blood, from under the skin or to inject medicine under the skin.

**HYPODERMIS (EXODERMIS; HYPODERM; SUBCUTANEOUS TISSUE; SUPERFICIAL FASCIA):** The outermost layer of cells in the plant cortex, lying immediately below the epidermis consisting of cells such as fibroblasts, adipose cells and macrophages. It is also a layer of cells lying immediately below the epidermis of plants.

**HYPOGEAL:** 1. A term used to describe seed germination in which the cotyledons remain inside the seed shell underground without photosynthesis. Examples of such germinations are found in dicotyledon plants such as pea, gram, broad bean, scarlet runner bean, mango and jack-fruit; and in monocotyledons such as maize, rice, wheat and coconut. The other type of germination is the **epigeal** in which the cotyledons come out of the underground and germinate. Examples of such plants include sunflower, castor, mustard and neem. 2. In water purification works, the **hypogeal layer** is a biological film just below the surface of slow sand filters. It contains microorganisms that remove bacteria and trap contaminant particles.

**HYPOGLOSSAL NERVE:** The twelfth cranial nerve leading to the tongue and controlling its movements. It arises in the medulla oblongata and leaves the skull through the hypoglossal canal in the occipital bone.

**HYPOGLYCAEMIA:** An abnormally low concentration of glucose in the blood plasma, causing weakness, dizziness and sweating.

In healthy people hypoglycaemic occurs only after a prolonged fasting or heavy exercise without adequate intake of food. Severe hypoglycaemia can occur in people with diabetes whose insulin dosage has been excessive relative to food intake or who has been involved in prolonged fasting or heavy exercise without adequate intake of food. Hypoglycaemia can be treated by eating sugars or by glucose infusion into a vein. If not treated the victim may lose consciousness, which may result in death.

**HYPOGYNY:** A floral arrangement in which the floral parts, such as sepals, petals and stamens, are borne on the receptacle beneath the ovary. The perianth and the stamens are said to be hypogynous with respect to the ovary as seen in tulip.

**HYPOLIMNION:** In a thermally stratified lake, it is the dense, bottom layer of water that lies below the thermocline. At this depth, it receives insufficient light for photosynthesis to occur.

**HYPOMAGNESAEMIA (GRASS STAGGERS):** An electrolyte disturbance caused by lack of magnesium in the bloodstream that makes animals shiver and stagger. Cattle may be affected shortly after being turned out onto spring pastures after having wintered indoors. It may also be caused by conditions such as inadequate intake of magnesium, chronic diarrhoea, malabsorption, alcoholism, chronic stress, diuretic use and other disorders.

**HYPONASTY:** A nastic movement in which the plant responds by more rapid growth on the lower side of an organ, resulting in a curving upwards of that part.

**HYPOPLASIA:** Underdevelopment as a result of disease or nutrient deficiency, giving dwarfed or stunted plants. In some cases particular parts of the plant may develop abnormally or not at all.

**HYPORHEIC ZONE:** The region beneath and alongside the bed of a stream or river, where the stream water and ground water mix.

**HYPOTASIS:** The situation in which the expression of one gene (the hypostatic gene) is dependent on a second gene, the epistatic gene. The genes need not be on the same chromosome.

**HYPOTRACUM:** The shell located below all other shell layers in some molluscs.

**HYPOTENSION (LOW BLOOD PRESSURE):** In medicine, an abnormally low blood pressure, usually characterised by dizziness, lightheadedness, lack of concentration, blurred vision, rapid and shallow breathing, fatigue, etc. Some of the causative factors include pregnancy, some heart conditions, anaemia, low blood sugar, dehydration, blood loss, severe infection such as septicemia (when an infection in the body enters the bloodstream). **Postural hypotension** refers to the sudden fall in blood pressure that can occur on standing up suddenly.

**HYPOTENUSE:** The longest side of a right-angled triangle, opposite the right angle or the side of a right-angled triangle that forms a square equal to the sum of the squares of the lengths of the other two sides. It is of particular application in Pythagoras' theorem.

**HYPOTHALAMUS:** A part of the brain of a vertebrate rich in ganglia, nerve fibres and synaptic connections that controls such things as hunger, thirst, body temperature, blood pressure, heartbeat, metabolism of fats and carbohydrates and sugar levels in the blood, sleeping and osmoregulation.

**HYPOTHERMIA:** Having an abnormally low body temperature, at least 2 °C (275 K). It occurs mainly in young and old people when body heat is lost more rapidly to the surroundings than it can be maintained.

**HYPOTHESIS:** 1. A temporary working explanation or supposition or a possible explanation for an observation or an idea concerning an

event, which can be tested by experiment. It is often a generalisation from the results of an experiment or observation and it is the first step on the way to formulating a theory. 2. An assumption that can be verified as true or false.

**HYPOTHESIS TESTING:** A general term for the procedure of accessing whether sample data is consistent or otherwise with statements made about the population.

**HYPOTHETICAL SYLLOGISM:** An argument of the form if  $P$  then  $Q$ , if  $Q$  then  $R$ , so if  $P$  then  $R$ .

**HYPOTHYROIDISM:** A condition in humans and in vertebrates caused by insufficient production of thyroid hormones by the thyroid gland or to a deficiency in thyroid stimulating hormones. In adults, it causes a decrease in basal metabolic rate with tissue swelling and weight gain. In children it results in cretinism.

**HYPOTONIC:** 1. A hypotonic solution is one in which the total molar concentration of all dissolved solutes is less than the concentration of these solutes in another solution. Also, a solution whose osmotic pressure is lower than the medium on the other side of a selectively permeable membrane. Such a solution will lose water by osmosis. 2. A low or diminished muscle tone or tension.

**HYPOTONIC SOLUTION:** A solution that has lower solutes or osmotic pressure than some other solution. In biology, a hypotonic solution refers to a solution that contains less solute (more water) compared to the cytoplasm of the cell. If the solution surrounding the cell is hypotonic, osmosis causes water to have a net flow into the cell, thus, resulting in the swelling and expansion of the cell. When an animal cell is set to a hypotonic environment the cell will eventually lyse (rupture) due to the osmotic pressure. In a plant, the cell will not lyse but become turgid because of its cell wall that prevents it from bursting. It is the osmotic pressure (or turgor pressure) that helps keep the plant from wilting and losing its shape.

**HYPOTROCHOID:** A curve formed by the path of a point attached to a point  $c$ , which is not on the circumference, of circle of radius  $b$  that rolls around the inside of a larger circle of radius  $a$ . The parametric equations for a hypotrochoid can be written as:

$$x = (a - b) \cos t + c \cos(a/b - 1)t, y = (a - b) \sin t - c \sin(a/b - 1)t.$$

The hypotrochoid is a generalised hypocycloid and comes in two varieties: a **prolate hypocycloid** if the starting point is outside the circumference of the rolling circle and a **curtate hypocycloid** if the starting point is inside the rolling circle. In the same family of curves as the hypotrochoid (and hypocycloid) are the epicycloid and epitrochoid.

**HYPOVENTILATION:** An insufficient exchange of gases due to a reduced amount of air entering the pulmonary alveoli resulting in decreased levels of oxygen and increased levels of carbon dioxide in the blood. It is usually characterised by shallow breathing, headaches, faintness, sleepiness at night and daytime sleepiness. It may be caused by obesity, defective thorax, injury to the respiratory centre and prolonged use of certain drugs. If not treated, it can lead to organ damage and death.

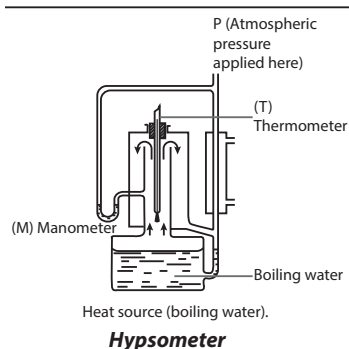
**HYPOXIA:** A deficiency of oxygen in the body tissues, which can result from living in an oxygen-deficient environment such as high altitudes, inadequate inspiration or deficiency of red blood cells or haemoglobin.

**HYSOCHROMIC SHIFT (BLUE SHIFT):** A reduction of absorption, reflectance, transmittance or emission spectrum in the wavelength (higher frequency) of a particular spectral band of a substance as a



result of chromophoric substitution changes in the medium and other physical conditions. Because the blue colour in the visible spectrum has a lower wavelength than most other colours, this effect is also commonly called a **blue shift**. Introduction of electron withdrawing groups into organic compounds shifts their spectral bands towards high energy by reducing the tendency for delocalisation.

**HYPSONETER:** An instrument used for calibrating or testing the accuracy of a thermometer at the boiling point of water at a known height above sea level. It is also used for estimating height above sea level by finding the temperature at which water boils. Liquids boil at progressively lower temperatures as the atmospheric pressure decreases and since atmospheric pressure decreases with altitude, the temperature at which the liquid boils is an indicator of the atmospheric pressure at that location and hence of the location's altitude. It comprises of an L-shaped frame with a plumb or level and an adjustable straightedge.



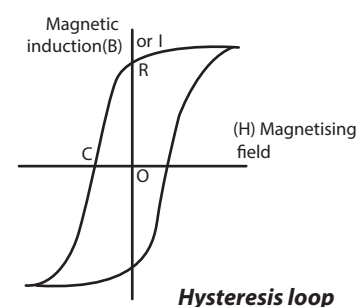
**HYRACOIDEA:** An order of the Mammalia that includes the tree and rock hyraxes, also known as dassies and coneys. They are small and similar in appearance to rodents but are related to the Perissodactyla and Proboscidea. The front feet have four functional toes and the back feet only three.

**HYSTERECTOMY:** Removal of the uterus, possibly with surrounding structures. It is performed to treat fibroids or cancer or to put an

end to heavy menstrual bleeding. A woman who has undergone hysterectomy does not menstruate and is no longer fertile.

**HYSTERESIS:** 1. A phenomenon in which two physical quantities are related in a manner that depends on whether one is increasing or decreasing in relation to the other. 2. A phenomenon observed in the elastic and electromagnetic behaviour of materials in which a lag occurs between the application or removal of a force or field and its effect. For example, if the magnetic field applied to a magnetic material is increased and then decreased back to its original value, the magnetic field inside the material does not return to its original value.

A **hysteresis loop** is a closed curve obtained by plotting a graph of the magnetic induction of a ferromagnetic substance



(as ordinate) against the external magnetic field. The shape of the curve is characteristic of the magnetic properties of the material and shows the ease with which it is magnetised and the ability to retain magnetisation.

**Hysteresis coefficient** is a constant, characteristic of a particular material in a formula for hysteresis loss. Hysteresis loss in a transformer is expressed by Steinmetz's empirical formula as,  $P_h = k_h f B_m^n W$ , where  $k_h$  is the hysteresis coefficient,  $B_m$  is the maximum flux density and  $n$  is the Steinmetz constant.

**HYSTERIA:** A mental disorder characterised by symptoms such as paralysis, blindness, loss of speech, recurrent cough, clenched teeth, swelling of the neck vomiting and general malaise. The causes include psychological disorders, physical disorders, brain disorders, dementia, brain tumour, alcoholism, schizophrenia, depression, use of some drugs, etc.



**i:** 1. Also written in upper case, a symbol representing the number 1 in the Roman numerals. 2. A notation in vectors, conventionally printed in bold typeface, which represents unit vector, normally taken as the  $x$ -axis or the horizontal vector. 3. The identity function, for which  $I(x) = x$  for all  $x$ . 4. It is used to denote the identity element of a group; although the symbol  $e$  is much more usual. 5. Unit of quaternions.

**i:** The symbol used to denote the imaginary number as the principal square root of  $-1$ .

**I:** 1. Also written in lower case, a symbol representing the number 1 in the Roman numerals. 2. Symbol for the identity matrix. For example, the  $n \times n$  unit matrix is denoted  $I_n$  to show the rank of the matrix. 3. Symbol for current in an electrical circuit.

**I-BAND (I BAND):** 1. Also known as **isotropic band**, it is the region of striated muscle fibre that contains only thin filaments. The thin filament is an assembly of three proteins: actin (the main one), tropomyosin and troponin. I bands are visible as light bands at either end of a sarcomere. Unlike the A-band, the I-band can vary in width depending upon the state of contraction of the muscle when fixed. **Striated muscle** also called **skeletal muscle**, **striped muscle** or **voluntary muscle** is attached to bone and produces all the movements of body parts in relation to each other. Unlike smooth and cardiac muscles, striated muscle is under voluntary control. 2. A range of radio frequencies from 8 GHz to 10 GHz in the electromagnetic spectrum, equal to wave lengths between 3.75 cm and 3 cm. Marine navigation radars often operate with I band frequencies.

**I-GENE (REGULATORY GENE):** A gene whose product can prevent or promote the transcription of other genes. The other genes may or may not be adjacent to the regulator gene.

**I/O (INPUT/OUTPUT):** In computer terminology, it refers to any program, operation or device that transfers data to or from a computer and to or from a peripheral device. Every transfer is an output from one device and an input into another. Keyboards and mouse devices are input-only, while devices such as printers are output-only. A writable CD-ROM is both an input and an output device. **I/O adapter** is an adapter that converts between the timing and protocol requirements of an intelligent device's memory bus and those of an I/O bus or network.

**IAEA (INTERNATIONAL ATOMIC ENERGY AGENCY):** Set up in 1957, it is an organisation within the United Nations for cooperation in the nuclear field to promote the safe, secure and peaceful use of nuclear technologies. Its headquarters is in Vienna, Austria.

**IATROCHEMISTRY:** An early branch of chemistry, having roots in alchemy, that tried to provide chemical remedies to diseases.

**IATROGENIC DISEASE:** Any condition, complication or symptoms that may be induced in a patient by medical treatment, the facilities or comments of the medical staff.

**IBM (INTERNATIONAL BUSINESS MACHINES):** A US multinational and consulting company, which is the world's largest information technology company. Its revenue in 1914 was USD \$93 billion. It was formed in 1911 as the Computing-Tabulating-Recording

Company (CTR) by a merger of The Tabulating Machine Company (formed in 1896 by US inventor Herman Hollerith to produce punch card tabulating machines) and some other companies. IBM adopted its present name in 1924. Its present products include hardware and software, storage products, microchips, network hosting and IT consulting services.

**ICACINACEAE:** A tropical family of flowering plants of the order Sapindales consisting of 60 genera, with about 320 species, occurring mainly in the tropics, mostly in rain forests. They are trees, shrubs and lianas. They have alternate leaves without stipules, flowers in panicles, and the fruits are mostly drupes. Some of the species in the family have some medicinal properties. For example, *Icacina trichantha* has antibacterial activities against microorganisms such as *Escherichia coli*. The plant is also used traditionally in some communities to treat ailments like soft tumours, poisoning, constipation, and malaria, to induce emesis and as an aphrodisiac.

**ICANN (INTERNET CORPORATION FOR ASSIGNED NAMES AND NUMBERS):** A non-profit organisation that oversees the assignment of IP addresses, managing the domain name system, allocation of Internet numbering resources and other services. These were previously performed by the Internet Assigned Numbers Authority (IANA).

**ICE:** The solid form of water and water vapour when it freezes or forms solid crystalline water with melting point of  $0^{\circ}\text{C}$  ( $273\text{ K}$ ) at 1 atmosphere of pressure. Water vapour forms frost on surfaces and snowflakes in clouds.

**ICE AGE (GLACIAL AGE):** A period in the Earth's history during which ice advanced towards the equator and a general lowering of temperatures occurred resulting in an expansion of continental ice sheets, polar ice sheets and alpine glaciers. The last major ice age ended about 10,000 years ago.

**ICE CAP (ICECAP):** A dome-shaped or plate-like body of ice that covers less than  $50,000\text{ km}^2$  making it larger than a glacier but smaller than an ice sheet (it covers more than  $50,000\text{ km}^2$ ). Ice caps form from the accumulation of snowfall over many years, followed by compression of the snow as the weight of new snow presses down on previous years' fall. Ice caps cover mountain ranges, such as the Alps or small islands. Glaciers often originate from icecaps. Few organisms have adapted to life on an ice cap, although many plants and animals live on its periphery.

**ICE POINT:** The temperature at which pure water freezes or pure ice melts without change in temperature. At this temperature, pure ice and water mixture are in equilibrium at standard atmospheric pressure. It is used as a fixed point ( $0^{\circ}\text{C}$ ) on the Celsius scale for defining temperature scales and for calibrating thermometers. To calibrate the ice point of a thermometer, crushed ice is placed inside a funnel that has a beaker or container placed below it to collect dripping water. The thermometer is pushed inside the ice until the top of the mercury thread in the thermometer is just above the ice. When the top of this mercury thread remains stationary for a long period of time, its position is marked on the stem of the thermometer and labelled  $0^{\circ}$  as the lower fixed point.

**ICE SHEET:** A body of ice covering a large land mass or continent for a long period of time that is more than  $50,000\text{ km}^2$ , making it larger than an ice cap. Ice sheets cover much of Antarctica and Greenland.

